

PRESERVING VERIFIED OUTPUT MODES IN RLVR: MODEBENCH AND REPLAYMAXRL

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ABSTRACT

Mathematical agents need candidate solutions for search, verification, and revision. Alternative routes may help when an approach fails or constraints change. Yet reinforcement learning with verifiable rewards (RLVR) can improve correctness while narrowing the model’s repertoire. We introduce MODEBENCH, spanning five executable domains whose validators report both correctness and a canonical solution identity. This makes the loss of verified alternatives measurable alongside $\text{pass}@K$. ReplayMaxRL retains policy-generated verified outputs by identity and replays them with balanced weights while optimizing success on fresh samples. A categorical analysis explains why binary on-policy updates amplify frequent correct modes and how recurrent replay counters this tendency. Across three model scales, canonical replay improves held-out correctness and retains a broader set of sampled outputs than verifier-only RL. At the two scales with complete MaxRL comparisons, replay also complements sampling-aware learning. ModeBench thus provides a controlled setting for studying two requirements of mathematical reasoning: discovering successful solutions and preserving alternatives an agent can still propose for verification and further search.

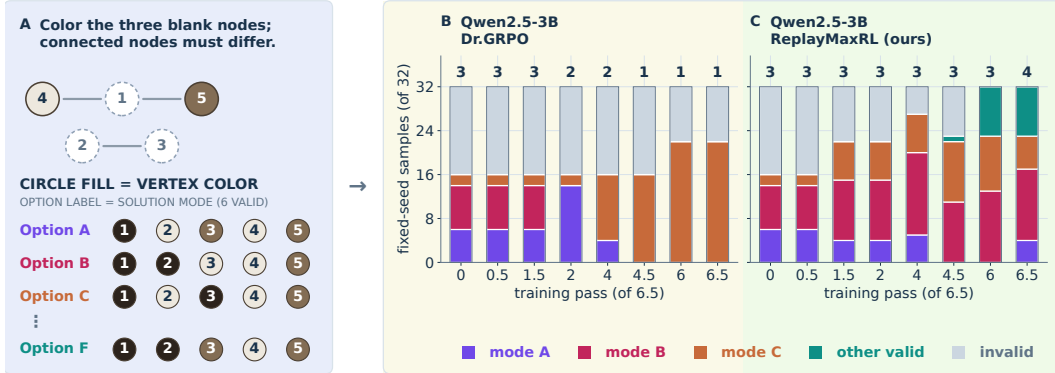


Figure 1: **Correctness can improve while verified solution support collapses.** (A) One prompt admits six valid colorings; four are shown. Circle fill marks vertex color and a dashed ring marks a vertex the prompt leaves blank; label color marks solution mode. (B–C) Each bar pools four fixed-seed $K = 8$ draws (32 samples) from matched Qwen2.5-3B-Instruct runs. Both start at 16/32 correct across three modes. By pass 4.5, Dr.GRPO has collapsed to one; at pass 6.5, it returns 22/32 correct from that mode, while ReplayMaxRL returns 32/32 across four. Numbers above bars are pooled unique verified keys at the corresponding training pass.

1 INTRODUCTION

Mathematical reasoning agents generate candidate proof or computation routes, check them with tools or verifiers, and revise after feedback (Brown et al., 2024; Snell et al., 2024). Alternative routes may support recovery when a familiar approach fails or constraints change, and give self-correction another option to try. A verifier can only select from the candidates the model still

supplies. This motivates a question about the model inside an agent: does training preserve the range of verified options on which search and revision can draw?

Reinforcement learning with verifiable rewards (RLVR) trains from executable correctness feedback (Guo et al., 2025). Reasoning RL can reduce policy entropy and output diversity (Cui et al., 2025; Dang et al., 2025). A correctness-only reward treats repeated and distinct successes alike; we show that accuracy can rise while verified-mode support contracts. Repeated sampling supplies proposals for verifier-guided selection (Brown et al., 2024; Snell et al., 2024; Cobbe et al., 2021; Lightman et al., 2024). Concentration leaves fewer verified alternatives, without necessarily reducing $\text{pass}@K$ at fixed per-sample correctness. Figure 1 exhibits this failure on one prompt; Appendix Figure 10 shows it is systematic. Eight passes of unmodified Dr.GRPO or GRPO remove 99% and 98% of the base model’s verified breadth beyond its first correct mode at Qwen2.5-0.5B, and 42–72% of it at the two larger scales. In an idealized model, our analysis shows that additional parameters can preserve more sampled alternatives at matched accuracy when they strengthen updates shared across correct outputs (Appendix A.3, Theorem A.5); size alone provides no guarantee.

Measuring this loss requires identifying which successful output occurred. Different strings can encode the same executed solution, while different computations can reach the same correct answer. ModeBench therefore asks which verified alternatives remain sampleable, alongside whether any sample succeeds. Its canonical keys make the same identity useful for measurement and memory.

The failure separates two requirements that are often treated as one. Training must continue to discover correct behaviors, but it must also preserve a useful behavior after on-policy sampling makes it rare. Sampling-aware objectives, entropy bonuses, and outcome exploration address continued discovery (Cheng et al., 2025; Song et al., 2025; Chen et al., 2025); replay and quality-diversity methods address the second (Lin, 1992; Mnih et al., 2015; Ecoffet et al., 2021; Mouret & Clune, 2015). A binary reward alone lacks the execution-grounded identity needed to support both: without knowing which correct behavior occurred, one cannot measure its loss or deliberately retain it. We make three contributions:

- **ModeBench measures verified output support.** We introduce five executable domain families with two matched difficulty levels each. Validators return both reward and a canonical key. Countdown and MathIR keys distinguish computation or derivation routes; Python keys distinguish observed input–output behaviors; Graph and PantryPlan keys distinguish valid configurations. This makes collapse auditable with $\text{distinct}@K$ alongside $\text{pass}@K$.
- **ReplayMaxRL explores and preserves.** MaxRL directs fresh-rollout learning toward success across sampling budgets (Tajwar et al., 2026); key-balanced verified replay retains one exemplar per discovered mode and revisits modes uniformly. A categorical mean-flow calculation connects the objectives to standard replicator dynamics: binary MaxRL inherits binary GRPO’s winner-take-all instability, while an idealized fixed replay bank adds a qualitative barrier against exact extinction.
- **Matched evidence isolates the value of memory.** Across five domains and three model scales, replay improves both endpoints in all 14 complete five-seed blocks and the remaining four-seed block; raw $\text{distinct}@8$ rises in all 74 admissible pairs. A Qwen2.5-0.5B comparison tests canonical replay against five direct alternatives. In the central factorial, ReplayMaxRL exceeds MaxRL on both endpoints in every Qwen2.5-0.5B domain. Falcon3-1B improves in all five domains on mean $\text{pass}@8$ and four on raw $\text{distinct}@8$, with positive averages across domains on both endpoints.

2 RELATED WORK

Verifiable learning and output support. Sampling-based selection depends on the proposal distribution (Brown et al., 2024; Snell et al., 2024). Prior work measures reduced output diversity under RLHF or reasoning RL (Kirk et al., 2024; Dang et al., 2025), declining policy entropy (Cui et al., 2025), and reduced high-budget problem coverage for tested RLVR models (Yue et al., 2025). These metrics differ from within-prompt execution-key breadth. Semantic clustering infers answer equivalence (Farquhar et al., 2024); MODEBENCH defines keys by accepting execution.

Sampling-aware exploration. Outcome bonuses, entropy control, set objectives, distribution matching, and uniform-correct objectives encourage broader verified-output support (Song et al., 2025; Chen et al., 2025; Li et al., 2025; 2026a;b; Lochab et al., 2026). MaxRL combines $\text{best}@k$ objectives

across sampling budgets (Tajwar et al., 2026). For our binary rewards, `best@k` equals `pass@k`; ReplayMaxRL tests whether explicit mode memory adds support beyond this fresh-rollout objective.

Replay and persistent memory. Replay reuses experience and can mitigate forgetting (Lin, 1992; Rolnick et al., 2019); RLEP replays successful reasoning trajectories (Zhang et al., 2025). ReplayDr.GRPO retains one exemplar per executed key and samples keys uniformly, regardless of discovery frequency. Novelty search rewards behavioral novelty (Lehman & Stanley, 2011); MAP-Elites retains one elite per descriptor cell (Mouret & Clune, 2015). Our replay bank uses execution keys as descriptors within policy-gradient updates.

3 CORRECTNESS AND VERIFIED SUPPORT

Correctness and solution support are distinct properties of a policy. For a prompt x with reference specification s_x , the validator maps a response to either failure or a canonical key:

$$V(\cdot, s_x) : \mathcal{Y} \rightarrow \mathcal{C}_x \cup \{\perp\}.$$

Here $\mathcal{C}_x$ is the set of valid solution keys and $\perp$ denotes failure. The learner observes a key only for a response it generates; neither the full set $\mathcal{C}_x$ nor its size is revealed.

For $Y \sim \pi_\theta(\cdot | x)$, the policy induces both a total probability of verified success and a distribution over how that probability is allocated:

$$P_\theta^+(x) = \Pr[V(Y, s_x) \neq \perp], \quad p_\theta^+(c | x) = \Pr[V(Y, s_x) = c | V(Y, s_x) \neq \perp].$$

The conditional distribution is defined only when $P_\theta^+(x) > 0$; when $P_\theta^+(x) = 0$, all three sampling metrics below are zero. The first quantity measures correctness. The second describes the distribution of verified behaviors conditional on being correct. We call concentration of p_θ^+ onto fewer keys *mode collapse*, particularly when P_θ^+ remains stable or increases.

3.1 MEASURING SUCCESS AND SUPPORT

For K independent samples $Y_{1:K}$, let

$$D_K = |\{V(Y_i, s_x) : i \leq K\} \setminus \{\perp\}|, \quad F_K = K^{-1} \sum_i \mathbf{1}[V(Y_i, s_x) \neq \perp].$$

ModeBench reports three averages with different units. `distinct@K` = $\mathbb{E}[D_K]$ is an expected count in $[0, K]$ of unique verified keys; `pass@K` = $\Pr[D_K > 0]$ and `mean@K` = $\mathbb{E}[F_K]$ are probabilities in $[0, 1]$. For every draw, $F_K \leq \mathbf{1}[D_K > 0] \leq D_K$; hence `mean@K` $\leq$ `pass@K` $\leq$ `distinct@K`. The inequality is valid, but equal numerical changes are not commensurate: .02 in `pass@8` is two probability points, whereas .02 in `distinct@8` is .02 expected verified keys.

For a fixed prompt, write $\mu_c(x) = \Pr[V(Y, s_x) = c]$. Independence gives

$$\text{pass@K}(x) = 1 - (1 - P_\theta^+(x))^K, \quad \text{distinct@K}(x) = \sum_{c \in \mathcal{C}_x} [1 - (1 - \mu_c(x))^K].$$

These identities apply before averaging over prompts; applying the nonlinear formula to an average success probability generally gives a different result.

Raw `distinct@K` measures sampled coverage, not the total number of positive-probability modes. For an uninformed reference, suppose a finite space has N actions and n_c actions map to valid key c . Uniform independent action sampling then gives $\sum_c [1 - (1 - n_c/N)^K]$ expected verified keys. The simpler expression $m_x[1 - (1 - 1/N)^K]$ applies when the m_x valid actions have distinct keys, as in our reported finite-action references. Open-ended program and derivation spaces have no canonical nominal “uniform” generator.

The registered terminal co-primary endpoints are `pass@K`, the probability of finding any correct output, and raw `distinct@K`, the number of verified keys found. We report both separately for each model–domain–level block. The derived contrast `distinct@K` – `pass@K` = $\mathbb{E}[D_K - \mathbf{1}[D_K > 0]]$ is an interpretive decomposition: it estimates the number of verified modes beyond the first. It is computable without knowing the full valid set $\mathcal{C}_x$ and remains coupled

to the number of successful draws. We use it as an interpretive decomposition rather than a fully success-matched diversity measure or a third primary endpoint. Because every headline evaluation fixes $K = 8$, reporting $\text{distinct}@8/8$ would add no information beyond a linear rescaling; we retain the raw count and label its units explicitly.

3.2 WHY BINARY REWARDS AMPLIFY COMMON MODES

GRPO (Shao et al., 2024) and Dr.GRPO (Liu et al., 2025) center binary rewards within sampled groups, giving correct responses equal task credit regardless of execution key. Common correct modes therefore appear in more updates; absent modes supply no direct signal.

A categorical reduction makes this feedback loop explicit. Let p_c denote the probability of correct mode c and $P = \sum_{c \in \mathcal{C}} p_c$ the total correct mass. Under Euclidean gradient ascent in independently trained softmax logits on expected binary reward, two correct modes c and d satisfy

$$\frac{d}{dt} \log \frac{p_c}{p_d} = (p_c - p_d)(1 - P).$$

Whenever incorrect mass remains, $p_c > p_d$ causes the log-odds gap to grow: binary reward magnifies an existing imbalance rather than correcting it. Finite on-policy groups add two sampling effects. An unsampled mode supplies no direct score term, while an all-correct group has zero centered task advantage. The full derivation and the GRPO/Dr.GRPO equivalence appear in Appendix A. The next section supplies an executable identity for correct behavior, and Section 5 turns that identity into persistent training state.

4 MODEBENCH: EXECUTABLE MODES ACROSS LEVELS

A domain is mode-verifiable when the same execution both accepts a response and produces its canonical identity. Cosmetic aliases must merge; different executed outcomes must remain distinct. MODEBENCH instantiates this contract in five domain families. Each family is generated rather than fixed: we release two disjoint levels with the same validator and canonical-key definition, each containing 384 training prompts and a fixed 128-prompt evaluation split.

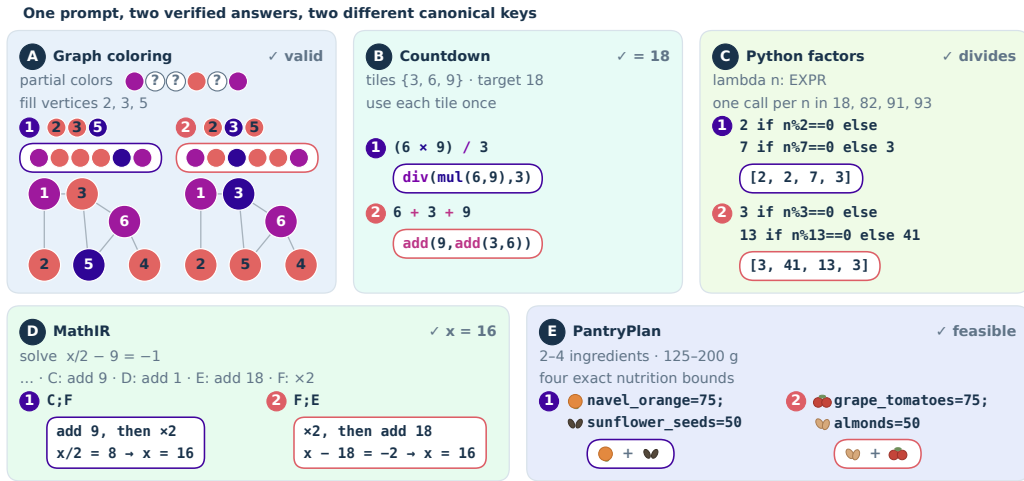


Figure 2: **MODEBENCH turns successful answers into executable modes.** Each panel shows two correct responses to one prompt and the distinct canonical keys produced by the accepting execution. Identity is a color vector in Graph, a normalized expression tree in Countdown, an executed return vector in Python, a state trajectory in MathIR, and an ingredient support in PantryPlan. Surface aliases merge, while execution-level behavioral differences remain distinct.

Figure 2 illustrates the five executable domains at Level 1:

- **Graph coloring** completes a partial three-color assignment. Execution checks every fixed color and edge; the full color vector is the key, yielding 6.27 valid modes per evaluation prompt on average across the fixed evaluation split of 128 prompts.
- **Countdown** builds an exact arithmetic expression from a fixed multiset of numbers. The key is its normalized executed syntax tree: commutative reorderings merge, while different computations remain distinct (4.52 modes per evaluation prompt).
- **Python factors** writes a restricted one-line function that is executed on frozen inputs. Its return vector is the key, so different source programs with identical behavior merge (229.44 modes).
- **MathIR** solves an equation by selecting executable algebra actions. The normalized state trajectory is the key, preserving different valid derivations of the same answer (5.00 modes).
- **PantryPlan** selects ingredients for a constrained plan; a trusted environment finds and validates exact feasible quantities. The selected ingredient support is the key, abstracting away quantity and order aliases (18.19 modes per evaluation prompt).

ModeBench targets task-defined executable output support. ReplayMaxRL stores one verified exemplar per observed task-defined key and samples uniformly over those keys. Appendices E, D, and F give the complete execution, alias, and split contracts.

Matched levels. Level 2 preserves each validator, key, and valid-mode-count histogram while adding structural difficulty: more hidden Graph vertices and Countdown operands, larger Python inputs; variable-on-both-sides MathIR equations; and tighter PantryPlan constraints. Frozen evaluation admits a level only when it is harder but solvable; Section 6 gives the design.

5 REPLAYMAXRL: EXPLORE AND PRESERVE VERIFIED MODES

ReplayMaxRL turns ModeBench keys into persistent training state without requiring the full valid support. Binary MaxRL learns from fresh rollouts across sampling budgets; key-balanced replay keeps previously verified modes available when they become rare or disappear from the current group.

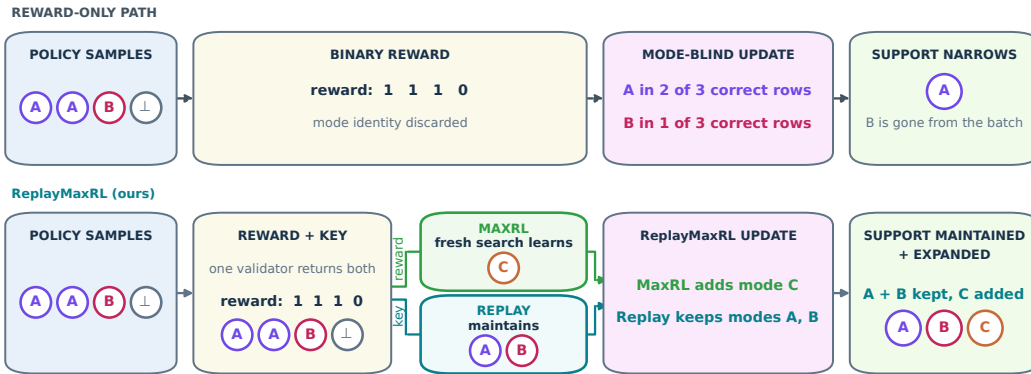


Figure 3: **ReplayMaxRL separates fresh search from retention.** Both paths begin with the same samples. Binary reward discards solution identity, allowing repeated modes to dominate. ReplayMaxRL instead sends reward to MaxRL and canonical keys to Replay. In this schematic, fresh learning adds C while replay revisits A and B. Endpoint gains support the design; tracking each stored mode throughout training would test its continued preservation directly.

5.1 EXPLORATION WITH BINARY MAXRL

For a fresh group of G binary rewards $r_{1:G}$, let $m = \sum_{i=1}^G r_i$. The implemented MaxRL task advantage is

$$A_i^{\text{MaxRL}} = \begin{cases} Gr_i/m - 1, & m > 0, \\ 0, & m = 0. \end{cases}$$

Applied only to fresh on-policy rows, this centered binary MaxRL estimator (Tajwar et al., 2026) assigns failures -1 when $m > 0$ and zero advantages to all-failure or all-success groups. Lemma A.2 derives its mean objective for $k = 1, \dots, G - 1$.

5.2 PRESERVATION WITH KEY-BALANCED VERIFIED REPLAY

Each prompt x owns an initially empty, capacity-16 bank $\mathcal{B}_x$. One policy-generated exemplar is chosen deterministically when a key is first seen; no gold solution or unverified proposal enters the bank. A round-robin cursor revisits one nonempty prompt bank per optimizer update.

For exemplar e_c of key c , with token length $|e_c|$, the replay loss is

$$L_{\text{replay}}(x) = -\frac{1}{|\mathcal{B}_x|} \sum_{c \in \mathcal{B}_x} \frac{\log \pi_{\theta}(e_c | x)}{|e_c|}. \quad (1)$$

Uniform sampling over keys prevents discovery frequency from determining replay frequency. To explain its relationship to the evaluation metric, we first analyze an equal-weight categorical surrogate, without length normalization. A stored response’s probability need not equal its key’s total probability. For $C = |\mathcal{B}_x|$ banked keys, write $p(c)$ for their probabilities and q_B for their total mass. Then

$$-\frac{1}{C} \sum_{c \in \mathcal{B}_x} \log p(c) = -\log q_B + \log C + \text{KL}(U_C \| p(c | c \in \mathcal{B}_x)). \quad (2)$$

The objective rewards greater banked verified mass and a more balanced conditional distribution, aligning with `pass@K` and `distinct@K`. The combined update need not improve both at every step. Indeed banked-key coverage contributes $\sum_c [1 - (1 - p(c))^K]$ to expected `distinct@K`; at fixed q_B and C , concavity makes it largest under uniform allocation, while $1 - (1 - q_B)^K$, the probability of any banked key, depends only on total mass. This deliberate objective–metric alignment motivates treating held-out `pass@K` as co-primary. Uniformity also maximizes the smallest conditional banked-key mass, making equal retention explicit. Banks contain only training-prompt responses; evaluation responses never enter training.

ReplayMaxRL combines the replay loss, with $\lambda_{\text{replay}} = .10$, with the fresh-rollout MaxRL update. Substituting Dr.GRPO for MaxRL yields ReplayDr.GRPO. Controls execute the same bank and scoring code with zero replay gradient; realized banks and workloads depend on the policy. Replay rows never enter the fresh task group. This separation supports the factorial comparison of objective and memory. Appendix G gives the implementation invariants.

6 EXPERIMENTAL DESIGN

Three experiments test replay retention and direct alternatives, cross fresh objective with memory, and transfer the factorial to a harder matched level. Appendix Table 13 reports comparator coverage.

Problem precheck: collapse without an intervention. We evaluate unmodified Dr.GRPO and GRPO at pass 0 and 8: 150 registered runs across 30 model–domain–method blocks of five seeds. Their five-domain macro-average of extra modes falls by 98.2%–99.4% at Qwen2.5-0.5B and 41.5%–71.6% at larger scales (Appendix Figure 10). `pass@8` falls in 10 of the 30 domain–method–scale comparisons, while macro `distinct@8` ends above its pass-0 reference at Qwen2.5-3B. The precheck characterizes the failure pattern before intervention comparison.

Shared protocol + evidence policy: Runs use 384 training prompts, group size 16, temperature 1, top- $p = 1$, one PPO epoch, $\beta_{\text{KL}} = 0$, and eight passes (3,072 updates). Evaluation uses a disjoint 128-prompt split, a separate temperature-0 greedy trace, and four fixed $K = 8$ draws at

temperature 1 and top- $p = 1$ at half-pass checkpoints. Samples are independent within a draw; draw $d \in \{0, 1, 2, 3\}$ uses the registered domain seed plus d . Paired methods share model revision, initialization, prompt order, optimizer, decoding, checkpoints, placement, and recovery rules.

The primary reporting unit is a five-seed model–domain–level terminal block. Complete blocks receive paired 95% Student- t intervals; smaller admissible intersections report descriptive means and exact n , without intervals. We average prompts and evaluation draws within each training seed before pairing. These are approximate, unadjusted intervals for training-seed variation conditional on the frozen evaluation prompts and seeds, not simultaneous bounds across comparisons. We do not substitute shallower checkpoints, impute final results, or pool models or levels. `pass@8` and `raw distinct@8` are co-primary; adjusted breadth is derived. Figure 4’s within-seed, equal-weighted domain Average is descriptive. Exploratory sign tests over complete block means assume independent block signs and use Holm adjustment. Appendix J gives the complete identities and fail-closed checks.

Exp. 1: We pair Dr.GRPO with ReplayDr.GRPO on five Level-1 domains, three model scales, and five seeds: 75 registered comparisons, of which 74 are admissible after one source-integrity exclusion (Appendix B.1). Methods differ only in the replay derivative; the control executes the same bank code with zero replay gradient. Fully matched Qwen2.5-0.5B blocks also provide direct alternatives: ordinary GRPO normalization; UCPO on-policy advantage redistribution (Lochab et al., 2026); our frequency-weighted RLEP-Dr adaptation (Zhang et al., 2025); memoryless binary MaxRL; and fixed Semantic-MaxEnt. Figure 4B compares them with ReplayDr.GRPO. The semantic estimator is defined only as a comparator in Appendix C.

Exp. 2: The central 2×2 factorial crosses the fresh objective (Dr.GRPO or binary MaxRL) with verified replay (absent or present), yielding Dr.GRPO, ReplayDr.GRPO, MaxRL, and ReplayMaxRL. The two within-objective replay contrasts identify memory; their difference is the replay-by-objective interaction. The MaxRL pair is registered for five seeds in all five Level-1 domains and three scales. Eleven MaxRL pair blocks are complete; the common four-arm intersection gives ten complete blocks and Falcon Countdown at four seeds. Claims use shared final pass-8 evaluations; partial blocks report exact n .

Exp. 3: ModeBench Level 2 preserves each domain’s validator, canonical key, split sizes, and split-wise valid-mode-count histogram while changing structure: four hidden Graph vertices, four Countdown operands, larger Python inputs, variable-on-both-sides MathIR equations, and tighter PantryPlan constraints. The disjoint Level-2 prompts receive the same four-method, five-seed Qwen2.5-0.5B factorial. We estimate ReplayDr.GRPO–Dr.GRPO, ReplayMaxRL–MaxRL, MaxRL–Dr.GRPO, and their interaction within level. Cross-level effect patterns compare the unpaired prompt sets.

7 RESULTS

The three experiments build one argument: replay retains verified support across scale, remains useful after stronger fresh-rollout optimization, and survives the first controlled increase in task difficulty. The main figures incorporate validated results through the September 6 analysis update. Terminal comparisons use their displayed matched seeds; the interim Level-1/Level-2 summary in Figure 6 compares matched training checkpoints.

7.1 EXPERIMENT 1: CANONICAL REPLAY RETAINS VERIFIED SUPPORT ACROSS SCALE

Replay improves success and sampled support across scales. Replay revisits a bank of verified solutions discovered during training. Both block-mean effects are positive in all 14 complete five-seed blocks. Falcon Countdown also favors replay in its four admissible pairs; its fifth replay endpoint is excluded for conflicting evaluations (Appendix B.1). Raw `distinct@8` rises in all 74 admissible pairs, so the result is not driven by one favorable domain or seed. At Qwen2.5-0.5B, ReplayDr.GRPO has the largest mean `pass@8` gain among all five direct alternatives in every domain.

On Countdown, replay raises correctness while preserving distinct routes to the same answer. At Qwen2.5-0.5B, `pass@8` rises from .480 to .672, while `distinct@8` more than triples from .487 to 1.637; adjusted breadth increases by +.959 (unadjusted 95% interval $[+.863, +1.054]$). Because Countdown keys encode computation trees, this is the clearest evidence for more sampled correct computation routes. It does not identify which previously discovered routes survived training. The other domains sharpen the boundary of that claim. Python improves from

.172 to .528 on pass@8, but adjusted breadth changes by only +.043 ($[-.077, +.164]$); MathIR improves from .512 to .796, with adjusted breadth +.030 ($[+.001, +.060]$). Thus the correctness benefit is broad, while the size and meaning of the support benefit depend on whether a key denotes a route, a functional behavior, or a valid configuration.

Balancing memory by key also matters. A matched ablation changes uniform replay weights to fresh-frequency weights. Uniform replay adds +.318 extra correct modes, measured by $\text{distinct@8} - \text{pass@8}$ and averaged across five Qwen2.5-0.5B domains (paired bootstrap 95% interval $[+.272, +.358]$). Graph, Countdown, and PantryPlan drive this advantage; Python and MathIR remain inconclusive (Appendix B.4). This tests key balancing while retaining the same bank construction and update code.

7.2 EXPERIMENT 2: REPLAY ADDS VALUE BEYOND MAXRL

Stronger fresh-rollout optimization does not eliminate the value of memory at either completed scale. Averaging the five domains within seed, ReplayMaxRL minus MaxRL is +.307 on pass@8 and +.991 raw distinct@8 at Qwen2.5-0.5B (unadjusted 95% intervals $[+.208, +.406]$ and $[+.815, +1.167]$). At Falcon3-1B, the corresponding effects remain positive at +.133 and +.228 ($[+.043, +.223]$, $[+.094, +.361]$). Because MaxRL already rewards success across sampling budgets, the remaining increment isolates memory: a verified key that disappears from fresh rollouts can still contribute through replay.

The gain follows the memory intervention, not a particular fresh objective. Both replay arrows move right on both cross-domain endpoints for the two completed factorial scales (Figure 5). On the Dr.GRPO track, replay adds (+.337, +.999) at Qwen2.5-0.5B and (+.442, +.774) at Falcon3-1B on pass@8/raw distinct@8; the latter uses four common admissible seeds without an interval. The completed Qwen2.5-3B Dr.GRPO track adds (+.279, +.559) across its five paired seeds. These cross-domain averages are post-hoc descriptive summaries. The domain breakdown in Appendix Figure 13 sharpens the boundary: ReplayMaxRL improves both endpoints in every Qwen2.5-0.5B domain and Falcon pass@8 in every domain, while Falcon raw support improves

A ReplayDr.GRPO (ours) – Dr.GRPO across three model scales

$\Delta \text{pass@8}$							$\Delta \text{distinct@8}$						
Qwen 0.5B	+0.65	+0.19	+0.36	+0.28	+0.21	+0.34	Qwen 0.5B	+2.11	+1.15	+0.40	+0.32	+1.02	+1.00
Falcon 1B	+0.15	+0.31 [†]	+0.98	+0.64	+0.12	+0.44 [†]	Falcon 1B	+0.68	+0.42 [†]	+1.11	+0.66	+0.94	+0.77 [†]
Qwen 3B	+0.15	+0.14	+0.59	+0.41	+0.11	+0.28	Qwen 3B	+0.58	+0.33	+0.78	+0.43	+0.67	+0.56
	Graph	Countdown	Python	MathIR	Pantry	Average		Graph	Countdown	Python	MathIR	Pantry	Average

B Initial reference + alternatives – terminal Dr.GRPO at Qwen2.5-0.5B

$\Delta \text{pass@8}$							$\Delta \text{distinct@8}$						
Before training	+0.13	-0.38	-0.17	-0.29	+0.38	-0.07	Before training	+0.24	-0.38	-0.17	-0.29	+1.90	+0.26
ReplayDr.GRPO (ours)	+0.65	+0.19	+0.36	+0.28	+0.21	+0.34	ReplayDr.GRPO (ours)	+2.11	+1.15	+0.40	+0.32	+1.02	+1.00
MaxRL (no replay)	+0.21	+0.10	+0.00	-0.07	+0.01	+0.05	MaxRL (no replay)	+0.35	+0.13	+0.00	-0.07	+0.01	+0.09
UCPO	-0.00	+0.13	+0.00	+0.03	+0.05	+0.04	UCPO	-0.00	+0.18	+0.00	+0.03	+0.05	+0.05
RLEP-Dr	+0.10	+0.10	+0.00	+0.13	+0.00	+0.07	RLEP-Dr	+0.15	+0.13	+0.00	+0.12	+0.00	+0.08
Semantic-MaxEnt	+0.00	+0.14	+0.00	-0.12	+0.20	+0.04	Semantic-MaxEnt	+0.00	+0.18	+0.00	-0.12	+1.12	+0.24
GRPO	+0.04	+0.16	+0.00	-0.05	+0.02	+0.03	GRPO	+0.05	+0.17	+0.00	-0.05	+0.02	+0.04
	Graph	Countdown	Python	MathIR	Pantry	Average		Graph	Countdown	Python	MathIR	Pantry	Average

Figure 4: **Canonical replay improves held-out correctness and verified output support.** (A) ReplayDr.GRPO minus terminal Dr.GRPO across scales. (B) Qwen2.5-0.5B references and alternatives minus terminal Dr.GRPO. Columns are co-primary $\Delta \text{pass@8}$ and $\Delta \text{distinct@8}$. They use different units and axis scales: probability points versus expected verified-key counts, so their numerical magnitudes should not be compared directly. The green Average is a within-seed domain macro-average. A dagger marks four admissible paired seeds in Falcon Countdown and its common-seed Average; all other cells use five. Only complete blocks receive unadjusted 95% estimation intervals. Purple identifies ReplayDr.GRPO (ours); bold marks Panel B column maxima.

in four of five. A newly completed Qwen2.5-3B Python block also gains $+.391$ raw `distinct@8` ($[+.079, +.704]$); its correctness effect remains inconclusive (Appendix Table 12).

7.3 EXPERIMENT 3: REPLAY AT MATCHED DIFFICULTY LEVELS

Replay improves both interim means at both difficulty levels. Figure 6B compares the latest valid checkpoint shared by both levels and all four methods for each domain and seed. Replay raises both mean endpoints under Dr.GRPO and MaxRL at each level. The five-domain summary includes 21 matched domain–seed cells: five per domain except PantryPlan, whose one matched seed contributes its initial evaluation. Appendix J.1 records the checkpoint selection and weighting.

The completed Graph block shows the benefit after eight passes. The matched construction lowers frozen-model `pass@8` from $.438$ to $.148$. After eight passes, Dr.GRPO/MaxRL reach $.202/.434$, versus $.763/.739$ with replay. Paired replay gains are $+.561$ and $+.305$ (unadjusted 95% intervals $[+.445, +.677]$ and $[+.250, +.360]$). Even the weakest replay run ($.711$) exceeds the strongest non-replay run ($.477$). Replay also adds $1.019/.668$ raw `distinct@8`, yielding $1.243/1.214$ with the two fresh objectives. These gains show improved success and sampled support on harder Graph; they do not establish that broader support causes the correctness gains.

8 CONCLUSION

ModeBench makes a missing part of evaluation measurable: the range of distinct, verified outcomes a model can produce for one problem. Accuracy tells us whether a model can succeed; diversity helps describe the alternatives still available to it. That distinction matters because different approaches can have different strengths. One may succeed where another fails, or remain useful when the problem or its constraints change. Knowing several ways to solve could therefore be valuable even when a familiar approach already works.

Our experiments show that replay complements learning from fresh samples. Remembering verified solutions improves held-out correctness and preserves more correct output alternatives across the completed comparisons. ModeBench lets us measure those alternatives through executable, task-defined identities. It provides a controlled starting point for understanding what training preserves, alongside what training learns to solve.

Limits and next steps. ModeBench provides a controlled setting for studying diverse solutions. Understanding how this diversity relates to broader mathematical reasoning and later prob-

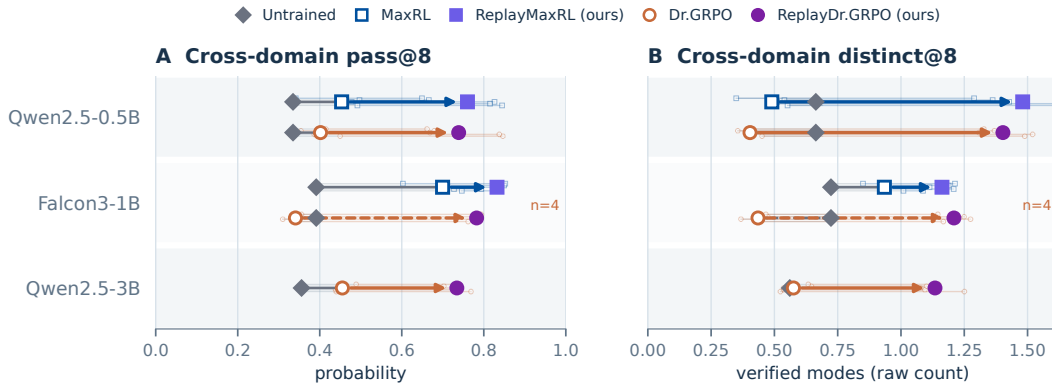


Figure 5: **Replay adds value across fresh objectives and model scales.** Each track averages domains within matched seeds; faint paths show seeds and bold paths show means. Falcon Dr.GRPO uses four common admissible seeds; the other tracks use five. Qwen2.5-3B shows its completed Dr.GRPO/replay track. Gray diamonds show initial means. Upper: Untrained–MaxRL (open blue square)–ReplayMaxRL (filled purple square); lower: Untrained–Dr.GRPO (open orange circle)–ReplayDr.GRPO (filled deep-purple circle). Gray segments show training without replay; arrows isolate the replay contrast. Per-domain results for both Qwen2.5-0.5B and Falcon3-1B appear in Appendix Figure 13 for both metrics, with domain-specific seed counts.

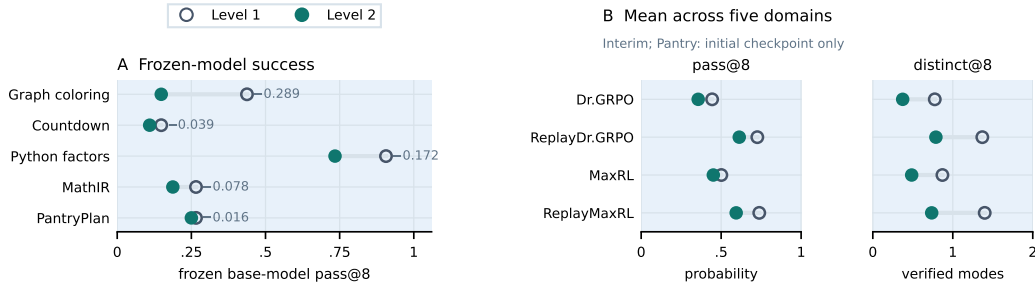


Figure 6: **Replay across difficulty levels with Qwen2.5-0.5B-Instruct.** Open/filled markers are Level 1/2. (A) Frozen-model success. (B) Interim means: average seeds within each domain, then the five domains equally. Both levels and all methods use matched checkpoints; Pantry contributes one seed at step 0 (Appendix J.1). All other domains contribute five matched seeds.

lem solving remains an open direction. Replay can retain only modes it discovers and stores. Individual-mode survival and size-dependent update geometry remain unmeasured (Appendices H and A.3); the optimizer and admission results are conditional. Theory studies idealized mean flows. We have not shown that broader support causes the accuracy gains. The MaxRL factorial has eleven complete paired blocks; Falcon Countdown’s Dr.GRPO replay comparison has four admissible seeds, and only Graph has a complete Level-2 factorial. Richer mathematical tasks should test which alternatives survive training, when a different approach succeeds after another fails, and whether preserving those options improves problem solving.

AI USE STATEMENT

Generative AI assisted manuscript preparation, mathematical analysis, and supporting code. The authors reviewed and verified the resulting content and take responsibility for the final manuscript.

ETHICS STATEMENT

The benchmark and experiment-specific data are synthetic; this study collects no human-subject or personal data. More verified modes are not inherently better: a distinct execution may be risky or costly even when it passes the validator, and `distinct@K` does not measure downstream value. Systems that evaluate or optimize breadth should report correctness, invalidity, and application-specific costs alongside mode counts.

REPRODUCIBILITY STATEMENT

Claims are computed from versioned result files; missing final results and aggregate drift cause the build to fail. Appendix J records protocols, hashes, validations, and regeneration commands; Appendix F records split identities. The AI Use Statement above discloses generative-AI assistance.

REFERENCES

- Alekh Agarwal, Sham M. Kakade, Jason D. Lee, and Gaurav Mahajan. On the theory of policy gradient methods: Optimality, approximation, and distribution shift. *Journal of Machine Learning Research*, 22(98):1–76, 2021. URL <https://jmlr.org/papers/v22/19-736.html>.
- Emmanuelle Anceaume, Yann Busnel, Ernst Schulte-Geers, and Bruno Sericola. Optimization results for a generalized coupon collector problem, 2015. URL <https://arxiv.org/abs/1504.03878>.
- Daniel Berend and Aryeh Kontorovich. On the concentration of the missing mass. *Electronic Communications in Probability*, 18(3):1–7, 2013. doi: 10.1214/ECP.v18-2359. URL <https://arxiv.org/abs/1210.3248>.
- Michael S. Branicky. Multiple Lyapunov functions and other analysis tools for switched and hybrid systems. *IEEE Transactions on Automatic Control*, 43(4):475–482, 1998. URL <https://people.ece.ubc.ca/moishi/eece571m/articles/Branicky98.pdf>.
- Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. Large language monkeys: Scaling inference compute with repeated sampling, 2024. URL <https://arxiv.org/abs/2407.21787>.
- Zhipeng Chen, Xiaobo Qin, Youbin Wu, Yue Ling, Qinghao Ye, Wayne Xin Zhao, and Guang Shi. Pass@k training for adaptively balancing exploration and exploitation of large reasoning models, 2025. URL <https://arxiv.org/abs/2508.10751>.
- Daixuan Cheng, Shaohan Huang, Xuekai Zhu, Bo Dai, Wayne Xin Zhao, Zhenliang Zhang, and Furu Wei. Reasoning with exploration: An entropy perspective, 2025. URL <https://arxiv.org/abs/2506.14758>.
- C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi: 10.1093/biomet/26.4.404. URL <https://academic.oup.com/biomet/article-abstract/26/4/404/291538>.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- Ganqu Cui, Yuchen Zhang, Jiacheng Chen, Lifan Yuan, Zhi Wang, Yuxin Zuo, Haozhan Li, Yuchen Fan, Huayu Chen, Weize Chen, Zhiyuan Liu, Hao Peng, Lei Bai, Wanli Ouyang, Yu Cheng, Bowen Zhou, and Ning Ding. The entropy mechanism of reinforcement learning for reasoning language models. *arXiv preprint arXiv:2505.22617*, 2025. URL <https://arxiv.org/abs/2505.22617>.

- Xingyu Dang, Christina Baek, J. Zico Kolter, and Aditi Raghunathan. Assessing diversity collapse in reasoning. In *ICLR Workshop on Scaling Self-Improving Foundation Models without Human Supervision*, 2025. URL <https://openreview.net/forum?id=AMiKsHLjQh>.
- Rick Durrett. *Probability: Theory and Examples*. Cambridge University Press, 5 edition, 2019. URL https://sites.math.duke.edu/~rtd/PTE/PTE5_011119.pdf.
- Adrien Ecoffet, Joost Huizinga, Joel Lehman, Kenneth O. Stanley, and Jeff Clune. First return, then explore. *Nature*, 590:580–586, 2021. doi: 10.1038/s41586-020-03157-9.
- Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, 2024. doi: 10.1038/s41586-024-07421-0.
- Aurélien Garivier and Olivier Cappé. The KL-UCB algorithm for bounded stochastic bandits and beyond. In *Proceedings of the 24th Annual Conference on Learning Theory*, volume 19 of *Proceedings of Machine Learning Research*, pp. 359–376. PMLR, 2011. URL <https://proceedings.mlr.press/v19/garivier11a.html>.
- Matthieu Geist, Bruno Scherrer, and Olivier Pietquin. A theory of regularized Markov decision processes. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, volume 97 of *Proceedings of Machine Learning Research*, pp. 2160–2169. PMLR, 2019. URL <https://proceedings.mlr.press/v97/geist19a.html>.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, et al. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature*, 645(8081):633–638, 2025. doi: 10.1038/s41586-025-09422-z.
- Marc Harper. Information geometry and evolutionary game theory. *arXiv preprint arXiv:0911.1383*, 2009. URL <https://arxiv.org/abs/0911.1383>.
- Josef Hofbauer and Karl Sigmund. *Evolutionary Games and Population Dynamics*. Cambridge University Press, Cambridge, United Kingdom, 1998.
- Steven R. Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. Time-uniform Chernoff bounds via nonnegative supermartingales. *Probability Surveys*, 17:257–317, 2020. doi: 10.1214/18-PS321. URL <https://doi.org/10.1214/18-PS321>.
- Steven R. Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. Time-uniform, nonparametric, nonasymptotic confidence sequences. *The Annals of Statistics*, 49(2):1055–1080, 2021. doi: 10.1214/20-AOS1991. URL <https://arxiv.org/abs/1810.08240>.
- Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and Roberta Raileanu. Understanding the effects of RLHF on LLM generalisation and diversity. In *International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=PXD3FAVHJT>.
- Joel Lehman and Kenneth O. Stanley. Abandoning objectives: Evolution through the search for novelty alone. *Evolutionary Computation*, 19(2):189–223, 2011.
- Chenyi Li, Yuan Zhang, Bo Wang, Guoqing Ma, Wei Tang, Haoyang Huang, and Nan Duan. SetPO: Set-level policy optimization for diversity-preserving LLM reasoning, 2026a. URL <https://arxiv.org/abs/2602.01062>.
- Long Li, Zhijian Zhou, Jiaran Hao, Jason Klein Liu, Yanting Miao, Wei Pang, Xiaoyu Tan, Wei Chu, Zhe Wang, Shirui Pan, Chao Qu, and Yuan Qi. The choice of divergence: A neglected key to mitigating diversity collapse in reinforcement learning with verifiable reward, 2025. URL <https://arxiv.org/abs/2509.07430>.
- Xiaozhe Li, Yang Li, Xinyu Fang, Shengyuan Ding, Peiji Li, Yongkang Chen, Yichuan Ma, Tianyi Lyu, Linyang Li, Dahua Lin, Qipeng Guo, Qingwen Liu, and Kai Chen. Beyond mode collapse: Distribution matching for diverse reasoning, 2026b. URL <https://arxiv.org/abs/2605.19461>.

- Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.
- Long-Ji Lin. Self-improving reactive agents based on reinforcement learning, planning and teaching. *Machine Learning*, 8:293–321, 1992. doi: 10.1007/BF00992699.
- Zichen Liu, Changyu Chen, Wenjun Li, Penghui Qi, Tianyu Pang, Chao Du, Wee Sun Lee, and Min Lin. Understanding R1-Zero-like training: A critical perspective. In *Conference on Language Modeling (COLM)*, 2025. URL <https://arxiv.org/abs/2503.20783>.
- Anamika Lochab, Bolian Li, and Ruqi Zhang. Uniform-correct policy optimization: Breaking RLVR’s indifference to diversity, 2026. URL <https://arxiv.org/abs/2605.00365>.
- David McAllester and Luis Ortiz. Concentration inequalities for the missing mass and for histogram rule error. *Journal of Machine Learning Research*, 4:895–911, 2003. URL <https://jmlr.org/papers/volume4/mcallester03a/mcallester03a.pdf>.
- Jincheng Mei, Chenjun Xiao, Csaba Szepesvári, and Dale Schuurmans. On the global convergence rates of softmax policy gradient methods. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pp. 6820–6829. PMLR, 2020. URL <https://proceedings.mlr.press/v119/mei20b.html>.
- Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A. Rusu, Joel Veness, Marc G. Bellemare, Alex Graves, Martin Riedmiller, Andreas K. Fidjeland, Georg Ostrovski, Stig Petersen, Charles Beattie, Amir Sadik, Ioannis Antonoglou, Helen King, Dharshan Kumaran, Daan Wierstra, Shane Legg, and Demis Hassabis. Human-level control through deep reinforcement learning. *Nature*, 518:529–533, 2015. doi: 10.1038/nature14236.
- Jean-Baptiste Mouret and Jeff Clune. Illuminating search spaces by mapping elites, 2015. URL <https://arxiv.org/abs/1504.04909>.
- Gabriel Pereyra, George Tucker, Jan Chorowski, Łukasz Kaiser, and Geoffrey Hinton. Regularizing neural networks by penalizing confident output distributions. *arXiv preprint arXiv:1701.06548*, 2017. URL <https://arxiv.org/abs/1701.06548>.
- Sashank J. Reddi, Satyen Kale, and Sanjiv Kumar. On the convergence of Adam and beyond. In *International Conference on Learning Representations*, 2018. URL <https://openreview.net/forum?id=ryQu7f-RZ>.
- Herbert Robbins and David Siegmund. A convergence theorem for non negative almost supermartingales and some applications. In *Optimizing Methods in Statistics*, pp. 233–257. Academic Press, 1971. doi: 10.1016/B978-0-12-604550-5.50015-8. URL <https://doi.org/10.1016/B978-0-12-604550-5.50015-8>.
- David Rolnick, Arun Ahuja, Jonathan Schwarz, Timothy P. Lillicrap, and Greg Wayne. Experience replay for continual learning. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL <https://arxiv.org/abs/1811.11682>.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- Abhijeet Sinha, Sundari Elango, and Dianbo Liu. Expected return causes outcome-level mode collapse in reinforcement learning and how to fix it with inverse probability scaling, 2026. URL <https://arxiv.org/abs/2601.21669>.
- Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling LLM test-time compute optimally can be more effective than scaling model parameters, 2024. URL <https://arxiv.org/abs/2408.03314>.
- Yuda Song, Julia Kempe, and Remi Munos. Outcome-based exploration for LLM reasoning, 2025. URL <https://arxiv.org/abs/2509.06941>.

- Fahim Tajwar, Guanning Zeng, Yueer Zhou, Yuda Song, Daman Arora, Yiding Jiang, Jeff Schneider, Ruslan Salakhutdinov, Haiwen Feng, and Andrea Zanette. Maximum likelihood reinforcement learning, 2026. URL <https://arxiv.org/abs/2602.02710v3>. Version 3.
- Tim van Erven and Peter Harremoës. Rényi divergence and Kullback–Leibler divergence. *IEEE Transactions on Information Theory*, 2014. doi: 10.1109/TIT.2014.2320500. URL <https://arxiv.org/abs/1206.2459v2>.
- Xiao Wang, Shiqian Ma, Donald Goldfarb, and Wei Liu. Stochastic quasi-newton methods for nonconvex stochastic optimization. *SIAM Journal on Optimization*, 27(2):927–956, 2017. doi: 10.1137/15M1053141. URL <https://doi.org/10.1137/15M1053141>.
- Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Yang Yue, Shiji Song, and Gao Huang. Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model?, 2025. URL <https://arxiv.org/abs/2504.13837>.
- Hongzhi Zhang, Jia Fu, Jingyuan Zhang, Kai Fu, Qi Wang, Fuzheng Zhang, and Guorui Zhou. RLEP: Reinforcement learning with experience replay for LLM reasoning, 2025. URL <https://arxiv.org/abs/2507.07451>.
- Junzi Zhang, Jongho Kim, Brendan O’Donoghue, and Stephen Boyd. Sample efficient reinforcement learning with REINFORCE. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pp. 10887–10895, 2021. doi: 10.1609/aaai.v35i12.17300. URL https://stanford.edu/~boyd/papers/conv_reinforce.html.

APPENDIX ORGANIZATION

The appendix follows the paper’s argument rather than the chronology of the project. Appendix A gives the formal collapse and retention analysis. Appendix B reports supporting measurements and direct comparators. Appendix C motivates Semantic MaxEnt and specifies its estimator and interpretation limits. Appendices D–F specify the executable benchmark, its prompts, two-level construction, and split identities. Appendix G gives the complete ReplayMaxRL state transition; Appendix H checks that the intended replay mechanism was actually active. Appendices I and J record the scale extensions and the frozen reproduction contract.

A MODE COLLAPSE AND VERIFIED REPLAY

Main-body link. This appendix supplies the formal claims invoked in Sections 5 and 7.

A binary verifier assigns the same reward to every correct mode, but an update can still amplify the most common one. We first explain that mechanism in a finite categorical model, then ask what changes when we alter the update geometry, add entropy, or replay earlier successes. Replay directly protects represented modes; discovery, admission, and reliable neural optimization remain separate requirements.

How to read the proof chain. Begin with the three-correct-mode example in Section A.1. The group calculations reveal why a correctness gradient need not balance correct alternatives, and the next section turns the local imbalance into an asymptotic collapse theorem. Sections A.3–A.6 then vary geometry and entropy: they show both how collapse can slow and why it is not a universal consequence of rewarding correctness.

The second part asks what replay adds. Sections A.7 and A.8 separate a missing fresh signal from a persistent bank signal, then turn bounded replay loss into protected probabilities. The remaining sections connect that argument to complete responses, controlled stochastic updates, and actual bank admission. Finally, Section A.12 shows how much those bounds can certify and what finite observations can establish about a named mode. Each proof states its assumptions and attributes the literature tools it uses.

All worked numbers below are illustrative calculations in the stated models, not measurements from the training runs. Unless a new example is introduced, the examples reuse the same three correct modes and one incorrect category. Comparisons keep the optimized objective, decoding law, and time variable explicit, since changing any of these can change the conclusion.

Here G counts training rollouts, K counts evaluation draws, m, n count correct and incorrect categories, and k is the bank size. In the mean-flow results, t is continuous optimization time and τ is a reparameterized learning clock. The later stochastic and admission results use explicitly defined update or opportunity indices. None is a wall-clock or GPU-runtime prediction. “Collapse” refers to limiting concentration: finite softmax logits already give positive probabilities at every finite time. A uniform lower bound for all future times is the stronger retention property proved below.

A.1 CATEGORICAL REDUCTION AND THE GRPO MEAN UPDATE

Fix one prompt. Let the finite response categories be $\mathcal{A} = \mathcal{C} \dot{\cup} \mathcal{N}$, where the $m \geq 2$ categories in $\mathcal{C}$ are correct execution modes and the $n \geq 1$ categories in $\mathcal{N}$ are incorrect. Let $p_a = \text{softmax}(z)_a$, $P = \sum_{c \in \mathcal{C}} p_c$, and $q_c = p_c/P$. Thus q is exactly the conditional correct-mode distribution $p_\theta^+(\cdot \mid x)$ of Section 3, and P is the correct mass $P_\theta^+(x)$.

Running example: three correct alternatives. Take three correct modes and one incorrect category, with

$$p = (0.30, 0.15, 0.05, 0.50), \quad P = 0.50, \quad q = (0.60, 0.30, 0.10).$$

Half of all draws succeed, but among successful draws the three modes appear in a 6:3:1 ratio. Increasing P and balancing q are therefore separate goals. We reuse this hypothetical distribution below to distinguish the effects of the fresh objective, optimizer geometry, and replay.

Draw a group $A_1, \dots, A_G \stackrel{\text{iid}}{\sim} p$ with $G \geq 2$, write $R_i = \mathbf{1}[A_i \in \mathcal{C}]$, and let $M = \sum_i R_i$. Both GRPO variants can be written, at the on-policy point where the PPO ratio equals one, as

$$\hat{g}_G = \frac{1}{G} \sum_{i=1}^G w(M) \left(R_i - \frac{M}{G} \right) \nabla_z \log p_{A_i}. \quad (3)$$

For Dr.GRPO, $w(M) = 1$ after absorbing the shared positive length scale. For binary-reward GRPO, $w(M)$ is the reciprocal group reward standard deviation on mixed groups. We absorb the same fixed response-length normalizer in both variants; response-dependent row normalization is outside this reduction. In either case $0 < w(\ell) < \infty$ for $\ell \in \{1, \dots, G-1\}$. We

define the entire update as zero on all-equal groups, without evaluating an undefined reciprocal standard deviation; likewise, $w(M)M(G - M)$ below is defined as zero when $M = 0$ or $M = G$. We take explicit reference KL to be zero, as in the experiments.

Lemma A.1 (A group baseline does not remove frequency amplification). *For $0 < P < 1$, the expected group update is*

$$\mathbb{E}[\hat{g}_G] = c_G(P)\nabla_z P, \quad c_G(P) = \frac{\mathbb{E}[w(M)M(G - M)]}{G^2 P(1 - P)} > 0. \quad (4)$$

For Dr.GRPO, $c_G(P) = (G - 1)/G$. Hence reward centering and binary reward standardization change the speed, but not the expected direction, of the binary-reward policy gradient.

Proof. Condition on the complete binary reward vector, whose sum is M . The score identities give

$$\mathbb{E}[\nabla_z \log p_{A_i} \mid R_i = 1] = \nabla_z \log P, \quad \mathbb{E}[\nabla_z \log p_{A_i} \mid R_i = 0] = \nabla_z \log(1 - P).$$

There are M correct rows with centered reward $(G - M)/G$ and $G - M$ incorrect rows with centered reward $-M/G$. Therefore the conditional expectation of the unaveraged sum in (3) is

$$w(M) \frac{M(G - M)}{G} (\nabla_z \log P - \nabla_z \log(1 - P)) = w(M) \frac{M(G - M)}{GP(1 - P)} \nabla_z P.$$

The outer factor $1/G$ and expectation over $M \sim \text{Binomial}(G, P)$ yield (4). For $w \equiv 1$, $\mathbb{E}[M(G - M)] = G(G - 1)P(1 - P)$, proving the Dr.GRPO value. $\square$

The binomial identity also gives the denominator-free expression

$$c_G(P) = \frac{G - 1}{G} \sum_{j=0}^{G-2} \binom{G-2}{j} w(j+1) P^j (1 - P)^{G-2-j}.$$

The binomial weights sum to one, so c_G lies between $(G - 1) \min_j w(j+1)/G$ and $(G - 1) \max_j w(j+1)/G$. This establishes the smooth, positive extension to $P = 0, 1$ used in the subsequent flow and convergence arguments.

The next lemma specializes the practical-estimator analysis of Tajwar et al. (2026, Section 4.3 and Appendix D, Theorem 5) to our categorical notation; we include the derivation for completeness.

Lemma A.2 (Binary MaxRL has the same correctness-gradient direction). *Let MaxRL assign $A_i^{\text{MaxRL}} = GR_i/M - 1$ when $M > 0$ and zero when $M = 0$, as in Section 5.1. At the on-policy point, its expected update is*

$$\mathbb{E}[\hat{g}_G^{\text{MaxRL}}] = c_G^{\text{MaxRL}}(P)\nabla_z P, \quad c_G^{\text{MaxRL}}(P) = \frac{1 - (1 - P)^{G-1}}{P} > 0, \quad (5)$$

with continuous endpoint values $c_G^{\text{MaxRL}}(0) = G - 1$ and $c_G^{\text{MaxRL}}(1) = 1$.

Proof. Condition on $M = m > 0$. Each successful row has advantage $(G - m)/m$ and each failed row has advantage -1 . Using the same conditional score identities as above, the expected unaveraged sum is

$$(G - m)(\nabla_z \log P - \nabla_z \log(1 - P)) = \frac{G - m}{P(1 - P)} \nabla_z P.$$

After the outer factor $1/G$ and averaging over $M \sim \text{Binomial}(G, P)$,

$$\mathbb{E}[(G - M)\mathbf{1}\{M > 0\}] = G(1 - P) - G\Pr(M = 0) = G[(1 - P) - (1 - P)^G].$$

Substitution gives Equation (5). Its endpoint values follow by continuity (or the finite geometric-series identity), which also gives the stated continuous extension at both boundaries. $\square$

The same finite-series identity gives the potential used below:

$$\Psi_{\text{MaxRL}}(P) := \int_0^P c_G^{\text{MaxRL}}(s) ds = \sum_{k=1}^{G-1} \frac{1 - (1 - P)^k}{k}. \quad (6)$$

In particular, $\Psi_{\text{MaxRL}}(1) = \sum_{k=1}^{G-1} 1/k$. For binary rewards, $\text{best}@k$ equals $\text{pass}@k$, so this is a harmonic mixture of sampling objectives for $k = 1, \dots, G-1$. The upper limit is $G-1$, not G : the implemented centered estimator drops the entire update on all-failure groups. Subtracting an unconditional score baseline even on those groups instead gives the order- G estimator. This distinction concerns the exact finite-group expectation at the on-policy point; it does not assert unbiasedness of subsequent clipped or adaptive-optimizer updates.

What the group normalization changes. For the running example with $G = 4$, $\nabla_z P = (0.15, 0.075, 0.025, -0.25)$. The two lemmas give

$$\begin{aligned}\mathbb{E}[\hat{g}_4^{\text{Dr.GRPO}}] &= 0.75 \nabla_z P, \\ \mathbb{E}[\hat{g}_4^{\text{MaxRL}}] &= 1.75 \nabla_z P.\end{aligned}$$

These are expectations over all possible groups, not updates from one particular sample. MaxRL moves faster along the same direction at this state, while the most common correct mode receives six times the logit increment of the rarest. The next section asks what repeated unequal increments do to the relative probabilities of equally rewarded modes.

A.2 WINNER-TAKE-ALL COLLAPSE UNDER GRPO, DR.GRPO, AND MAXRL

Prior work studies objective-induced outcome collapse and inverse-probability correction (Sinha et al., 2026); UCPO analyzes correctness-indifferent collapse through sampling bias and gradient amplification (Lochab et al., 2026, Section 4). Here we connect our group estimators to a standard categorical flow and then study persistent replay over execution-defined keys. Consider the infinitesimal expected-update flow $\dot{z} = \mathbb{E}[\hat{g}_G] = c_G(P) \nabla_z P$. This is the clean mean-field limit of fresh on-policy groups and the first-order PPO update. By Lemmas A.1 and A.2, it covers GRPO, Dr.GRPO, and binary MaxRL with their respective positive functions c_G . It deliberately removes sampling noise; the result below therefore identifies a structural instability rather than blaming finite batches.

The conditional correct-mode flow derived below is a standard frequency-dependent replicator system with fitness $f_c(q) = q_c$ (Hofbauer & Sigmund, 1998); the following theorem is not claimed as a new replicator-dynamics result. Its role is to show that the expected GRPO, Dr.GRPO, and binary-MaxRL updates reduce to that familiar winner-take-all flow under the paper’s categorical assumptions.

A local view of the instability. For $q = (0.60, 0.30, 0.10)$ in the running example, $\sum_c q_c^2 = 0.46$. In the effective time τ defined in the proof, Equation (7) gives

$$\frac{dq}{d\tau} = (0.084, -0.048, -0.036), \quad \frac{d}{d\tau} \log \frac{q_1}{q_2} = 0.30.$$

The two less common modes lose their share of successful outputs even while correctness increases. This calculation is local; the theorem establishes that the imbalance persists all the way to the single-mode limit.

Theorem A.3 (Standard replicator specialization for binary-reward RL). *Start from finite logits with $0 < P(0) < 1$. If the conditional correct-mode distribution $q(0)$ has a unique largest coordinate c^* , then*

$$P(t) \longrightarrow 1, \quad q_{c^*}(t) \longrightarrow 1, \quad q_c(t) \longrightarrow 0 \quad (c \neq c^*).$$

Thus these GRPO, Dr.GRPO, and binary MaxRL mean flows converge to a single correct execution mode for generic initial conditions, even though every mode in $\mathcal{C}$ has exactly the same reward.

Proof. For a correct mode c and incorrect category a ,

$$\dot{z}_c = c_G(P) p_c (1 - P), \quad \dot{z}_a = -c_G(P) p_a P.$$

Also $\dot{P} = c_G(P) \|\nabla_z P\|_2^2 > 0$. If P converged to an interior value, continuity and positivity of c_G , together with

$$\|\nabla_z P\|_2^2 \geq P^2 (1 - P)^2 \left(\frac{1}{m} + \frac{1}{n} \right),$$

would keep $\dot{P}$ bounded away from zero, a contradiction. Hence $P \rightarrow 1$.

Define the effective optimization time

$$\tau(t) = \int_0^t c_G(P(s))P(s)(1 - P(s)) ds.$$

If $\tau(\infty) < \infty$, every correct and incorrect logit would have finite total variation, because the absolute velocity of each is at most $\dot{\tau}$. All logits would then converge to finite values, contradicting $P \rightarrow 1$. Therefore $\tau(t) \rightarrow \infty$, leaving infinite effective time for the correct-simplex dynamics below.

On the correct simplex, changing from t to τ gives the replicator flow

$$\frac{dq_c}{d\tau} = q_c \left(q_c - \sum_{d \in \mathcal{C}} q_d^2 \right), \quad \frac{d}{d\tau} \log \frac{q_c}{q_d} = q_c - q_d. \quad (7)$$

The initially unique maximum remains the unique maximum. For $c \neq c^*$, set $\xi_c = q_c/q_{c^*}$, so $\xi_c(0) < 1$ and ξ_c is nonincreasing. Since $q_{c^*} \geq 1/m$,

$$\frac{d \log \xi_c}{d\tau} = -q_{c^*}(1 - \xi_c) \leq -\frac{1 - \xi_c(0)}{m}.$$

Hence $\xi_c(\tau) \leq \xi_c(0) \exp[-(1 - \xi_c(0))\tau/m] \rightarrow 0$. Every competing mode vanishes relative to c^* ; normalization then yields $q_{c^*} \rightarrow 1$, establishing the claimed single-mode limit. $\square$

Corollary A.4 (Ties are nongeneric). *If exactly $k < m$ correct modes tie for the largest initial probability, symmetry preserves the tie and the flow converges to the uniform distribution on those k modes while eliminating the rest. Only the exactly uniform initialization preserves all m correct modes in the limit; a tie among $1 < k < m$ maxima yields partial collapse onto those k modes. Any perturbation creating a unique maximum gives single-mode collapse by Theorem A.3.*

An exact tie is a different initial condition. Starting instead from $q = (0.45, 0.45, 0.10)$ gives the limit $(0.50, 0.50, 0)$. Changing the initialization to $(0.451, 0.449, 0.10)$ makes the first mode the unique winner. Thus an exactly symmetric two-mode limit does not imply robustness to a small imbalance in the initial policy.

Finite groups also introduce sampling effects: a mode absent from a group has no sampled response of its own carrying a score term, all-correct groups have zero task advantage, and sampling noise can break exact ties. Softmax normalization and shared parameters can still change an unsampled mode’s probability. These facts are not needed for the theorem.

A.3 HOW MODEL PARAMETERIZATION CHANGES COLLAPSE

The cross-scale losses raise a further question: can a larger model learn correctness while preserving more correct alternatives? The preceding theorem already allows every logit to vary independently, so representational capacity alone cannot answer this question. We give a conditional mechanism: directions that raise correct logits together can improve correctness with less amplification of differences among correct modes.

For a smooth parameterization $z = z(\theta)$, write $J_\theta = \partial z / \partial \theta$ and $K_\theta = J_\theta J_\theta^\top$. The chain rule extends Lemmas A.1 and A.2 to Euclidean parameter gradient flow:

$$\dot{z} = c_G(P)K_\theta \nabla_z P, \quad \frac{d}{dt} \log \frac{q_c}{q_d} = c_G(P)(\mathbf{e}_c - \mathbf{e}_d)^\top K_\theta \nabla_z P, \quad (8)$$

where $\mathbf{e}_c$ is a coordinate basis vector. Thus the logit-gradient geometry, not the number of columns of J_θ alone, governs mode amplification. The independent-logit theorem has $K_\theta = I$.

To isolate shared correctness learning, let $v = \mathbf{1}_C$ be the vector that is one on correct categories and zero on incorrect categories, and consider the constant geometry

$$K_\lambda = I + \lambda v v^\top, \quad \lambda \geq 0. \quad (9)$$

The added direction moves all correct logits equally, so it improves their mass without directly changing their ratios. For example, $z = u + \alpha \sum_{j=1}^s h_j v$, with trainable u and scalar h_j , realizes this geometry with $\lambda = s\alpha^2$. More such coordinates strengthen the shared update if α is fixed; normalizing $\alpha = 1/\sqrt{s}$ removes that dependence. These coordinates are redundant for expressivity and assume perfect correctness alignment. They describe an optimization mechanism, not a feature that an arbitrary larger language model is guaranteed to possess.

Theorem A.5 (Shared correctness learning preserves more sampled modes at matched accuracy). *Compare the flows (8) with $K_\theta = K_\lambda$, the same finite initial logits, and any of the three functions c_G above. Fix a target correct mass $P_\star \in (P(0), 1)$. At the first time each flow reaches $P_\star$, $\text{distinct}@K$ is nondecreasing in λ for every integer $K \geq 1$, while $\text{pass}@K$ is identical. The increase is strict for $K \geq 2$ when $q(0)$ is nonuniform. In addition, every correct mode satisfies*

$$q_c^{(\lambda)}(P_\star) \geq q_c(0) \exp\left(-\frac{L}{\lambda + 1/m + 1/n}\right), \quad L = \log \frac{P_\star}{1 - P_\star} - \log \frac{P(0)}{1 - P(0)}. \quad (10)$$

Here m, n count output categories, not model parameters. For every fixed finite λ , a unique initial maximum still implies eventual single-mode collapse as $t \rightarrow \infty$.

Proof. Let $r_a = p_a/(1 - P)$ for $a \in \mathcal{N}$, and write $S_q = \sum_c q_c^2$, $S_r = \sum_a r_a^2$. In the effective time $d\tau = c_G(P)P(1 - P) dt$, the logit velocities are $dz_c/d\tau = q_c + \lambda$ and $dz_a/d\tau = -r_a$. Hence

$$\frac{dq_c}{d\tau} = q_c(q_c - S_q), \quad \frac{dr_a}{d\tau} = -r_a(r_a - S_r), \quad \frac{d}{d\tau} \log \frac{P}{1 - P} = S_q + S_r + \lambda. \quad (11)$$

The paths $q(\tau), r(\tau)$ therefore do not depend on λ . Also $\dot{P} = c_G(P)P^2(1 - P)^2(S_q + S_r + \lambda) > 0$; the same interior-limit argument as in Theorem A.3 gives $P \rightarrow 1$. At $P_\star$, the effective time τ_λ uniquely solves

$$L = \lambda\tau_\lambda + \int_0^{\tau_\lambda} (S_q(u) + S_r(u)) du.$$

The right-hand side increases strictly with both its positive time argument and λ , so τ_λ decreases strictly with λ . Since $S_q \geq 1/m$ and $S_r \geq 1/n$, $\tau_\lambda \leq L/(\lambda + 1/m + 1/n)$. Integrating $d \log q_c/d\tau = q_c - S_q \geq -1$ over this bounded interval gives the probability floor in Equation (10).

At fixed $P_\star$, define $\mathcal{D}_K(q) = \sum_c [1 - (1 - P_\star q_c)^K]$, the expected number of correct modes in K independent draws. For $a(u) = KP_\star(1 - P_\star u)^{K-1}$, the replicator equation gives

$$\frac{d\mathcal{D}_K}{d\tau} = \frac{1}{2} \sum_{c,d} q_c q_d (q_c - q_d) (a(q_c) - a(q_d)) \leq 0.$$

The inequality holds because a is nonincreasing; it is strict for $K \geq 2$ and nonuniform q . Thus the smaller τ_λ gives larger $\text{distinct}@K$, whereas $\text{pass}@K = 1 - (1 - P_\star)^K$ is unchanged. Finally, $S_q + S_r + \lambda \leq 2 + \lambda$ and $P \rightarrow 1$ imply $\tau \rightarrow \infty$. The unchanged correct-mode replicator flow then gives eventual collapse under a unique initial maximum. $\square$

Same accuracy, different remaining breadth. Start from the running example and stop each idealized flow when P first reaches 0.90. Both policies then have $\text{pass}@8 = 1 - 10^{-8}$. Numerical integration of Equation (11) gives the following illustrative values; the probability floors are calculated separately from Equation (10) with $L = \log 9$.

λ	Conditional mass q_3	Guaranteed q_3 floor	$\text{distinct}@8$
0	0.05255	0.01924	2.125
4	0.08582	0.06623	2.374

The trajectory values are rounded approximations; the displayed floors are rounded down. The shared direction preserves more of the initially rare mode at the same correctness, but its mass still falls below its initial 0.10. A better finite-accuracy trajectory can therefore coexist with eventual loss, as the next corollary makes precise.

Corollary A.6 (Collapse rate as correctness approaches one). *Under Theorem A.5, fix finite $\lambda \geq 0$ and suppose the initial correct-mode maximum is unique. For every other correct mode c ,*

$$\lim_{P \uparrow 1} \frac{\log q_c^{(\lambda)}(P)}{\log(1 - P)} = \frac{1}{\lambda + 1 + 1/n}. \quad (12)$$

Thus $q_c^{(\lambda)}(P) = (1 - P)^{1/(\lambda + 1 + 1/n) + o(1)}$: larger fixed λ reduces the decay exponent, but every minority mode still vanishes. The exponent describes the limiting ratio of logarithms; finite-accuracy probabilities also depend on the initial distribution.

Proof. Theorem A.5 gives $q \rightarrow \mathbf{e}_{c^*}$, so $S_q \rightarrow 1$. The incorrect conditional flow in Equation (11) preserves ordering and makes $r_{\min}$ nondecreasing. For $h = \log(r_{\max}/r_{\min})$,

$$h' = -(r_{\max} - r_{\min}) = -r_{\min}(e^h - 1) \leq -r_{\min}(0)h.$$

Hence $r \rightarrow (1/n, \dots, 1/n)$ and $S_r \rightarrow 1/n$, including trivially $n = 1$. Integrating $d \log q_c / d\tau = q_c - S_q$ and the correctness-clock equation and taking time averages yields

$$\frac{\log q_c(\tau)}{\tau} \rightarrow -1, \quad \frac{\log(P(\tau)/(1 - P(\tau)))}{\tau} \rightarrow \lambda + 1 + 1/n.$$

Since $\log(1 - P) = \log P - \log(P/(1 - P))$ and $\log P \rightarrow 0$, their ratio proves Equation (12). $\square$

A smaller exponent still permits extinction. With one incorrect category, the minority-mode exponents are $1/2$ at $\lambda = 0$ and $1/6$ at $\lambda = 4$. Both are positive, so both limits lose minority modes as $P \uparrow 1$. These are asymptotic logarithmic slopes; they do not replace the finite-accuracy values in the preceding table.

Correctness alone does not imply collapse. The independent-logit component matters. With purely shared geometry $\mathbf{K} = \lambda \mathbf{v} \mathbf{v}^\top$, $\lambda > 0$, all correct logits move equally, so $q(t) = q(0)$ while $\dot{P} = \lambda c_G(P) P^2 (1 - P)^2 > 0$ and $P \rightarrow 1$. For $\mathbf{K} = a\mathbf{I} + b\mathbf{v} \mathbf{v}^\top$ with fixed $a > 0$ and $b \geq 0$, time rescaling instead recovers Theorem A.5 with $\lambda = b/a$. Thus persistent mode-specific amplification, not increasing correctness by itself, drives this collapse result.

Why parameter count alone is insufficient. For an integer $h \geq 1$, replace each independent logit by $z_a = h^{-\gamma} \sum_{j=1}^h \theta_{aj}$, initializing $\theta_{aj}(0) = h^{\gamma-1} z_a(0)$. Every added parameter affects the logits, and every model represents the same categorical policies, but $\mathbf{K} = h^{1-2\gamma} \mathbf{I}$. Its policy follows exactly the original collapse trajectory with time multiplied by $h^{1-2\gamma}$. Increasing the parameter count therefore makes collapse faster, unchanged, or slower for $\gamma = 0, 1/2, 1$, respectively. For example, using $h = 4$ multiplies the speed by 4, 1, or $1/4$ under those three normalizations, despite quadrupling the parameter count in every case. A size claim therefore requires assumptions about how additional parameters change the update geometry.

What this explains and what remains untested. Theorem A.5 connects a specified kind of shared learning to the paper’s sampling metrics at matched accuracy. It also clarifies why a slower loss can coexist with eventual collapse, leaving a role for replay. It does not identify the cause of the empirical cross-scale pattern: those runs start from different pretrained distributions, use different learning-rate schedules, and compare equal training passes rather than matched accuracy. Falcon also changes model family. Neither the assumed geometry nor its dependence on size is measured here. Establishing that link would require controlled within-family scaling and measurements of shared versus mode-specific updates. The result retains prompt isolation, exact mean updates, and zero reference KL; it does not cover shared-prompt interference, adaptive optimizer effects, or finite-step stochastic training.

A.4 AN EXACT OPTIMIZER COMPARISON FROM INFORMATION GEOMETRY

Changing the optimization metric lets us isolate correctness learning from changes among correct alternatives. The next result shows both effects in separate equations for P and q .

Natural gradient on the categorical simplex uses the Fisher metric, also called the Shahshahani metric in evolutionary dynamics (Harper, 2009, Sections 2.3 and 3.1). Its gradient formula gives an exact comparison with the Euclidean-logit collapse result in Theorem A.3. We derive the specialization below so that the dependence on optimizer geometry is explicit.

Theorem A.7 (Natural-gradient selection and replay). *Retain the finite categorical setup and start from an interior distribution $p(0)$ with $0 < P(0) < 1$. Let $\Psi \in C^2([0, 1])$ satisfy $c(P) := \Psi'(P) > 0$ on $[0, 1]$. Choose a nonempty fixed bank $\mathcal{B} \subseteq \mathcal{C}$, positive weights w_b summing to one over $\mathcal{B}$, and extend them by zero to a vector $\bar{w}$ on $\mathcal{A}$. Exact Fisher natural-gradient ascent of*

$$\mathcal{J}(p) = \Psi(P) + \rho \sum_{b \in \mathcal{B}} w_b \log p_b, \quad \rho \geq 0,$$

induces

$$\dot{p}_i = c(P)p_i(\mathbf{1}\{i \in \mathcal{C}\} - P) + \rho(\bar{w}_i - p_i), \quad (13)$$

$$\dot{P} = (1 - P)(c(P)P + \rho), \quad \dot{q}_c = \frac{\rho}{P}(\bar{w}_c - q_c). \quad (14)$$

For $\rho = 0$, $P(t) \rightarrow 1$ while $q(t) = q(0)$: correctness learning preserves all relative probabilities among correct modes. For $\rho > 0$, define $A(t) = \int_0^t \rho/P(s) ds$. Every correct mode satisfies

$$q_c(t) = \bar{w}_c + (q_c(0) - \bar{w}_c)e^{-A(t)}, \quad A(t) \geq \rho t. \quad (15)$$

Furthermore,

$$1 - P(t) \leq (1 - P(0))e^{-\rho t}, \quad p_b(t) \geq P(0) \min\{q_b(0), w_b\} > 0 \quad (b \in \mathcal{B}, t \geq 0). \quad (16)$$

Thus full-support replay sends the conditional distribution to its bank target, whereas an incomplete bank still loses unbanked correct modes in the limit.

Proof. For tangent vectors u, v with zero coordinate sum, the categorical Fisher metric is $g_p(u, v) = \sum_i u_i v_i / p_i$. The tangent vector

$$v_i = p_i \left(\partial_i \mathcal{J} - \sum_j p_j \partial_j \mathcal{J} \right)$$

has zero sum and satisfies $g_p(v, u) = \sum_i \partial_i \mathcal{J} u_i$ for every tangent u . It is therefore the natural gradient. Here

$$\partial_i \mathcal{J} = c(P)\mathbf{1}\{i \in \mathcal{C}\} + \rho \bar{w}_i / p_i, \quad \sum_i p_i \partial_i \mathcal{J} = c(P)P + \rho,$$

which proves Equation (13). Summing over correct categories gives the P equation. Differentiating $q_c = p_c/P$ then gives the q equation in (14); the correctness terms cancel.

The P equation makes P nondecreasing. If $\rho = 0$, q is constant; if $\rho > 0$, solving its linear equation gives Equation (15). Since $P \leq 1$, this also gives $A(t) \geq \rho t$. Incorrect coordinates satisfy $\dot{p}_i = -(c(P)P + \rho)p_i$ and remain positive at every finite time. Correct conditional coordinates remain positive by the explicit solution, and $P \geq P(0) > 0$. Because c is bounded on $[0, 1]$ and $\rho/P \leq \rho/P(0)$, these formulas extend the interior solution to every finite time.

For $\rho = 0$, P increases and cannot tend to a limit below one: at any such limit $(1 - P)c(P)P$ would stay positive. For $\rho > 0$, $\dot{P} \geq \rho(1 - P)$ yields the correctness bound in Equation (16). Each banked $q_b(t)$ lies between $q_b(0)$ and w_b , so multiplication by $P(t) \geq P(0)$ proves the probability floor. Finally, $A(t) \rightarrow \infty$ gives $q_c(t) \rightarrow \bar{w}_c$, so unbanked correct modes have zero limiting mass. $\square$

Worked example: the metric changes which modes grow. At the running example's $P(0) = 0.5$, $q(0) = (0.6, 0.3, 0.1)$, and $G = 4$, Dr.GRPO has $c(P) = 3/4$. Without replay, the Fisher flow gives $\dot{P}(0) = 0.1875$ and $\dot{q}(0) = 0$: the model learns correctness while keeping its correct-mode proportions. With the uniform three-mode target $w = (1/3, 1/3, 1/3)$ and $\rho = 0.1$, the same initial distribution instead gives

$$\dot{P}(0) = 0.2375, \quad \dot{q}(0) \approx (-0.05333, 0.00667, 0.04667).$$

Replay now increases the rare third mode's share among correct responses. Equation (16) also guarantees $p_3(t) \geq 0.05$ for all time in this ideal flow. These are instantaneous derivatives and a trajectory bound, not the result of taking a discrete optimizer step.

The smooth potentials with positive derivatives in Lemmas A.1 and A.2 satisfy the premises. The Fisher-gradient formula is established information geometry; the result above is its elementary specialization to correctness selection plus replay toward a fixed target. It makes the scope of the collapse claim concrete: increasing correctness alone does not force a loss of conditional support under every optimization geometry. The theorem concerns the exact categorical Fisher flow. It gives no guarantee for approximate natural gradient, a shared neural Fisher matrix, PPO, or AdamW, and its time variable is an ideal optimization clock rather than elapsed training time.

A.5 EXACT ENTROPY AND SAMPLED SURPRISAL HAVE DIFFERENT GUARANTEES

To interpret an entropy ablation, we must specify whether the update differentiates exact entropy or applies a bounded score to sampled responses. The two lemmas below make that distinction explicit.

The preceding collapse theorem has no entropy term. It does not imply collapse under exact Shannon-entropy regularization, which has a different vector field.

Lemma A.8 (Exact categorical entropy prevents extinction). *Let $d = m + n$, $p = \text{softmax}(z)$, and $J(z) = \Psi(P(z)) + \beta H(p(z))$, where $\Psi \in C^2([0, 1])$, $\beta > 0$, and $0 \leq \Psi'(P) \leq M < \infty$. Under the exact independent-logit gradient flow $\dot{z} = \nabla_z J$, put*

$$\bar{z} = \frac{1}{d} \sum_i z_i(0), \quad \ell = \min_i \{z_i(0), \bar{z} - M/\beta\}.$$

For every $t \geq 0$ and every category i ,

$$p_i(t) \geq \frac{1}{d} \exp(d(\ell - \bar{z})) > 0.$$

If Ψ is also concave, the policy converges to the unique interior maximizer of $\Psi(P) + \beta H(p)$. In particular, for Dr.GRPO with $a = (G - 1)/G$, the limit is

$$p_i^* = \frac{\exp(a \mathbf{1}\{i \in \mathcal{C}\}/\beta)}{m \exp(a/\beta) + n}.$$

Thus the conditional correct-mode distribution is uniform. The full entropy objective also retains positive incorrect mass, so its limiting correctness is below one for fixed $\beta > 0$.

Proof. The gradient components are

$$\dot{z}_i = p_i [\Psi'(P)(\mathbf{1}\{i \in \mathcal{C}\} - P) + \beta(-\log p_i - H(p))].$$

Their sum is zero, so $\bar{z}$ is conserved. Jensen’s inequality and $H(p) \leq \log d$ give $-\log p_i - H(p) \geq \bar{z} - z_i$. Consequently, at any coordinate with $z_i = \ell$, its velocity is at least $p_i[-M + \beta(\bar{z} - \ell)] \geq 0$. The region $z_i \geq \ell$ for every i is therefore forward invariant. Conservation of the sum also gives $z_i \leq d\bar{z} - (d - 1)\ell$, proving the probability bound.

These bounds make the fixed-mean logit trajectory compact. Since $\dot{J} = \|\nabla_z J\|_2^2$ and J is bounded on that compact set, the gradient has a finite squared-norm integral. Bounded derivatives on the set then give $\nabla_z J \rightarrow 0$, so all limit points are stationary. Interior softmax stationarity is equivalent to simplex stationarity. Concavity of Ψ and strict concavity of βH give a unique maximizer; it is interior because the entropy gain from adding a missing coordinate dominates the bounded first-order change of Ψ . Hence the entire policy converges to that maximizer. For linear $\Psi(P) = aP$, the simplex first-order conditions give the stated Gibbs distribution. $\square$

Worked example: exact full entropy preserves incorrect mass too. For a separate hypothetical fixture with two correct categories and one incorrect category, take $G = 4$, $\Psi(P) = 3P/4$, and $\beta = 1/4$. The reward-to-entropy ratio is $a/\beta = 3$, so the lemma gives

$$p^* = \frac{(e^3, e^3, 1)}{2e^3 + 1} \approx (0.48786, 0.48786, 0.02429), \quad P^* \approx 0.97571.$$

Both correct modes survive equally, but the incorrect category also retains about 2.43% probability. This is the limit of the exact full-output entropy flow under the lemma’s assumptions; it is not a prediction for the historical sampled semantic score.

This result concerns exact entropy over the finite output categories in the reduction. Entropy over completion strings, conditional verified keys, and sampled token distributions are different objectives. It does not certify a clipped surrogate or an adaptive neural optimizer. The concavity condition holds for Dr.GRPO and the implemented binary-MaxRL potential.

Lemma A.9 (Bounded sampled scores have no probability-independent boundary force). *Condition on the training history, draw $A_1, \dots, A_G$ independently from p , and let a_i be any detached, possibly group-dependent advantages with $|a_i| \leq B$. At the on-policy point define*

$$\hat{g} = \frac{1}{G} \sum_{i=1}^G a_i (e_{A_i} - p).$$

For every category b ,

$$|\mathbb{E}[\hat{g}_b]| \leq \mathbb{E}[\hat{g}_b] \leq 2Bp_b(1 - p_b).$$

On a group with no draw of b , exactly $\hat{g}_b = -p_b G^{-1} \sum_i a_i$; it is zero if the applied advantages sum to zero. By comparison, the verified-replay component for a fixed target w has logit velocity $\rho(w_b - p_b)$, which approaches $\rho w_b > 0$ for a banked category as $p_b \rightarrow 0$.

Proof. Use the triangle inequality and $|a_i| \leq B$. Each marginal draw satisfies $\mathbb{E}[\mathbf{1}\{A_i = b\} - p_b] = 2p_b(1 - p_b)$, irrespective of the dependence of a_i on the group. On the absence event all indicator terms are zero. The replay formula follows by differentiating $-\sum_b w_b \log p_b$. $\square$

The fixed historical semantic advantage has $B = \eta = .10$, including its unit pseudocount, reserved unseen bucket, and clipped surprisal. The bound does not prove inevitable collapse or rule out asymptotic retention by a particular bounded estimator. Unsampled probabilities can also change through softmax normalization or shared parameters. Exact entropy is outside the uniformly bounded-score premise because its surprisal is unbounded near zero.

A.6 EXACT CONDITIONAL ENTROPY AND REPLAY TARGET THE SAME OPTIMUM

Two regularizers can favor the same uniform target while assigning different penalties to rare modes. The following identities locate that difference.

The distinction between confidence penalties and label smoothing is a known choice of KL direction (Pereyra et al., 2017, Section 3.2). Log-barrier regularization has also been studied in policy-gradient theory (Agarwal et al., 2021, Section 5.2). Here we specialize those identities to a target supported on verified correct modes. The resulting comparison separates an objective’s maximizer from what its value certifies about a rare mode.

Let $u_c = 1/m$ on all $m \geq 2$ correct categories, retain $P > 0$ and $q_c = p_c/P$, and write $H(q) = -\sum_c q_c \log q_c$, with $0 \log 0 = 0$.

Lemma A.10 (Full-support objective comparison). *For a bank containing every correct category with a uniform categorical replay target,*

$$H(q) = \log m - \text{KL}(q||u), \tag{17}$$

$$R_u(p) := -\frac{1}{m} \sum_{c \in \mathcal{C}} \log p_c = \log m - \log P + \text{KL}(u||q). \tag{18}$$

If Ψ is continuous and strictly increasing on $[0, 1]$, then for all $\beta, \rho > 0$, both objectives

$$\Psi(P) + \beta H(q) \quad \text{and} \quad \Psi(P) - \rho R_u(p)$$

have the unique maximizing categorical distribution $P = 1, q = u$. Maximization is over the closed simplex with $P > 0$; replay loss is $+\infty$ if a protected coordinate is zero. This identifies the optimum; convergence of conditional-entropy flow or stochastic updates is not asserted.

Proof. Expanding the KL divergences gives the first identity. Substitution of $p_c = Pq_c$ into R_u gives the second. In the first objective, $H(q) \leq \log m$, with equality only at $q = u$; after choosing this q , strict increase of Ψ selects $P = 1$. The second objective is

$$\Psi(P) + \rho \log P - \rho \log m - \rho \text{KL}(u||q).$$

Its terms depending on P increase strictly, while the last term is maximized only at $q = u$. At $P = 1$ all incorrect probabilities are zero, so the resulting full categorical distribution is unique. $\square$

Worked example: separate correctness from conditional spread. For the running distribution $p = (0.30, 0.15, 0.05, 0.50)$, the uniform target on the three correct modes gives

$$H(q) \approx \log 3 - 0.20067 \approx 0.89795,$$

$$R_u(p) \approx \log 3 - \log 0.5 + 0.24052 \approx 2.03228.$$

The subtracted 0.20067 is $\text{KL}(q\|u)$; the added 0.24052 is $\text{KL}(u\|q)$. Raising P from 0.5 to 0.9 while holding q fixed leaves $H(q)$ unchanged but reduces replay loss by $\log(0.9/0.5) \approx 0.58779$. Thus replay’s loss measures both correct mass and allocation within it; conditional entropy measures only the latter. The correctness potential $\Psi(P)$ supplies the first objective’s incentive to increase P .

Full-output entropy $H(p)$ is a different comparator. Its Gibbs optimum in Lemma A.8 follows the standard entropy-conjugacy calculation (Geist et al., 2019, Section 2.2). That optimum assigns positive incorrect mass for fixed $\beta > 0$ under linear correctness reward. By contrast, Lemma A.10 shows that exact *conditional* entropy has no such loss of correctness at its global optimum. The historical semantic score used in the experiments is not an exact gradient of either entropy objective.

Existing literature also gives discrete-time guarantees for an exact comparator. Under their finite-bandit setup and step-size conditions, Mei et al. (2020, Lemma 13 and Theorem 5) prove positive probability floors and geometric convergence for exact tabular entropy-regularized policy gradient. The Dr.GRPO reduction has the linear reward vector $a\mathbf{1}_C$ with $a = (G - 1)/G \in (0, 1)$, which satisfies their bounded-reward premise. This linear-reward result does not directly cover the nonlinear finite-group MaxRL potential, the historical semantic estimator, a clipped PPO surrogate, or AdamW.

The KL directions explain a difference between certificates: $\text{KL}(u\|q)$ diverges if any protected coordinate vanishes, whereas $\text{KL}(q\|u)$ stays finite. In particular, losing one mode and allocating mass uniformly to the remaining modes leaves $H(q) = \log(m - 1)$. Section A.12 gives the sharp entropy threshold and stronger replay-based probability bounds. This comparison concerns objective level sets; it does not assert that exact entropy dynamics lose a mode.

A.7 WHEN REPLAY SUPPLIES A SIGNAL THAT FRESH GROUPS LACK

High accuracy can remove the reward contrasts that fresh-group learning needs. A stored target can still supply a learning signal in that regime.

The collapse theorem concerns an expected direction. A separate question is how often a finite group supplies any task signal, and whether a stored mode must reappear in that group before receiving a nonzero update from the replay objective.

Lemma A.11 (Fresh-group starvation and replay signal). *Fix a categorical policy with correctness probability P , and draw $G \geq 2$ fresh responses independently. For GRPO, Dr.GRPO, and the implemented centered MaxRL estimator, the probability of a mixed-reward group is*

$$h_G(P) = 1 - P^G - (1 - P)^G \leq G \min\{P, 1 - P\}.$$

Every all-equal group has zero fresh task gradient. Thus h_G is the probability of an opportunity for a nonzero task gradient, rather than a lower bound on the gradient magnitude.

For a fixed bank with normalized positive weights w_b , let $R_w = -\sum_b w_b \log p_b$ and $\rho_w > 0$. Extend w by zero outside the bank to obtain $\bar{w}$. The deterministic replay ascent vector is

$$v^{\text{replay}} = -\rho_w \nabla R_w = \rho_w (\bar{w} - p).$$

It vanishes only at $p = \bar{w}$. In particular, if $p_b \leq w_b/2$ for a banked mode, then $v_b^{\text{replay}} \geq \rho_w w_b/2$, even when b is absent from the fresh group. For a uniform bank of $k \geq 2$ modes, as $p \rightarrow \mathbf{e}_b$ at one banked correct mode, $h_G(P) \rightarrow 0$ whereas $\|v^{\text{replay}}\|_2 \rightarrow \rho_w \sqrt{1 - 1/k} > 0$. The replay norm therefore stays positive as fresh task-signal opportunities vanish.

Proof. The all-correct and all-incorrect events are disjoint and have probabilities P^G and $(1 - P)^G$. A mixed group requires both a correct and an incorrect sample; the union bound gives the two bounds GP and $G(1 - P)$. For $R_w = -\sum_b w_b \log p_b$, the softmax score identity gives $\partial R_w / \partial z_a = p_a - \bar{w}_a$, proving the replay formula and coordinate bound. Finally,

$$\|\bar{w} - \mathbf{e}_b\|_2^2 = (1 - 1/k)^2 + (k - 1)/k^2 = 1 - 1/k,$$

which gives the strictly positive limiting norm for $k \geq 2$. $\square$

Worked example: a rare key need not reappear to receive replay. Consider the hypothetical policy $p = (0.600, 0.389, 0.001, 0.010)$, with the first three categories correct, and take $G = 16$. Its $P = 0.99$ gives

$$h_{16}(0.99) = 1 - 0.99^{16} - 0.01^{16} \approx 0.14854.$$

Only about 14.85% of groups offer a fresh task-gradient opportunity. The third key is absent with probability $0.999^{16} \approx 0.98412$. Nevertheless, uniform replay over the three correct keys with $\rho_w = 0.1$ has third logit coordinate $v_3^{\text{replay}} = 0.1(1/3 - 0.001) \approx 0.03323$, including on groups that miss that key. The two sampling probabilities describe different events; the replay value is the deterministic full-bank vector from the lemma.

Here the replay vector averages the fixed bank; a stochastic draw of one exemplar has this vector only in expectation. Its nonzero logit coordinate does not guarantee a nonzero neural-parameter gradient after multiplication by a possibly degenerate Jacobian. Nor does replay necessarily increase the next mixed-group probability: h_G is largest at $P = 1/2$, so increasing an already high P decreases it. The precise benefit is a separate verified learning signal when the fresh task gradient is unavailable.

A.8 WHY UNIFORM VERIFIED REPLAY PREVENTS EXTINCTION

A persistent replay signal becomes a retention guarantee when vanishing probability would force a bounded loss to diverge. The next argument makes that loss barrier explicit, then shows what changes with unequal weights and incomplete coverage.

Let $\mathcal{B} \subseteq \mathcal{C}$ be a nonempty fixed bank of k modes that the policy has produced and the validator has accepted. Uniform verified replay uses one exemplar per banked mode and, in the categorical reduction, minimizes

$$R_{\mathcal{B}}(z) = -\frac{1}{k} \sum_{b \in \mathcal{B}} \log p_b. \quad (19)$$

This is ordinary cross-entropy against the distribution that assigns mass $1/k$ to each banked mode and zero elsewhere. It is not conditioned on the bank: decreasing (19) can raise total bank mass as well as redistribute mass within the bank.

In an averaged continuous-time model, absorb a fixed round-robin replay frequency and loss scale into an effective dose $\rho > 0$. This assumes a fixed eligible-bank set and constant relative weighting; it does not identify finite alternating updates with simultaneous gradient flow. The prompt-isolated categorical mean flow for any of the three fresh binary-reward objectives above plus verified replay is

$$\dot{z} = c_G(P) \nabla_z P - \rho \nabla_z R_{\mathcal{B}}(z). \quad (20)$$

Let $\Psi(P) = \int_0^P c_G(s) ds$. For the finite-group estimators in Lemmas A.1 and A.2, c_G has a finite positive extension to $[0, 1]$; in particular, Dr.GRPO has $c_G = (G - 1)/G$ and MaxRL has $c_G = [1 - (1 - P)^{G-1}]/P$. Define

$$F_{\mathcal{B}}(z) = \rho R_{\mathcal{B}}(z) - \Psi(P(z)). \quad (21)$$

Then (20) is exactly gradient flow $\dot{z} = -\nabla_z F_{\mathcal{B}}$.

Theorem A.12 (Verified replay prevents banked-mode extinction). *Suppose that after time T the nonempty verified bank $\mathcal{B}$ is fixed, fits within replay capacity, and receives a fixed positive effective replay dose ρ . Under (20), define*

$$C_T = R_{\mathcal{B}}(z(T)) + \frac{\Psi(1) - \Psi(P(T))}{\rho}.$$

Then for every $b \in \mathcal{B}$ and every $t \geq T$,

$$p_b(t) \geq \exp(-kC_T) > 0. \quad (22)$$

Thus each banked probability is uniformly bounded away from zero for all $t \geq T$, including asymptotically, even when the bank does not cover every correct mode. This is stronger than finite-time positivity, which already follows from finite softmax logits without replay.

Proof. Along the flow,

$$\frac{d}{dt} F_{\mathcal{B}}(z(t)) = -\|\nabla_z F_{\mathcal{B}}(z(t))\|_2^2 \leq 0.$$

Since $P(t) \leq 1$ and Ψ is increasing,

$$R_{\mathcal{B}}(z(t)) \leq R_{\mathcal{B}}(z(T)) + \frac{\Psi(1) - \Psi(P(T))}{\rho} = C_T.$$

Every $-\log p_b$ is nonnegative and their sum is $kR_{\mathcal{B}}$. Therefore $-\log p_b(t) \leq kC_T$ for every banked mode, which is exactly Equation (22) for every time after T . $\square$

Worked example: a rare mode cannot hide inside a bounded loss. In the running example, bank the first and third correct modes, whose probabilities are .30 and .05. Their uniform replay loss is $R_{\mathcal{B}} = -\frac{1}{2} \log(.30 \cdot .05) \approx 2.09985$. For the illustrative Dr.GRPO choices $G = 4$ and $\rho = .30$, $C_T = R_{\mathcal{B}} + (.75)(1 - .50)/.30 \approx 3.34985$, giving a floor $p_b \geq e^{-2C_T} \approx 0.00123127$. The point is the constraint on the loss: if either banked probability fell to 10^{-4} , its term alone would contribute $-\frac{1}{2} \log(10^{-4}) \approx 4.60517 > C_T$. The other term is nonnegative and cannot cancel this increase. These are hypothetical categorical values; the argument explains the barrier without making its conservative floor an experimental prediction.

This is a qualitative positivity statement, not a useful numerical guarantee. If a prompt receives fresh updates with frequency $\nu_x > 0$ and replay with frequency $\mu_x > 0$, normalizing time by its fresh-update frequency gives $\rho = \rho_{\text{raw}} \mu_x / \nu_x$. Equal fresh and replay frequencies cancel; round-robin scheduling alone does not imply dilution of the relative dose. As an illustrative equal-frequency categorical calculation, the registered $G = 16$ coefficient and per-rollout scale give $\rho = .10(15/256) = 5.86 \times 10^{-3}$. With $k = 16$, the worst-case Dr.GRPO potential term alone permits a bound on the order of $\exp(-2560)$ (and $R_{\mathcal{B}}(z(T))$ can only weaken it). The bound is therefore vacuous at machine and experimental scales; it establishes only that an idealized continuous-time, fixed-bank coordinate cannot vanish asymptotically.

The retention bound also holds under smooth parameterization. The same argument applies to $\dot{\theta} = -\nabla_{\theta} F_{\mathcal{B}}(z(\theta))$:

$$\frac{d}{dt} F_{\mathcal{B}}(z(\theta(t))) = -\|J_{\theta}^{\top} \nabla_z F_{\mathcal{B}}\|_2^2 \leq 0.$$

Thus Equation (22) holds regardless of parameter count for this fixed-bank, prompt-isolated parameter gradient flow. It does not require the shared-correctness geometry in Theorem A.5. This extends the qualitative retention bound, not convergence to a uniform distribution: a restricted or degenerate parameterization can stall. The next corollary retains independent logits.

Corollary A.13 (Conditional full-coverage categorical limit). *Assume the conditions of Theorem A.12 and that by time T the bank covers the full correct support, $\mathcal{B} = \mathcal{C}$. Let $u_c = 1/m$ on $\mathcal{C}$. Then the categorical mean flow converges to*

$$P(t) \longrightarrow 1, \quad q(t) \longrightarrow u.$$

Consequently

$$\min_{c \in \mathcal{C}} q_c(t) \longrightarrow \frac{1}{m}, \quad \mathcal{H}(q(t)) \longrightarrow \log m,$$

and conditional correct-mode collapse is impossible.

Proof. When $\mathcal{B} = \mathcal{C}$,

$$R_{\mathcal{C}} = \log m - \log P + \text{KL}(u\|q). \quad (23)$$

Thus one uniform verified-likelihood loss already contains both a verified-mass term $-\log P$ and a within-support uniformity term $\text{KL}(u\|q)$. The potential (21) becomes

$$F_{\mathcal{C}} = \rho \log m + \rho \text{KL}(u\|q) - \rho \log P - \Psi(P).$$

For fixed P , its unique minimizer is $q = u$. Its derivative in P is $-\rho/P - c_G(P) < 0$, so its unique minimizing distribution on the closed simplex is $P = 1, q = u$, placing all limiting probability uniformly on the correct support and zero mass on every incorrect category in the categorical limit.

For completeness, Theorem A.12 keeps every correct coordinate away from zero. The decreasing potential gives $\int_T^\infty \|\nabla F_C\|_2^2 dt < \infty$. The finite-group c_G is smooth on $[0, 1]$; softmax derivatives and the cross-entropy gradient are bounded. Thus the Hessian of F_C and the velocity $\dot{z} = -\nabla F_C$ are bounded, making $\nabla F_C(z(t))$ uniformly continuous. Together with square integrability, this implies $\nabla F_C \rightarrow 0$: recurring gradients of fixed positive size would persist over intervals of a uniform minimum length, contradicting the finite energy integral. At any limiting distribution the correct-coordinate stationarity equations are

$$0 = \rho(p_c - 1/m) - c_G(P)p_c(1 - P),$$

so all p_c are equal. Substituting $p_c = P/m$ gives $0 = -(1 - P)(\rho + c_G(P)P)/m$, hence $P = 1$. Each incorrect coordinate satisfies $0 = p_a(\rho + c_G(P)P)$ and therefore has $p_a = 0$. The limiting distribution is unique, proving convergence from every initialization covered by the corollary. $\square$

Positive replay weights and exemplar length. Uniformity of the target is separate from protection against extinction. For fixed weights $w_b > 0$ with $\sum_{b \in \mathcal{B}} w_b = 1$, define $R_w = -\sum_b w_b \log p_b$ and use replay coefficient $\rho_w > 0$. The same potential argument gives, for all $t \geq T$,

$$p_b(t) \geq \exp(-C_{T,w}/w_b), \quad C_{T,w} = R_w(z(T)) + \frac{\Psi(1) - \Psi(P(T))}{\rho_w}.$$

If $\mathcal{B} = \mathcal{C}$, then $R_w = H(w) - \log P + \text{KL}(w||q)$, and the independent-logit limit is $P \rightarrow 1$, $q \rightarrow w$. The preceding stationarity proof applies with $1/m$ replaced by w_c : summing the correct-coordinate equations forces $P = 1$, then each equation forces $p_c = w_c$.

This distinction matters for the implemented mean-token exemplar loss. Even if each key were represented by exactly one categorical exemplar of fixed length L_b , its loss would be AR_w , where

$$A = \frac{1}{k} \sum_{b \in \mathcal{B}} \frac{1}{L_b}, \quad w_b = \frac{L_b^{-1}}{\sum_{j \in \mathcal{B}} L_j^{-1}}.$$

Its effective categorical replay coefficient would be $\rho_w = \rho_{\text{LM}}A$. The full-coverage limit would therefore weight shorter exemplars more, and would be uniform only for equal lengths. For an actual language model, $\pi_\theta(e_b | x)$ is only one contribution to the total key probability. Equal averaging of per-token exemplar scores does not imply uniform learned key probabilities; the categorical uniformity corollary is not that claim.

Worked example: equal key sampling need not imply a uniform target. For three singleton categorical exemplars of lengths 10, 20, 40, equal averaging of mean-token losses gives $A = 7/120$ and $w = (4/7, 2/7, 1/7)$. All three target weights are positive, so each mode is protected, but the full-coverage categorical limit assigns the shortest exemplar four times the probability of the longest. Equal lengths would instead give $(1/3, 1/3, 1/3)$. This example concerns the loss's induced categorical target, not an asserted distribution over actual LM keys.

The premise $\mathcal{B} = \mathcal{C}$ is not a description of the released Python-factors experiments. Their prompts have 16–3,600 valid execution modes (mean 229.44), while the replay bank has capacity 16. Thus the corollary is inapplicable to Python factors as a domain-wide guarantee. More generally it applies only to prompts whose entire correct support fits in the bank and has actually been discovered; neither condition follows from replay.

Lemma A.14 (Fixed-bank limits and deterministic finite steps). *Retain finite independent Euclidean logits, a nonempty fixed bank $\mathcal{B} \subseteq \mathcal{C}$, fixed positive normalized weights w_b , and $\rho_w > 0$. Let $F = \rho_w R_w - \Psi(P)$, with $\Psi \in C^2([0, 1])$ and $\Psi' = c_G \geq 0$. Set*

$$M_1 = \sup_{[0,1]} |\Psi'|, \quad M_2 = \sup_{[0,1]} |\Psi''|, \quad L = \frac{\rho_w}{2} + \frac{M_1}{2} + \frac{M_2}{8}.$$

For constant $0 < \eta < 2/L$, deterministic updates $z_{n+1} = z_n - \eta \nabla F(z_n)$ preserve, for every $n \geq 0$,

$$p_b(z_n) \geq \exp(-C_{0,w}/w_b), \quad C_{0,w} = R_w(z_0) + \frac{\Psi(1) - \Psi(P(z_0))}{\rho_w}.$$

Moreover, $p(z_n) \rightarrow \bar{w}$, where $\bar{w}_b = w_b$ on the bank and $\bar{w}_a = 0$ elsewhere. The same probability limit holds for the corresponding continuous mean flow. The logits need not converge: zero limiting probabilities require divergent logit differences relative to banked coordinates.

Proof. For a logit direction u , let $U = u_a$ and $Y = \mathbf{1}[a \in \mathcal{C}]$ under $a \sim p$. Differentiating along that direction gives

$$u^\top \nabla^2 P u = \mathbb{E}[(Y - P)(U - \mathbb{E}U)^2], \quad u^\top \nabla^2 R_w u = \text{Var}(U).$$

Since $|Y - P| \leq 1$, the first quadratic form is bounded in absolute value by $\text{Var}(U)$. The bounded-range variance inequality gives

$$\text{Var}(U) \leq \frac{(\max_a u_a - \min_a u_a)^2}{4} \leq \frac{\|u\|_2^2}{2}.$$

Both Hessians therefore have operator norm at most $1/2$. Also $\|\nabla P\|_2^2 \leq 2P^2(1 - P)^2 \leq 1/8$. The chain rule therefore bounds $\|\nabla^2 F\|_{\text{op}} \leq L$ globally. The descent lemma yields

$$F(z_{n+1}) \leq F(z_n) - \eta(1 - L\eta/2)\|\nabla F(z_n)\|_2^2.$$

Energy monotonicity gives the preceding replay-loss bound and hence the stated retention floor. Since $F \geq -\Psi(1)$, the squared gradients are summable and $\nabla F(z_n) \rightarrow 0$. Its coordinate expression,

$$[\nabla F]_a = \rho_w(p_a - \bar{w}_a) - c_G(P)p_a(\mathbf{1}[a \in \mathcal{C}] - P),$$

extends continuously to the closed probability simplex. Every subsequential probability limit is therefore stationary. On an incorrect coordinate, stationarity gives $(\rho_w + c_G(P)P)p_a = 0$, forcing all incorrect mass to zero and $P = 1$. The correct-coordinate equations then give $p_a = \bar{w}_a$. Compactness and this unique limit prove convergence of the whole probability sequence. For the mean flow, bounded Hessian and velocity together with energy descent give vanishing gradient as in the preceding corollary; the same stationarity argument applies. $\square$

For Dr.GRPO, $M_1 = (G - 1)/G$ and $M_2 = 0$. For the implemented finite-group MaxRL estimator, $M_1 = G - 1$ and $M_2 = (G - 1)(G - 2)/2$, so one sufficient smoothness constant is $L = \rho_w/2 + (G - 1)(G + 6)/16$. These bounds concern categorical logit steps, not the neural optimizer's learning rate. With an incomplete frozen bank, the categorical limit assigns zero mass even to correct modes outside the bank: fixed-bank retention is not a guarantee of discovering or preserving all correct modes. The fixed-bank limit also gives

$$\lim_{t \rightarrow \infty} \text{distinct@}K = \sum_{b \in \mathcal{B}} [1 - (1 - w_b)^K] \leq k,$$

with the same expression along deterministic iterates. Thus the bank itself sets an asymptotic support ceiling in this model. The later stochastic and changing-bank results give conditional retention bounds under additional energy assumptions; they do not extend this fixed-bank convergence limit to arbitrary neural training.

Worked example: a safe logit step and an incomplete bank. Return to the two-mode bank above with $\rho_w = .30$, $G = 4$, and Dr.GRPO. Here $L = .525$, so $\eta = 1$ satisfies $0 < \eta < 2/L \approx 3.81$. One deterministic logit-gradient step changes the running distribution approximately as

$$(.30, .15, .05, .50) \mapsto (.3861, .1643, .0632, .3864),$$

while the potential falls from about .25496 to .09689. Both banked modes gain probability. The unbanked second correct mode also gains on this first step, but the lemma's eventual limit is $(.50, 0, .50, 0)$: early improvement of an unbanked mode does not make it protected. The step size here belongs to the independent-logit example; it is not a recommendation for the neural optimizer.

A.9 FROM CATEGORICAL MODES TO COMPLETE REPLAYED RESPONSES

The categorical analysis protects an execution key, whereas the learner scores a particular response string. The bridge is event containment: one complete verified response is one way to produce its key. A bound on its sequence surprisal therefore protects the key even when many other strings realize the same behavior. Length normalization changes the numerical loss scale, so its factor must remain explicit in that bound.

Lemma A.15 (Retention from a joint exemplar potential). *Fix finitely many verified complete-response exemplars (x_j, e_j) , each realizing key b_j , with lengths $L_j > 0$ and fixed weights $\alpha_j > 0$. Let*

$$F(\theta) = \sum_j \alpha_j \frac{-\log \pi_\theta(e_j | x_j)}{L_j} - J(\theta), \quad J(\theta) \leq J_{\max} < \infty.$$

Suppose $F(\theta(T))$ is finite and $F(\theta(t)) \leq F(\theta(T))$ for every $t \geq T$. With $D = F(\theta(T)) + J_{\max}$, every protected exemplar satisfies

$$p_\theta(b_j | x_j) \geq \pi_\theta(e_j | x_j) \geq \exp\left(-\frac{L_j D}{\alpha_j}\right) > 0 \quad (t \geq T).$$

Proof. The weighted exemplar loss equals $F(\theta(t)) + J(\theta(t)) \leq D$. Each summand is nonnegative, so $-\log \pi_\theta(e_j | x_j) \leq L_j D / \alpha_j$. The complete-response event is a subset of the event producing key b_j , which proves the stated lower bound on every protected key’s probability. $\square$

Example: a token average is not a response probability. Suppose a complete response has four scored tokens, including termination, and each token has conditional probability $1/2$ along that response. Its mean log likelihood is $-\log 2$, but its complete-response probability is $(1/2)^4 = 1/16$; exponentiating the mean gives $1/2$, the geometric mean of token probabilities. For a single protected exemplar with $L = 4$, $\alpha = 1$, and joint energy ceiling $D = \log 2$, the lemma gives exactly the $1/16$ floor. If a different complete response with the same key has probability $1/32$, the two disjoint response events together give key probability at least $1/16 + 1/32 = 3/32$. The exemplar bound can thus be conservative even when its sequence probability is known exactly.

Exact gradient flow of a smooth joint potential is one sufficient condition for this monotonicity; the lemma requires only the stated energy inequality. Shared parameters and multiple prompts therefore need not invalidate retention. For example, the exact categorical mean model admits $J(\theta) = \sum_x \nu_x \Psi_x(P_x(\theta))$ with fixed nonnegative prompt weights and $J_{\max} = \sum_x \nu_x \Psi_x(1)$. The bound does not require uniform key weights or imply convergence to a uniform policy. It does not establish joint energy descent for the implemented stochastic, clipped, alternating optimizer. The likelihood must describe the complete response under the sampling law being bounded, including its termination event or a fixed deterministic horizon. A prefix likelihood alone, or a score under a different temperature or token-truncation rule, is insufficient.

Retention and finite evaluation visibility. At a fixed checkpoint, suppose k distinct banked keys for one prompt have probability floors $p_b \geq \delta_b > 0$. For K independent draws from that same policy, the banked distinct count $D_{\mathcal{B}, K}$ satisfies

$$\mathbb{E}[D_{\mathcal{B}, K}] \geq \sum_{b \in \mathcal{B}} [1 - (1 - \delta_b)^K],$$

$$\Pr(\text{some banked key is unseen}) \leq \sum_{b \in \mathcal{B}} (1 - \delta_b)^K \leq k e^{-K \delta_{\min}},$$

where $\delta_{\min} = \min_b \delta_b$. The first statement follows from indicator expectations and the second from the union bound. Thus $K \geq \log(k/\varepsilon)/\delta_{\min}$ suffices to observe all banked keys with probability at least $1 - \varepsilon$, for $0 < \varepsilon < 1$. The energy floors can make this budget impractically large; asymptotic retention does not imply visibility in an eight-draw evaluation.

Example: protection can still be hard to observe. A key with probability exactly $1/16$ appears in eight independent draws with probability $1 - (15/16)^8 \approx 0.403$. If four keys each have a $1/16$ floor, the sufficient budget above is $\lceil 16 \log(4/0.05) \rceil = 71$ draws for at least 95% probability of seeing all four. These hypothetical probabilities distinguish retaining a mode from reliably observing it in a short evaluation.

A.10 RETENTION UNDER DISCRETE STOCHASTIC OPTIMIZER UPDATES

For discrete noisy updates, the useful quantity is the total upward drift allowed in the joint loss. Controlling that budget permits individual steps to increase the loss while keeping a high-probability ceiling on the loss over a stated update horizon.

Lemma A.15 converts a deterministic energy ceiling into complete-exemplar protection. We next replace pathwise descent with a controlled conditional drift, allowing shared parameters, noise, bias, and predictable preconditioning. The descent estimate follows the variable-metric analysis of Wang et al. (2017, Section 2); the trajectory bound uses Ville’s nonnegative-supermartingale inequality, as stated by Howard et al. (2020, Lemma 1). These are standard ingredients; the result below applies them to the verified-response potential.

Theorem A.16 (Conditional stochastic exemplar retention). *Fix finitely many complete verified responses (x_j, e_j) realizing keys b_j , with lengths $L_j > 0$ and weights $\alpha_j > 0$. Define*

$$R(\theta) = \sum_j \frac{\alpha_j}{L_j} [-\log \pi_\theta(e_j | x_j)], \quad F(\theta) = R(\theta) - J(\theta), \quad J(\theta) \leq J_{\max} < \infty.$$

Assume F is differentiable with globally L_F -Lipschitz gradient and θ_0 is deterministic with finite $F(\theta_0)$. Relative to the history filtration $(\mathcal{F}_t)$, consider

$$\theta_{t+1} = \theta_t - \eta_t H_t(g_t + b_t + \xi_t), \quad g_t = \nabla F(\theta_t).$$

The variables $\theta_t, \eta_t, H_t, b_t$ are $\mathcal{F}_t$ -measurable, $0 \preceq H_t \preceq MI$, $\eta_t > 0$, and $\eta_t L_F M \leq 1$. The noise satisfies $\mathbb{E}[\xi_t | \mathcal{F}_t] = 0$ and has finite conditional second moment. Suppose deterministic $d_t \geq 0$ satisfy

$$\frac{\eta_t}{2} \|b_t\|_{H_t}^2 + \frac{L_F \eta_t^2}{2} \mathbb{E}[\|H_t \xi_t\|_2^2 | \mathcal{F}_t] \leq d_t, \quad \|u\|_H^2 = u^\top H u. \quad (24)$$

For $N \in \mathbb{N} \cup \{\infty\}$ with $B_N = \sum_{t < N} d_t < \infty$, write $D_0 = F(\theta_0) + J_{\max} \geq 0$. For every $0 < \delta < 1$, with probability at least $1 - \delta$, the following holds simultaneously for all integer $t \leq N$ and all protected exemplars:

$$p_{\theta_t}(b_j | x_j) \geq \pi_{\theta_t}(e_j | x_j) \geq \exp \left[-\frac{L_j(D_0 + B_N)}{\alpha_j \delta} \right]. \quad (25)$$

For $N = \infty$, each protected probability also has an almost surely positive, potentially random, infimum over all iterations. The likelihood must include the complete response and its termination event, or a fixed deterministic horizon, under the sampling law whose key probability is bounded.

Proof. Smoothness, conditional centering of ξ_t , and $H_t^2 \preceq M H_t$ imply

$$\begin{aligned} \mathbb{E}[F(\theta_{t+1}) | \mathcal{F}_t] &\leq F(\theta_t) - \eta_t g_t^\top H_t(g_t + b_t) + \frac{\eta_t}{2} \|g_t + b_t\|_{H_t}^2 \\ &\quad + \frac{L_F \eta_t^2}{2} \mathbb{E}[\|H_t \xi_t\|_2^2 | \mathcal{F}_t] \\ &\leq F(\theta_t) - \frac{\eta_t}{2} \|g_t\|_{H_t}^2 + d_t. \end{aligned}$$

The bias cross term cancels when the square is expanded. Since $F + J_{\max} \geq R \geq 0$, the process

$$X_t = F(\theta_t) + J_{\max} + \sum_{s=t}^{N-1} d_s$$

is a nonnegative supermartingale, using the convergent infinite tail when $N = \infty$. Its initial value is $D_0 + B_N$. For $D_0 + B_N > 0$, Ville’s inequality gives $\Pr(\sup_{t \leq N} X_t \geq (D_0 + B_N)/\delta) \leq \delta$. If $D_0 + B_N = 0$, nonnegativity gives $X_t = 0$ almost surely instead. Every nonnegative summand of $R(\theta_t) \leq X_t$ obeys the same ceiling. Exponentiation therefore yields (25) without a union factor over exemplars. A complete-exemplar event is contained in its verified-key event. For $N = \infty$, the same maximal bound gives $\Pr(\sup_t X_t \geq u) \leq (D_0 + B_\infty)/u$ for $u > 0$. Letting $u \rightarrow \infty$ proves $\sup_t X_t < \infty$ almost surely and hence random positive all-time floors. $\square$

Finite and infinite horizons. With zero bias and uniformly bounded noise variance, the cost in (24) is $O(\eta_t^2)$, so square-summable steps suffice for an infinite-horizon budget. Bias instead contributes $O(\eta_t \|b_t\|^2)$. Constant steps with persistent variance yield the finite-horizon result, with a budget that can grow with N ; this argument does not give infinite-horizon protection in that regime. The classical almost-supermartingale framework of Robbins & Siegmund (1971) supplies related convergence tools, but the proof above needs only the displayed drift and maximal inequality. No positive lower eigenvalue of H_t is needed for retention alone, and the theorem does not assert stationarity, parameter convergence, or a useful evaluation-scale probability floor.

Example: a noise budget for one protected response. Consider a hypothetical fixed one-token horizon with two possible complete responses and $\pi_\theta(e) = (1 + e^{-\theta})^{-1}$, $L = \alpha = 1$, and $F = R = -\log \pi_\theta(e)$, with $J = J_{\max} = 0$. Start at $\theta_0 = 0$, so $D_0 = \log 2$. This scalar objective has $L_F = 1/4$ and $|g_t| \leq 1$. Take $H_t = 1$, zero bias, and independent noise $\xi_t \in \{-0.1, 0.1\}$ with equal probabilities. At constant step $\eta_t = 0.1$, the theorem permits $d_t = 1/80000$, giving $B_{100} = 0.00125$ but $B_\infty = \infty$. With $\eta_t = 0.1/(t+1)$ instead, $B_\infty = \pi^2/480000 \approx 2.06 \times 10^{-5}$, which supplies the infinite-horizon condition. For the constant-step, 100-update case, the 95% probability floor from (25) is only $\exp[-(\log 2 + 0.00125)/0.05] \approx 9.30 \times 10^{-7}$. The next result uses bounded noise to improve this confidence dependence.

Corollary A.17 (Sharper confidence dependence under bounded noise). *Under the assumptions of Theorem A.16, suppose $b_t = 0$, $N < \infty$, and η_t is deterministic. Let deterministic bounds $G_t, s_t < \infty$ satisfy $\|g_t\|_2 \leq G_t$ and $\|\xi_t\|_2 \leq s_t$ almost surely. Define*

$$c_t = \eta_t M G_t s_t, \quad V_N = \sum_{t < N} c_t^2, \quad Q_N = \frac{L_F M^2}{2} \sum_{t < N} \eta_t^2 s_t^2.$$

For every $0 < \delta < 1$, with probability at least $1 - \delta$, simultaneously for all $t \leq N$ and j ,

$$p_{\theta_t}(b_j | x_j) \geq \pi_{\theta_t}(e_j | x_j) \geq \exp \left[-\frac{L_j}{\alpha_j} \left(D_0 + Q_N + \sqrt{2V_N \log(1/\delta)} \right) \right]. \quad (26)$$

Proof. The pathwise smoothness estimate gives

$$F(\theta_{t+1}) - F(\theta_t) \leq -\frac{\eta_t}{2} \|g_t\|_{H_t}^2 + Z_{t+1} + \frac{L_F \eta_t^2 M^2 s_t^2}{2},$$

where

$$Z_{t+1} = -\eta_t g_t^\top H_t (I - L_F \eta_t H_t) \xi_t.$$

Predictability makes Z_{t+1} a martingale difference. The spectral bounds imply $0 \preceq I - L_F \eta_t H_t \preceq I$, so $|Z_{t+1}| \leq c_t$. Conditional Hoeffding's lemma therefore makes

$$\exp \left(\lambda \sum_{s < t} Z_{s+1} - \frac{\lambda^2}{2} \sum_{s < t} c_s^2 \right), \quad \lambda > 0,$$

a nonnegative supermartingale starting at one. This standard exponential construction is reviewed by Howard et al. (2020). Ville's inequality and optimization over λ yield

$$\Pr \left(\max_{t \leq N} \sum_{s < t} Z_{s+1} > \sqrt{2V_N \log(1/\delta)} \right) \leq \delta$$

when $V_N > 0$; when $V_N = 0$, the sum is identically zero. Sum the drift inequalities, drop the nonpositive gradient terms, and apply the summand bound from Theorem A.16. $\square$

In the same 100-update example, $G_t = M = 1$ and $s_t = 0.1$ give $V_{100} = 0.01$ and $Q_{100} = 0.00125$. The sharper 95% floor is therefore

$$\exp \left[-\log 2 - 0.00125 - \sqrt{0.02 \log 20} \right] \approx 0.391.$$

Both bounds apply simultaneously to the toy trajectory's first 100 updates; the improvement comes from stronger noise assumptions. These toy constants are assigned for the calculation; they do not estimate noise or curvature in the language-model runs.

Relation to stochastic policy-gradient guarantees. Zhang et al. (2021) analyze stochastic tabular REINFORCE with a state-action log barrier, phased step and regularization schedules, and policy mixing. Their regret guarantees provide a related stochastic policy-gradient precedent; they do not provide a fixed per-response floor for our replay bank. For the implemented optimizer, a current-gradient Adam second moment is not predictable relative to that gradient's noise. Momentum, clipping, prompt scheduling, and changing banks likewise require separate bias or energy control. The counterexamples of Reddi et al. (2018) explain why an unrestricted Adam guarantee cannot be assumed; they do not establish failure of these runs. The hypotheses above have not been established for the experiments' AdamW/PPO updates. In particular, positive replay weight and a small nominal learning rate do not verify the smoothness, predictability, or accumulated-error conditions.

A.11 DISCOVERY, ADMISSION, AND RETENTION THROUGH BANK CHANGES

Discovery and retention require two separate budgets: enough opportunities to admit a missing key, and enough persistent loss weight to protect it after admission. Bank growth connects the two because admitting a key can also change the objective used to protect earlier keys.

Retention begins only after a verified key enters active memory. Archive exploration separates discovery from reliable return to a discovered state (Ecoffet et al., 2021); replay with behavioral cloning supplies a related empirical mechanism for reducing forgetting (Rolnick et al., 2019). Neither result is a theorem that every execution key enters a finite bank or maintains a positive probability thereafter. We state the additional admission and energy conditions explicitly.

Actual insertion is the relevant event. Fix a prompt and count opportunities according to a specified schedule. An opportunity can be a prompt-local query or a global update; the resulting bounds concern that opportunity count. The admission probability includes verification, capacity, scheduling, and candidate selection. For example, G conditionally independent samples with unconditional accepted-key probability p_b contain key b with probability $1 - (1 - p_b)^G$. This is its admission probability only if the key remains eligible and insertion is guaranteed whenever it appears. The following argument permits adaptive histories; its probability foundation is conditional Borel–Cantelli (Durrett, 2019, Theorem 4.3.4).

Lemma A.18 (Conditional admission hazards). *Fix a key b not initially banked. Let $(\mathcal{F}_n)_{n \geq 0}$ contain the complete history after each opportunity, and let τ_b be the first opportunity at which a verified representative of b is inserted into active replay memory, with $\tau_b = \infty$ if this never happens. Define*

$$h_{b,n} = \Pr(\tau_b = n \mid \mathcal{F}_{n-1}), \quad H_{b,N} = \sum_{n=1}^N h_{b,n}.$$

The stopped hazard $h_{b,n}$ is zero after an earlier admission. For $a \geq 0$,

$$\Pr(\tau_b > N, H_{b,N} \geq a) \leq e^{-a}, \quad \Pr(\tau_b = \infty, H_{b,\infty} = \infty) = 0. \quad (27)$$

If deterministic $\underline{h}_{b,n} \in [0, 1]$ satisfy $h_{b,n} \geq \underline{h}_{b,n}$ on $\{\tau_b \geq n\}$, then

$$\Pr(\tau_b > N) \leq \prod_{n=1}^N (1 - \underline{h}_{b,n}) \leq \exp\left(-\sum_{n=1}^N \underline{h}_{b,n}\right). \quad (28)$$

For finitely many targets, summing over initially missing keys bounds non-admission by N .

Proof. Set $Z_N = \mathbf{1}\{\tau_b > N\} \exp(H_{b,N})$ and $Z_0 = 1$. Earlier admission is known at time $n - 1$, and both $h_{b,n}$ and Z_{n-1} vanish there. On survival through $n - 1$,

$$\mathbb{E}[Z_n \mid \mathcal{F}_{n-1}] = Z_{n-1} e^{h_{b,n}} (1 - h_{b,n}) \leq Z_{n-1}.$$

Thus Z_n is a nonnegative supermartingale. Markov’s inequality proves the finite-hazard bound. Since $\mathbb{E}Z_N \leq 1$, Fatou’s lemma excludes a positive-probability event on which survival continues forever and $H_{b,N} \rightarrow \infty$, which would force $Z_N \rightarrow \infty$. With deterministic hazard floors, conditioning gives

$$\Pr(\tau_b > n) \leq (1 - \underline{h}_{b,n}) \Pr(\tau_b > n - 1).$$

Iteration and a union bound prove the claims without independence across opportunities. $\square$

Example: turn an admission floor into a budget. Suppose a prompt has four possible correct keys and at least four bank slots. At each opportunity, every missing key is generated with probability at least 0.04 and inserted, conditional on appearing, with probability at least 1/2. Its actual admission hazard is therefore at least 0.02. The lemma gives $4e^{-0.02N}$ as a failure bound, so $N = \lceil \log(4/0.05)/0.02 \rceil = 220$ opportunities suffice for at least 95% probability of full admission. With only two non-evicting slots, full admission of those four keys is impossible; after the bank fills, the missing keys’ admission hazards become zero despite continued generation.

Positive probability at every opportunity is insufficient. Independent opportunities with admission probabilities $(n + 1)^{-2} > 0$ give

$$\Pr(\tau = \infty) = \prod_{n=1}^{\infty} \left(1 - \frac{1}{(n+1)^2}\right) = \frac{1}{2}.$$

A non-evicting capacity- K_{cap} bank also gives every unbanked key zero admission hazard once full. For a fixed sampling law with no such gates, discovery reduces to coupon collection, with incorrect responses acting as null coupons. Anceaume et al. (2015, Theorems 2–3) show that, at fixed total non-null mass, uniform independent sampling minimizes collection time in stochastic order. This comparison concerns a fixed independent sampling law; the adaptive admission bounds above supply the appropriate statement when the training history changes sampling or eligibility.

Charge changes in the replay objective. A key’s admission must preserve a representative with positive loss weight. The energy can change when bank membership or weights change, so descent between admissions alone is insufficient. We account for those changes explicitly, following the general switched-energy viewpoint of Branicky (1998). The following elementary bound is proved directly and does not invoke an equilibrium-stability theorem for neural training.

Lemma A.19 (Retention with a finite switching-energy budget). *Let $t_0 < t_1 < \dots$ be switching times, with finitely many switches on every bounded time interval. On interval r , define*

$$F_r(\theta) = \sum_{j: d_{rj} > 0} d_{rj} \ell_j(\theta) - J(\theta),$$

$$\ell_j(\theta) = -\log \pi_\theta(e_j | x_j), \quad J(\theta) \leq J_{\max} < \infty.$$

Each interval uses a finite collection of complete verified exemplars with nonnegative coefficients; zero coefficients denote absent exemplars. Assume finite initial energy, continuity of parameters at switches, nonincrease of F_r within interval r , and the finite positive-jump budget

$$A := \sum_{r \geq 1} [F_r(\theta(t_r)) - F_{r-1}(\theta(t_r))]_+ < \infty. \quad (29)$$

If exemplar j is protected from interval r_j onward and $d_{rj} \geq \underline{d}_j > 0$ for all $r \geq r_j$, let $D = F_0(\theta(t_0)) + A + J_{\max}$. Then

$$p_{\theta(t)}(b_j | x_j) \geq \pi_{\theta(t)}(e_j | x_j) \geq \exp(-D/\underline{d}_j) > 0 \quad (t \geq t_{r_j}). \quad (30)$$

Each likelihood uses the complete verified-response probability under the sampling law being bounded. For mean-token loss, $d_{rj} = \alpha_{rj}/L_j$, with fixed length $L_j > 0$.

Proof. Induction over interval descents and switch jumps gives $F_r(\theta(t)) \leq F_0(\theta(t_0)) + A$. Consequently $\sum_j d_{rj} \ell_j(\theta(t)) \leq D$. Each summand is nonnegative, so a protected exemplar satisfies $\ell_j(\theta(t)) \leq D/\underline{d}_j$. The key event contains its exemplar event, so exponentiation proves the bound. $\square$

If admission preserves old coefficients, the jump equals the weighted surprisal of the newly added exemplars at their admission probabilities. Uniform categorical replay with fixed total dose ρ instead dilutes old weights. For a size- k bank, write $R_k = -k^{-1} \sum_{b \in \mathcal{B}_k} \log p_b$; adding one new key gives

$$\Delta F = \frac{\rho}{k+1} [-\log p_{\text{new}} - R_k].$$

Old coefficients decrease from ρ/k to $\rho/(k+1)$ but stay at least ρ/K_{cap} under capacity K_{cap} . Finitely many admissions with positive admission-time probabilities and no eviction therefore have finite total switching cost in the exact descent model. Changing a representative must also be charged to the objective jump. A key-level version of Lemma A.19 follows by requiring at least one complete representative of that key with coefficient bounded below on every subsequent interval; the same nonnegative-summand proof applies even if the representative changes.

Example: adding a rare key has a calculable energy cost. Take $\rho = 0.3$ and a two-key categorical bank whose keys each have probability $1/4$, so $R_2 = \log 4$. Add a third key with probability $1/16$, without changing the policy at that instant. Then

$$\Delta F = \frac{0.3}{3} (\log 16 - \log 4) = 0.1 \log 4 \approx 0.1386.$$

The earlier keys’ coefficients decrease from 0.15 to 0.10. A capacity-three bank still supplies a 0.10 coefficient floor, but the positive jump 0.1386 must enter the energy budget. This is why retention through admission needs both the coefficient floor and the switch cost.

Corollary A.20 (Conditional discovery followed by retention). *Suppose one prompt’s entire correct support $\mathcal{C}$ is finite, of size m no larger than bank capacity. Admissions do not evict keys, and every missing key b has conditional admission hazard at least ϵr_b at every opportunity, for constants $\epsilon > 0$, $r_b > 0$, and $\epsilon r_b \leq 1$. Assume the switching-energy conditions of Lemma A.19, including a finite positive-jump budget and persistent positive coefficients for admitted complete exemplars. Then full correct coverage is reached in finitely many opportunities almost surely. Thereafter each correct key has a uniform-in-time positive probability floor. At horizon N ,*

$$\Pr(\text{full coverage by } N) \geq 1 - \sum_{b \in \mathcal{C}} e^{-N\epsilon r_b}. \quad (31)$$

If every $r_b \geq \gamma > 0$, this is at least $1 - me^{-N\epsilon\gamma}$.

Proof. Apply Lemma A.18 to the initially missing keys and take a union bound. The failure bound tends to zero. Finite support and non-eviction imply a finite almost-sure opportunity index of simultaneous full coverage. Lemma A.19 protects each admitted complete exemplar and hence its key; after full coverage this applies to all correct keys. $\square$

A fixed full-support proposal mixture supplies the assumed hazard floor if it is queried at every counted opportunity and insertion is guaranteed whenever a missing key appears. Capacity must accommodate the entire admissible correct support, including keys competing for the same slots. Finite-temperature sampling, bounded search attempts, or recurrent replay scheduling alone do not establish this admission floor or the energy budget. The probability bound (31) controls admission by a finite opportunity count; the subsequent retention floors can depend on the realized training history. The corollary establishes neither uniform key probabilities nor convergence of the neural optimizer.

A.12 SHARP PER-MODE CERTIFICATES AND FINITE-SAMPLE INTERPRETATION

The energy bounds above can be sharpened by using probability normalization. We give self-contained applications of relative-entropy data processing and Pinsker’s inequality (van Erven & Harremoës, 2014, Theorems 9 and 31); binary-KL inversion is also a standard statistical-learning tool (Garivier & Cappé, 2011). These certificates concern a specified probability distribution. Applying one at a checkpoint requires a valid loss bound there; applying it throughout training additionally requires a uniform energy bound. Neither application establishes descent for the implemented neural optimizer. All logarithms are natural.

A.12.1 A SHARP CERTIFICATE FROM NORMALIZED REPLAY CROSS ENTROPY

Write $H(w) = -\sum_b w_b \log w_b$, $h_2(a) = H((a, 1-a))$, and

$$\text{kl}(a, r) = a \log \frac{a}{r} + (1-a) \log \frac{1-a}{1-r},$$

with the usual endpoint conventions. For $0 < a < 1$ and $D \geq 0$, let $\ell_-(a, D)$ be the unique $r \in (0, a]$ satisfying $\text{kl}(a, r) = D$; set $\ell_-(1, D) = e^{-D}$.

Theorem A.21 (Sharp normalized replay certificate). *Let $\mathcal{B}$ be a finite nonempty bank, let $w_b > 0$ with $\sum_{b \in \mathcal{B}} w_b = 1$, and let p be a probability distribution on an alphabet containing $\mathcal{B}$. Suppose*

$$R_w(p) = -\sum_{b \in \mathcal{B}} w_b \log p_b \leq C < \infty, \quad D = C - H(w).$$

Then $D \geq 0$ and, for every $b \in \mathcal{B}$,

$$p_b \geq \ell_-(w_b, D) > 0. \quad (32)$$

This coordinate bound is sharp on the closed probability simplex whenever an outcome other than b is available. Moreover,

$$p(\mathcal{B}) \geq e^{-D}, \quad p_b \geq \max\{0, w_b - \sqrt{D/2}\}.$$

For $w_b < 1$, the positive bound $p_b \geq \exp[-(D + h_2(w_b))/w_b]$ is also valid.

Proof. Extend w by zero off the bank, obtaining $\bar{w}$. Then

$$0 \leq \text{KL}(\bar{w} \| p) = R_w(p) - H(w) \leq D.$$

For $a = w_b$ and $r = p_b$, the log-sum inequality on the complementary outcomes gives

$$\sum_{j \neq b} \bar{w}_j \log \frac{\bar{w}_j}{p_j} \geq (1 - a) \log \frac{1 - a}{1 - r},$$

so $\text{kl}(a, r) \leq D$. For $0 < a < 1$, its derivative in r is $(r - a)/(r(1 - r)) < 0$ on $(0, a)$; it diverges at zero and vanishes at $r = a$. Thus $r < \ell_-(a, D)$ is impossible. Equality is attained by choosing $r = \ell_-(a, D)$, $p_j = (1 - r)w_j/(1 - a)$ on the remaining banked outcomes, and zero mass off the bank. For $a = 1$, the inequality is $-\log p_b \leq D$, with equality obtained by placing the remaining mass on another outcome. If the entire alphabet has one outcome, necessarily $p_b = 1$. For $D > 0$, equality can be approached with positive off-bank probabilities by increasing r slightly and sending off-bank mass to zero. At $D = 0$, feasibility instead forces $p = \bar{w}$ exactly.

Let $s = p(\mathcal{B})$ and $\tilde{p}_b = p_b/s$. The decomposition

$$\text{KL}(\bar{w} \| p) = -\log s + \text{KL}(w \| \tilde{p}) \geq -\log s$$

proves the total-mass bound. For fixed $r \in (0, 1)$, $\partial_a^2 \text{kl}(a, r) = 1/[a(1 - a)] \geq 4$, whereas its value and first derivative vanish at $a = r$. Hence $\text{kl}(a, r) \geq 2(a - r)^2$, including endpoints by continuity, which proves the Pinsker bound. Finally,

$$\text{kl}(a, r) = -h_2(a) - a \log r - (1 - a) \log(1 - r) \geq -h_2(a) - a \log r$$

Combining this inequality with $\text{kl}(a, r) \leq D$ yields the positive exponential coordinate bound. $\square$

For the categorical potential argument behind Eq. (22) and its positive-weight extension, one may substitute

$$C = R_w(p(T)) + \frac{\Psi(1) - \Psi(P(T))}{\rho_w}.$$

Normalization improves the termwise certificate $p_b \geq e^{-C/w_b}$.

Worked example: normalization turns a tiny floor into a useful one. Suppose hypothetically that a uniform bank of 16 has the certified loss bound $C = \log 16 + 0.01$: only .01 nat above the target's minimum cross entropy. The earlier termwise argument gives approximately 4.62×10^{-20} per mode. Accounting for the other fifteen probabilities instead gives $p_b \geq 0.0339712$, about 3.4% for every mode, against a uniform target of 6.25%. It also certifies total bank mass at least $e^{-.01} \approx 99.00\%$. The improvement uses normalization: one probability cannot become very small while the others absorb arbitrary mass and keep this loss almost minimal. These are mathematical inputs, not measured replay losses. A large excess cross entropy can still make even the sharp bound too small to be useful.

A.12.2 WHAT A SCALAR ENTROPY VALUE CAN CERTIFY

Theorem A.22 (Sharp entropy threshold for all-mode retention). *Let q be a distribution on a known alphabet of $m \geq 2$ possible modes, allowing zero coordinates. A scalar bound $H(q) \geq h_0$, with $0 \leq h_0 \leq \log m$, implies a strictly positive uniform coordinate floor if and only if $h_0 > \log(m - 1)$. In that case the sharp floor is $q_b \geq r_m(h_0)$ for every b , where $r_m(h_0) \in (0, 1/m]$ uniquely solves*

$$h_2(r) + (1 - r) \log(m - 1) = h_0,$$

where m counts all possible modes, including those absent from the evaluation sample.

Proof. Fix $r = q_b < 1$. Splitting off that coordinate gives

$$H(q) = h_2(r) + (1 - r)H\left(\frac{q - b}{1 - r}\right) \leq h_2(r) + (1 - r) \log(m - 1) =: f_m(r).$$

Uniform remaining coordinates attain equality. On $0 < r < 1/m$, $f'_m(r) = \log[(1 - r)/(r(m - 1))] > 0$, with $f_m(0) = \log(m - 1)$ and $f_m(1/m) = \log m$. Thus a coordinate below the stated root contradicts $H(q) \geq h_0$; a coordinate equal to one already exceeds the root. The distribution with one coordinate at the root and the others uniform proves sharpness. Conversely, if $h_0 \leq \log(m - 1)$, one zero coordinate with the rest uniform satisfies the entropy bound. Even restricting to strictly positive coordinates permits arbitrarily small values by approximation. $\square$

Worked example: a reported entropy of 97.7% can hide a missing mode. On a known sixteen-mode alphabet, let one mode have probability zero and the other fifteen share mass equally. Then

$$H(q) = \log 15, \quad \frac{H(q)}{\log 16} \approx 0.97672.$$

Thus entropy rounds to 97.7% of its maximum although one mode is entirely absent. Its forward-KL gap to uniform is only $\log(16/15) \approx 0.06454$ nat. The theorem requires a strict entropy bound above $\log 15$ to exclude this example; a nearly maximal scalar entropy by itself does not certify all sixteen coordinates.

For uniform u , $H(q) = \log m - \text{KL}(q\|u)$, and data processing gives $\text{kl}(q_b, 1/m) \leq \text{KL}(q\|u)$. A forward-KL budget therefore excludes every missing coordinate precisely when it is below $\log[m/(m-1)]$. At the threshold, one missing coordinate is possible, whereas $\text{KL}(u\|q) = \infty$. This compares level sets; it does not assert that an exact-entropy gradient flow loses modes. Conditional entropy alone also leaves total correct mass unconstrained, and entropy estimated from a finite sample is not the exact $H(q)$ of a known alphabet.

A.12.3 FROM COMPLETE-RESPONSE LIKELIHOOD TO AN EXECUTION KEY

Corollary A.23 (Response-to-key certificate). *Fix one prompt x , one checkpoint, and distinct complete responses e_j with verified keys b_j . Give them weights $w_j > 0$ summing to one, and put $W_b = \sum_{j:b_j=b} w_j$. Under a fixed decoding law, suppose*

$$-\sum_j w_j \log \pi(e_j | x) \leq C < \infty.$$

For every represented key b ,

$$\text{kl}(W_b, p(b | x)) \leq C - H(w), \quad p(b | x) \geq \ell_-(W_b, C - H(w)) > 0.$$

Individually, $p(b_j | x) \geq \pi(e_j | x)$.

Proof. The distinct complete responses are disjoint events. Extend their target weights by zero to the full response alphabet; include invalid outcomes and, if necessary, nontermination as additional outcomes so that the policy law is normalized. The resulting response-level relative entropy is $-\sum_j w_j \log \pi(e_j | x) - H(w) \leq C - H(w)$. Apply the log-sum inequality within each execution-key event and then within its binary partition into key b and all other outcomes. The target assigns mass W_b to the first event and the policy assigns $p(b | x)$, so the resulting divergence is $\text{kl}(W_b, p(b | x))$. The inverse argument of Theorem A.21 proves the floor. Finally, each complete-response event is contained in its verified-key event, proving the individual bound used in Lemma A.15. $\square$

Worked example: two responses can protect the same key. At a hypothetical checkpoint, two distinct complete responses for key A each have probability .20, a response for key B has probability .40, and the remaining .20 is invalid. Give these three responses target weights $w = (1/4, 1/4, 1/2)$. Their cross entropy satisfies $C - H(w) = \log(1.25) \approx .22314$. For key A , combine the first two target weights to obtain $W_A = 1/2$; the certificate then gives $p(A) \geq .20$, since $\text{kl}(.50, .20) = \log(1.25)$. Certifying each response separately and adding the two floors would give only about .1009. The actual key probability in this constructed example is $.20 + .20 = .40$. Grouping complete responses by their shared execution key therefore strengthens the certificate without treating either response as the whole key.

The probability being scored must include EOS or another specified termination event, or a fixed deterministic horizon. It must use the same prompt formatting, temperature, vocabulary mask, top- p /top- k rule, response-truncation rule, and verifier as the probability being bounded. A prefix likelihood is insufficient because its continuations can produce other keys or invalid outputs. All likelihoods entering one cross entropy must use the same checkpoint. For a complete sequence of length L_j and mean-token log score $\bar{\ell}_j$, its probability is $\exp(L_j \bar{\ell}_j)$, not $\exp(\bar{\ell}_j)$.

Length-normalized loss requires normalization before the theorem applies: with $a_j = \alpha_j/L_j$, $A = \sum_j a_j$, and $w_j = a_j/A$,

$$\sum_j \alpha_j \frac{-\log \pi(e_j | x)}{L_j} = AR_w.$$

Thus an upper bound D_0 on this loss supplies $C = D_0/A$; uniform weights on mean-token scores generally give inverse-length target weights on complete responses. Different prompts require separate conditional calculations or an explicitly normalized joint prompt mixture. A change in one exemplar’s likelihood need not equal the change in its whole key’s probability. Approximate log scores need controlled numerical error to supply rigorous numerical certificates.

A.12.4 FINITE OBSERVATIONS AND REPEATED CHECKPOINT MEASUREMENT

At a frozen checkpoint, for a key specified before sampling, N independent draws under the same decoding law give $X \sim \text{Binomial}(N, p_b)$. Exact binomial-tail inversion gives one-sided confidence bounds (Clopper & Pearson, 1934). If $X = 0$, the one-sided $(1 - \alpha)$ upper bound and lower bound are respectively

$$U_0 = 1 - \alpha^{1/N}, \quad 0.$$

Indeed, $\Pr_{p_b}(X = 0) = (1 - p_b)^N < \alpha$ whenever $p_b > U_0$; for nonzero counts, the full confidence procedure uses the corresponding binomial-tail inversion.

Worked example: unseen is not the same as absent. For one key selected before sampling, zero sightings in 32 draws give a one-sided 95% upper confidence bound of about 8.94%, with lower bound zero. The sample therefore does not establish exact extinction. If the claim must cover 16 prespecified keys simultaneously, Bonferroni allocates .05/16 error to each key; the zero-count upper bound becomes about 16.50%. Requiring simultaneous coverage makes the certificate weaker. Conversely, two sightings in 32 draws have empirical frequency 6.25%, but their one-sided 95% lower confidence bound is only about 1.12%. These hypothetical upper and lower examples are separate one-sided limits; a joint interval requires allocation to both tails.

For repeated inspection of draws from a fixed policy, a confidence sequence can provide validity under optional stopping (Howard et al., 2021). For fixed beta-mixture parameters $a_0, b_0 > 0$, let S_t be the number of successes in t independent Bernoulli draws and let B denote the beta function. Define

$$M_t(p) = \frac{B(S_t + a_0, t - S_t + b_0)}{B(a_0, b_0)p^{S_t}(1 - p)^{t - S_t}}, \quad \mathcal{I}_t = \{p \in (0, 1) : M_t(p) < 1/\alpha\}.$$

For each true $p \in (0, 1)$, the numerator integrates the likelihood under a fixed beta distribution. Hence $M_t(p)$ is a mixture of likelihood-ratio martingales, is nonnegative, and has initial value one. The nonnegative martingale maximal inequality therefore gives $\Pr_p(\exists t : p \notin \mathcal{I}_t) \leq \alpha$. Error allocations across a fixed roster of keys make coverage simultaneous by a union bound; independence across keys is unnecessary.

This construction requires a fixed Bernoulli probability. Counts from different prompts, seeds, or changing checkpoints cannot be pooled to certify the current checkpoint’s probability. Separate frozen-checkpoint measurements can instead allocate error across prespecified prompt–key–checkpoint statements. Repeated deterministic teacher-forced scores are not independent Bernoulli observations, and choosing identities after inspecting the same samples also needs selection-aware inference. A probability bound at observed checkpoints alone makes no claim about unobserved intervening or subsequent training states.

Worked example: repeated checkpoints spend an error budget. Suppose the plan specifies 16 keys at each of ten frozen checkpoints, with 32 new draws per checkpoint. There are 160 key–checkpoint statements. Allocating .05/160 to each gives simultaneous 95% coverage by a union bound, without requiring the key counts to be independent. A zero count then has an upper bound of about 22.29%. Pooling all 320 draws would not fix this weakness: the ten checkpoints can have different key probabilities. The confidence sequence above instead permits repeated inspection and a data-dependent stopping time while sampling one fixed checkpoint.

For an unknown catalogue, missing mass is the total probability of categories not yet observed under a fixed iid law. Concentration results address this quantity (McAllester & Ortiz, 2003; Berend & Kontorovich, 2013); it is neither the number of missing modes nor proof of their extinction. Such bounds do not certify each named key, and stationary missing-mass formulas do not directly apply to a history generated by changing policies. Named-key binomial counts provide the direct

measurement of per-mode visibility. For example, a hypothetical missing mass of .10 could consist of one unseen mode with probability .10 or one hundred unseen modes of probability .001 each; the same aggregate mass does not identify how many keys are missing.

Remark A.24 (Exact scope of the ReplayMaxRL guarantee). The categorical replay theorem protects represented bank modes after discovery; it is not an unconditional end-to-end guarantee for a language model. Without full coverage it protects only $\mathcal{B}$, and Lemma A.14 shows that a frozen incomplete bank can exclude unbanked correct modes in the categorical limit. Capacity 16 can therefore truncate protected support. The admission results above identify sufficient coverage and switching conditions; they do not establish those conditions for the implemented proposal and insertion machinery.

The categorical proof uses $\log p_b$, whereas the implementation uses length-normalized teacher-forced exemplar scores. The complete-exemplar bridge and stochastic extension permit shared parameters under explicit joint-energy and update-error assumptions. They do not prove uniform neural key probabilities or verify those assumptions for the current clipped, alternating AdamW updates. The sharper certificates require correctly normalized complete likelihoods under the decoding law being bounded. Finally, exact entropy and exact categorical natural gradient have their own protection guarantees; collapse is a property of the stated update model, not of every procedure that optimizes correctness or entropy. These results motivate direct per-mode survival measurements.

B SUPPORTING MEASUREMENTS AND DIRECT COMPARATORS

This appendix collects the comparator design, complete and partial UCPO and sparse RLEP-Dr blocks, and measurements re-derived from registered run logs. In the primary analyses, exact prefixes are not promoted to five-seed estimands and receive no block-level uncertainty intervals. Here and in the figure labels, *correctness-adjusted breadth* denotes extra verified modes, `distinct@8-pass@8`, which remains coupled to correctness.

B.1 SOURCE INTEGRITY AND ADMISSIBLE COMPARISONS

The Falcon3-1B Countdown ReplayDr.GRPO source for seed 59 (job 30269051) contains two complete four-draw terminal evaluation sequences with matching request identities but conflicting metrics. The recorded E112-R1 integrity amendment excludes this source; we apply that exclusion to every efficacy endpoint, checkpoint, paired contrast, and aggregate that reuses the run (`paper/preregistration/e112r1_two_scale_49pair_terminal_integrity_amendment_20260828.md`). We select neither the first sequence, the last, nor an average. The source failure concerns ReplayDr.GRPO only: all 150 verifier-only Dr.GRPO/GRPO precheck runs remain admissible.

The core replay contrast therefore contains 74 admissible pairs: 14 complete five-seed model-domain blocks and Falcon Countdown at $n = 4$ (seeds 55–58). All 74 raw `distinct@8` differences are positive. The primary replay and factorial analyses report four-seed means descriptively, without confidence intervals. The Falcon Dr.GRPO replay macro-average uses those same four seeds across all five domains, yielding `+441699` on `pass@8` and `+773926` on `distinct@8`. The eleven complete MaxRL–ReplayMaxRL blocks retain all five seeds; the full four-arm factorial has ten complete blocks and one four-seed block. Interactions require the common admissible intersection. The separately reported historical E112-R1 comparison follows its exploratory amendment, including a Student- t interval with three degrees of freedom for its four-seed Falcon Countdown block; that interval is not primary evidence.

All 14 complete core blocks have positive mean effects on each endpoint. Assuming independent block signs, the two-sided exact sign test gives $p = 2/2^{14} = 0.0001220703125$ per endpoint; Holm adjustment across the two endpoints gives $p_{\text{Holm}} = 0.000244140625$ for each. These are nominal, exploratory summaries, not confirmatory evidence or a correction for dependence among model-domain blocks.

Trajectory readers use registered source jobs and documented replacements, excluding superseded attempts. A separate ambiguity affects the initial `mean@8` of Qwen2.5-3B MathIR ReplayDr.GRPO seed 71. We omit that descriptive checkpoint without selecting a retry; its terminal endpoint remains valid. All 17 registered checkpoints for the 50 Qwen2.5-0.5B core runs are unambiguous, so the

reported trajectory AUC is unchanged. Source identities, row hashes, and recovery authorities are recorded in `paper/audits/initial_auc_source_integrity_20260904.json`.

B.2 SUSTAINED EFFECTS THROUGH TRAINING

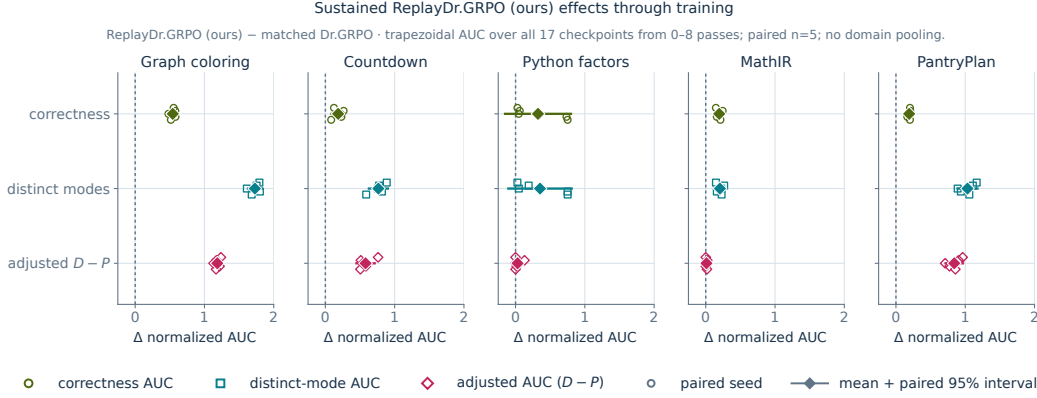


Figure 7: **Sustained ReplayDr.GRPO effects through training.** Qwen2.5-0.5B ReplayDr.GRPO is compared with matched Dr.GRPO separately on each static domain. Effects are trapezoidal trajectory AUCs over all 17 registered half-pass checkpoints from 0 through 8 passes, divided by the eight-pass horizon. Open markers are the five paired seeds 43–47, diamonds are paired means, and bars are unadjusted descriptive two-sided 95% Student- t estimation intervals. The adjusted effect subtracts correctness AUC from distinct-mode AUC per seed. No significance decision is made for an individual model–domain comparison, and no checkpoint, domain, or metric magnitude is pooled. An early-collapsed control can remain flat for most of the horizon, so AUC partly measures time spent in that state and is a sensitivity analysis, not independent mechanism evidence.

B.3 COMPARATOR FINAL EFFECTS AND TRAINING PROGRESS

Table 1: **Direct alternatives to canonical mode-balanced replay.** Each method inherits the same eight-pass evaluation contract. Status counts source-admissible model–domain–seed runs at the analysis snapshot; incomplete blocks are excluded from complete-block inference.

Method	Active distinction	Question isolated	Coverage
Dr.GRPO	Group-centered binary reward; no auxiliary derivative	verifier-only reference	75/75
GRPO	Reward-standardized group-relative update	estimator normalization	75/75
ReplayDr.GRPO	One retained exemplar per verified key, sampled uniformly over keys	persistent canonical memory	74/75
UCPO	Reallocates positive advantage within the current rollout group	on-policy diversity incentive without memory	50/75
Sparse RLEP-Dr	Replays verified successes in observed frequency, without canonical balancing	success memory without key balancing	47/75
MaxRL	Optimizes best@ k jointly across sampling budgets, without replay	sampling-aware fresh search without persistent memory	11/15 blocks
Fixed semantic entropy	Adds detached surprisal over verified canonical outcomes to Dr.GRPO	explicit within-rollout diversity incentive	10/15 blocks

Figure 8 gives the final-training comparison used to assess whether generic alternatives recover the retention pattern; the paragraphs below give exact domain-level values and the companion trajectories.

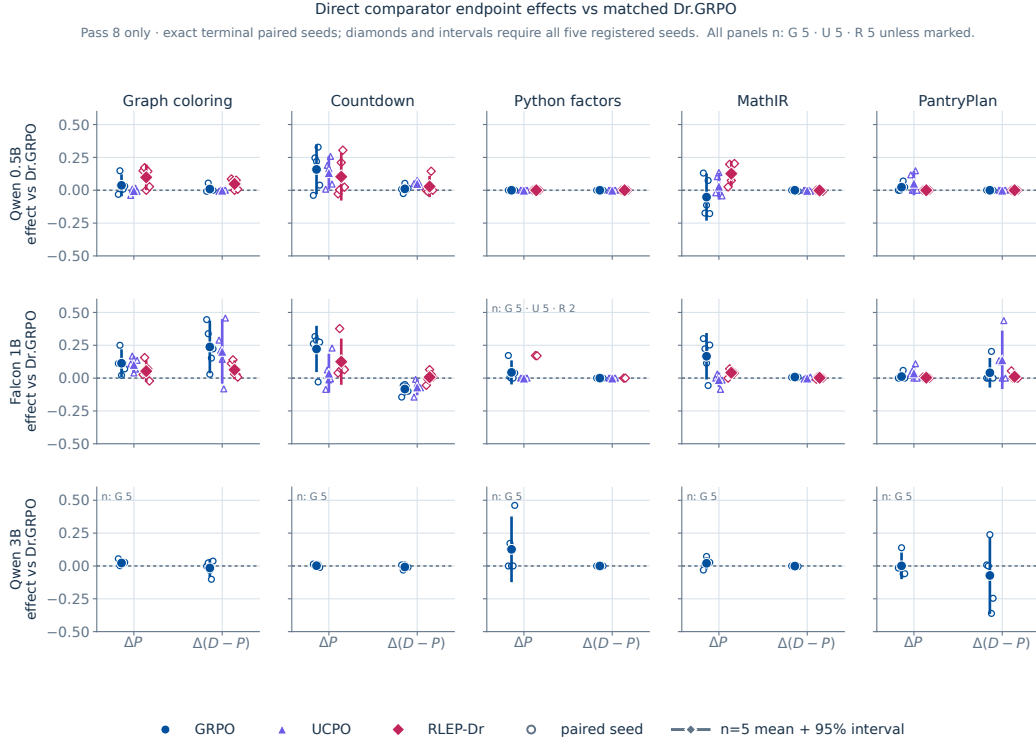


Figure 8: Direct alternatives produce local rather than uniform final gains. Each effect subtracts the same-seed Dr.GRPO control from the comparator after eight passes. Effects are shown for `pass@8` and correctness-adjusted breadth. Circles are paired seeds; diamonds and 95% Student-*t* intervals appear only for complete five-seed blocks. Exact *n* is printed, and unsupported comparisons remain blank rather than being extrapolated or imputed at any seed.

All five Qwen2.5-3B plain-GRPO blocks are now complete. Their `pass@8` effects on Graph, Countdown, Python, MathIR, and PantryPlan are $+0.023 [-0.002, +0.048]$, $+0.002 [-0.123, +0.376]$, $+0.020 [-0.026, +0.066]$, and $+0.001 [-0.100, +0.102]$. The corresponding correctness-adjusted-breadth effects are $-0.015 [-0.084, +0.053]$, $-0.008 [-0.028, +0.013]$, exactly zero, $-0.001 [-0.003, +0.001]$, and $-0.072 [-0.365, +0.220]$. All five correctness means are positive, but none of their individual 95% intervals excludes zero; the breadth intervals likewise include zero.

Complete Qwen Countdown UCPO has a mean `pass@8` effect of $+0.133 [+0.005, +0.262]$ and adjusted-breadth effect of $+0.051 [+0.031, +0.071]$. The completed Qwen MathIR UCPO block remains neutral: $+0.031 [-0.071, +0.133]$ on `pass@8` and $-0.002 [-0.006, +0.003]$ on adjusted breadth. The completed Falcon Pantry RLEP-Dr block is inconclusive: $+0.003 [-0.002, +0.008]$ on `pass@8` and $+0.011 [-0.019, +0.041]$ on adjusted breadth. Qwen Countdown RLEP-Dr is likewise inconclusive: $+0.103 [-0.078, +0.285]$ and $+0.029 [-0.052, +0.109]$. In contrast, Qwen MathIR RLEP-Dr has a positive correctness effect, $+0.126 [+0.030, +0.222]$, with neutral adjusted breadth, $-0.002 [-0.006, +0.003]$. All complete blocks use five paired seeds and are not pooled across domains. The RLEP offline pool used temperature .7 and top-*p* = .95 only during its frozen experience-collection job. Every reported endpoint for RLEP and the other arms uses the shared temperature-1, top-*p* = 1 evaluation. Thus there is no evaluation-decoding advantage for RLEP; its frozen offline pool versus our online bank remains a substantive algorithmic difference.

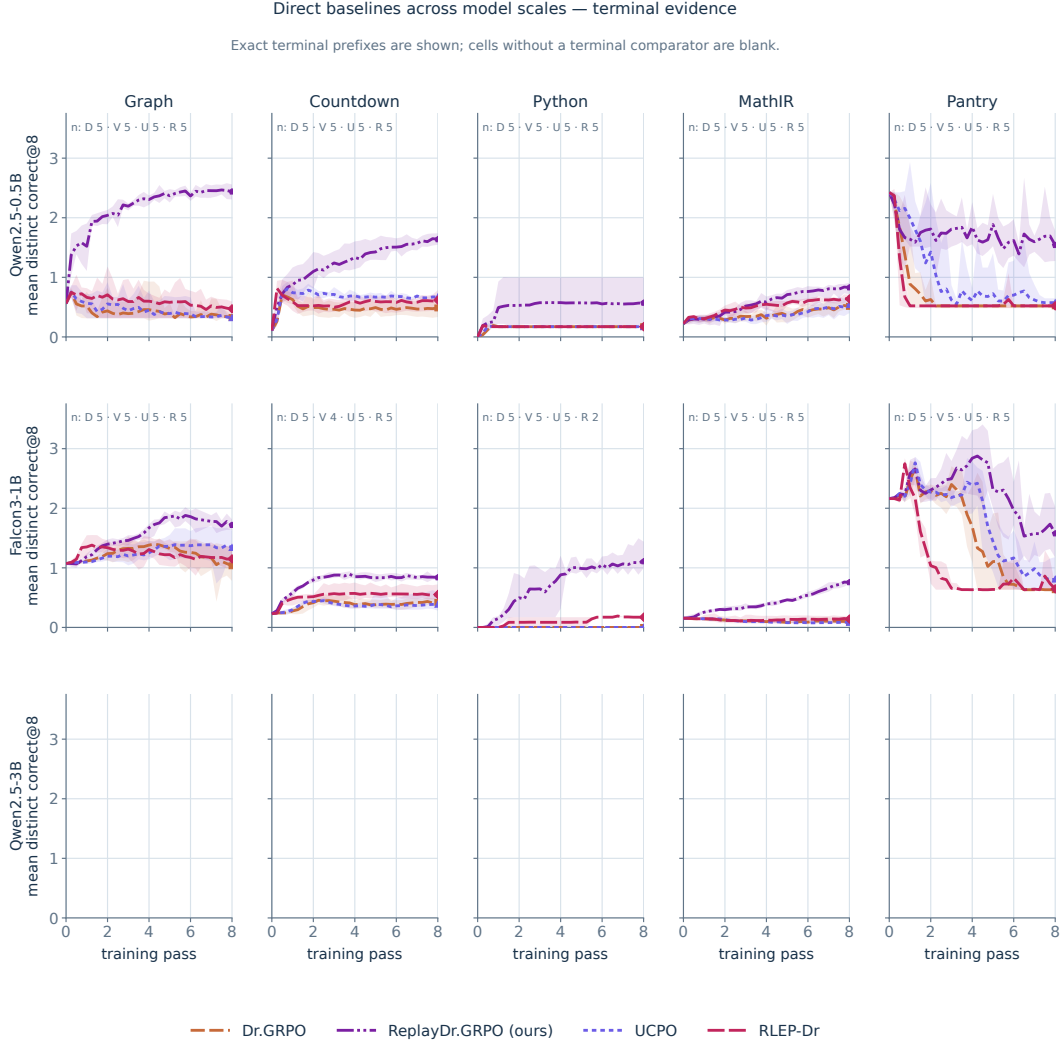


Figure 9: **Terminal direct-baseline learning trajectories in the invariant model grid.** Physical rows are Qwen2.5-0.5B, Falcon3-1B, and Qwen2.5-3B; columns are the five static domains. Each panel shows every terminal UCPO or sparse RLEP-Dr series together with matched Dr.GRPO and ReplayDr.GRPO, using its exact seed set and n . UCPO fills all ten smaller-model blocks at $n = 5$. RLEP-Dr fills all five Qwen domains and Falcon Graph/Countdown/MathIR/Pantry at $n = 5$, plus Falcon Python at $n = 2$. Models and domains without a final comparator remain blank; lines are descriptive means and bands are seed ranges. The core replay contrast appears in Figure 4.

B.4 UNIFORM VERSUS FRESH-FREQUENCY REPLAY: REGISTERED BREADTH ANALYSIS

E120-R1 changes the within-bank target weights: fresh-frequency replay weights retained keys in proportion to their validator-positive fresh-rollout counts, while matched ReplayDr.GRPO weights retained keys uniformly. Both arms use the same bank construction and update code; their subsequent policy-dependent bank contents can differ. We retain the snapshot generated on September 4, 2026 at 15:40:26 UTC (11:40:26 EDT): 25/45 treatment cells have exact step-3,072 endpoints, comprising all five Qwen2.5-0.5B domains on seeds 43–47. The larger-model extensions are outside this frozen analysis.

Registered estimands and paired uncertainty. Write $P = \text{pass@8}$, $D = \text{distinct@8}$, and $B = D - P$. For each successful eight-sample set, B counts verified modes beyond its first correct mode; failed sets contribute zero. It is an unconditional extra-mode count,

not diversity conditioned on equal correctness. The preregistered primary contrast is $\Delta B = B_{\text{uniform}} - B_{\text{frequency}}$; $\Delta P = P_{\text{uniform}} - P_{\text{frequency}}$ is the co-primary correctness endpoint. Positive effects favor uniform weighting. The registration also specifies all five seed values and the full five-domain mean (paper/preregistration/e120_frequency_weighted_replay_ablation_20260902.md). We compute each contrast within its matched seed. For the five-domain summary, we first average the five domain effects within each seed and then average the five resulting seed means, giving each domain equal weight.

Table 2 reports paired percentile bootstrap 95% intervals over training seeds. We enumerate all $5^5 = 3,125$ ordered resamples of the five paired seed records, including repetitions, and use linearly interpolated 2.5th and 97.5th percentiles. For the domain average, a resampled seed carries its entire five-domain vector, preserving the observed pairing across domains. Endpoints and the five-domain average were preregistered; the seed-resampling unit, exact enumeration, and percentile convention are implementation choices documented in this analysis. These nominal intervals use only five seeds, are not adjusted across domains or endpoints, and do not resample evaluation prompts or domains. They quantify seed variability within the fixed benchmark rather than uncertainty over a population of reasoning tasks.

Table 2: **Uniform replay retains more extra correct modes.** Every row uses five paired Qwen2.5-0.5B seeds at pass 8. Effects are uniform minus fresh-frequency replay; ΔB is the registered extra-mode contrast and ΔP its co-primary correctness contrast. Intervals are paired seed-bootstrap percentile intervals. The final row averages domains within seed before estimating uncertainty.

Domain	ΔB	95% interval	ΔP	95% interval
Graph coloring	+ .596	[+.513, +.687]	+ .066	[+.061, +.072]
Countdown	+ .639	[+.544, +.733]	+ .022	[+.004, +.040]
Python factors	+ .037	[-.012, +.123]	- .077	[-.706, +.552]
MathIR	+ .019	[-.001, +.038]	- .005	[-.084, +.075]
PantryPlan	+ .298	[+.174, +.400]	+ .043	[+.006, +.091]
Five-domain mean	+ .318	[+.272, +.358]	+ .010	[-.119, +.139]

Uniform replay increases extra modes by +.596 on Graph, +.639 on Countdown, and +.298 on PantryPlan; all three bootstrap intervals exclude zero and all five seeds have positive breadth effects in each domain. MathIR (+.019) and Python (+.037) remain inconclusive. The registered five-domain mean is +.318 extra modes per prompt, with interval [+ .272, +.358]; all five within-seed domain averages are positive. The corresponding mean correctness difference is +.010 ([-.119, +.139]). That wide interval establishes neither a correctness gain nor noninferiority or equivalence. Table 3 exposes the variation: Python correctness effects range from -.781 to +.828, whereas its extra-mode effects are small in four seeds. Thus uniform weighting has a breadth advantage concentrated in Graph, Countdown, and PantryPlan; these results do not establish universal accuracy safety or downstream utility.

Table 3: **All registered paired seed effects for the weighting ablation.** Each entry is $\Delta B/\Delta P$, uniform minus fresh-frequency replay. The final row averages the five domains within the indicated seed. Three-decimal rounding is for display; the accompanying JSON retains every full-precision effect and bootstrap interval, including the registered five-domain average.

Domain	Seed 43	Seed 44	Seed 45	Seed 46	Seed 47
Graph coloring	+ .533/ + .064	+ .676/ + .070	+ .467/ + .062	+ .564/ + .057	+ .742/ + .076
Countdown	+ .506/ + .000	+ .539/ + .000	+ .631/ + .031	+ .727/ + .021	+ .791/ + .057
Python factors	+ .000/ + .828	+ .000/ + .771	- .020/ - .781	+ .205/ - .766	+ .000/ - .438
MathIR	+ .014/ - .129	+ .016/ + .135	+ .035/ + .051	+ .049/ + .004	- .018/ - .084
PantryPlan	+ .363/ + .135	+ .465/ + .033	+ .096/ - .006	+ .219/ + .004	+ .346/ + .049
Five-domain mean	+ .283/ + .180	+ .339/ + .202	+ .242/ - .129	+ .353/ - .136	+ .372/ - .068

Supporting raw endpoints. For continuity with the frozen result artifact, Table 4 retains the original *frequency-minus-uniform* sign for raw P and D ; its sign is opposite to the primary tables above. These raw endpoints and their existing Student- t intervals are supporting descriptions, not substitutes for the registered breadth estimand and bootstrap analysis.

Table 4: **Supporting raw endpoints in the original contrast direction.** Effects are fresh-frequency minus uniform replay on the displayed five seeds. ΔP is the raw `pass@8` difference and ΔD the raw `distinct@8` difference. The frozen machine-readable artifact also retains paired, unadjusted 95% Student- t intervals for all five terminal domain blocks.

Domain	n	Seeds	ΔP	ΔD
Graph coloring	5	43,44,45,46,47	-.066	-.662
Countdown	5	43,44,45,46,47	-.022	-.661
Python factors	5	43,44,45,46,47	+.077	+.040
MathIR	5	43,44,45,46,47	+.005	-.014
PantryPlan	5	43,44,45,46,47	-.043	-.341

On this original scale, fresh-frequency replay reduces raw `distinct@8` on Graph by $-.6625$ ($[-.808, -.517]$), Countdown by $-.661$ ($[-.838, -.483]$), and PantryPlan by $-.341$ ($[-.564, -.117]$). MathIR is near zero, $-.014$ ($[-.160, +.132]$), while Python is heterogeneous, $+.040$ ($[-.926, +1.006]$). The registered primary analysis above separates the extra-mode component of these raw differences. It supports key balancing for retained sampled breadth in the stated domains, without identifying the survival history of any individual banked mode. Falcon3-1B and Qwen2.5-3B have no terminal E120-R1 cells in this frozen snapshot.

Newly completed Falcon3-1B Graph ablation (September 6). The registered Falcon Graph block now has all five paired seeds, 55–59, at step 3,072. Each endpoint has a completion receipt and all four required sampled evaluation draws. The replay audit passes for all 16,253 persisted applied-update telemetry rows across the frequency arm’s original and recovery attempts. Table 5 adds this completed block separately from the September 4 Qwen2.5-0.5B snapshot above, using the same paired percentile bootstrap convention.

Uniform weighting adds $+.233$ extra modes per prompt (95% interval $[+.182, +.320]$), with a positive effect in every seed. Its `pass@8` effect is $+.0453$ ($[-.0004, +.0910]$); the interval includes zero and establishes neither a correctness gain nor noninferiority. The supporting mean sampled-correctness metric, `mean@8`, decreases by $-.0152$ ($[-.0214, -.0090]$). Thus the breadth benefit coexists with lower per-sample correctness; greater sampled support does not imply a uniform accuracy improvement. This is one registered model–domain block, with nominal, unadjusted intervals and an unfinished larger-model grid.

Table 5: **Completed Falcon3-1B Graph replay-weight ablation.** All five paired seeds at pass 8; positive effects favor uniform replay. Extra modes $B = D - P$ and correctness P are the registered co-primary endpoints. Raw distinct modes D and mean sampled correctness are supporting metrics. Intervals resample paired training seeds within this registered block.

Endpoint	Uniform	Frequency	Effect	95% interval
Extra modes $D - P$	.908	.675	+.233	$[+.182, +.320]$
Correctness P	.809	.763	+.0453	$[-.0004, +.0910]$
Raw distinct modes D	1.717	1.439	+.278	$[+.184, +.404]$
Per-sample correctness	.334	.350	-.0152	$[-.0214, -.0090]$

Table 6: **All paired seeds in the Falcon Graph extension.** U/F denote uniform/fresh-frequency replay; effects are U minus F. Full-precision endpoints, all supporting metrics, evaluation origins, completion receipts, and telemetry checks accompany the result in `e120_falcon_graph_20260906.json` for all five seeds.

Seed	P_U	P_F	B_U	B_F	ΔP	ΔB
55	.803	.721	.836	.652	+.082	+.184
56	.783	.809	.957	.783	-.025	+.174
57	.838	.723	.977	.572	+.115	+.404
58	.830	.770	.857	.654	+.061	+.203
59	.789	.795	.914	.715	-.006	+.199

B.5 WHEN FRESH DR.GRPO GRADIENTS VANISH

Dr.GRPO centres advantages within a rollout group, so a group whose members all receive the same reward contributes *exactly zero* fresh task gradient. Only a *mixed* group — at least one correct and at least one incorrect member — can supply a nonzero fresh task gradient. The mixed-group fraction therefore measures how often a group can supply fresh reward information. A zero current task gradient does not imply an unchanged policy: AdamW momentum and weight decay can still move parameters.

Measured over all 3,072 updates of the Qwen2.5-0.5B cohort, five seeds for each domain:

Domain	matched Dr.GRPO	ReplayDr.GRPO
Graph coloring	.144	.954
Countdown	.140	.456
Python factors	.015	.185
MathIR	.364	.453
PantryPlan	.123	.789

On Python factors, 98.5% of control updates therefore have zero fresh task gradient. The effect sharpens with scale: the Qwen2.5-3B control on the same domain logged 7 mixed groups in 3,073 updates, versus 843 for its paired ReplayDr.GRPO run. Under the registered $G = 16$, $\beta_{KL} = 0$, no-token-entropy-bonus protocol, the fresh task gradient is nearly always zero in this cell. Replay supplies a verified-likelihood gradient directly and, after changing the policy, can also make later fresh groups mixed again. This is evidence that replay is robust to this sparse binary-reward regime, not that it dominates methods that keep the fresh RL signal active.

Direct output-collapse audit. Exact endpoint equality is a useful screening signature: in the control arm, `mean@8`, `pass@8`, and `distinct@8` are identical in 14/15 Python seeds and 12/15 PantryPlan seeds across the three scales. It is not our collapse criterion. We instead audited the stored raw responses in all five E78 Qwen2.5-0.5B Python control seeds. At pass 0, a sampled $K = 8$ draw contains 7.99 unique raw completions per prompt on average. By step 288–480 (0.75–1.25 passes), every seed shows identical observed completions: for each of 128 prompts, all 32 temperature-1 samples are byte-identical to the separately logged temperature-0 greedy completion. This persists through step 3,072, so the last 6.75–7.25 passes record persistence of this sampled-output pattern. At the terminal checkpoint, a control draw contains exactly 1.00 unique raw completion per prompt, versus 1.97 for ReplayDr.GRPO, averaged over the five seeds. When the singleton completion is correct it supplies one verified key; otherwise it supplies none. This is observed collapse to one completion per prompt, not one correct key on every prompt.

All three aggregate endpoint draw standard deviations are zero throughout the flat portion of these control traces. Zero draw spread is only corroborating: it is also zero at pass 0, when the varied completions are all invalid and raw within-draw uniqueness is 7.99. Raw per-prompt completion equality, rather than `draw_std` or equality of aggregate endpoints, establishes this observed collapse; it does not rule out unobserved low-probability outputs.

The logs also record masked-mean policy token entropy on sampled response tokens. The following descriptive values first average the 384 updates within each pass and then the five matched seeds:

Method	Training pass							
	1	2	3	4	5	6	7	8
Dr.GRPO	.679	.005	.003	.002	.002	.002	.002	.002
ReplayDr.GRPO	.325	.082	.088	.098	.087	.080	.072	.062

The first-pass mean includes the control’s rapid transition; after pass 1 its token entropy is near zero, whereas replay retains a larger, nonzero value. Token entropy is response-length dependent and is not canonical-key entropy, so we treat it as mechanism corroboration rather than a task-support endpoint.

Dynamic sampling that resamples until a group is mixed, larger G , reference KL, and token-entropy regularization directly target the same zero-gradient or collapse pathology. They are not present in the registered factorial and remain necessary controls for deciding whether replay adds value once fresh on-policy signal is kept alive. Likewise, the current arms have no

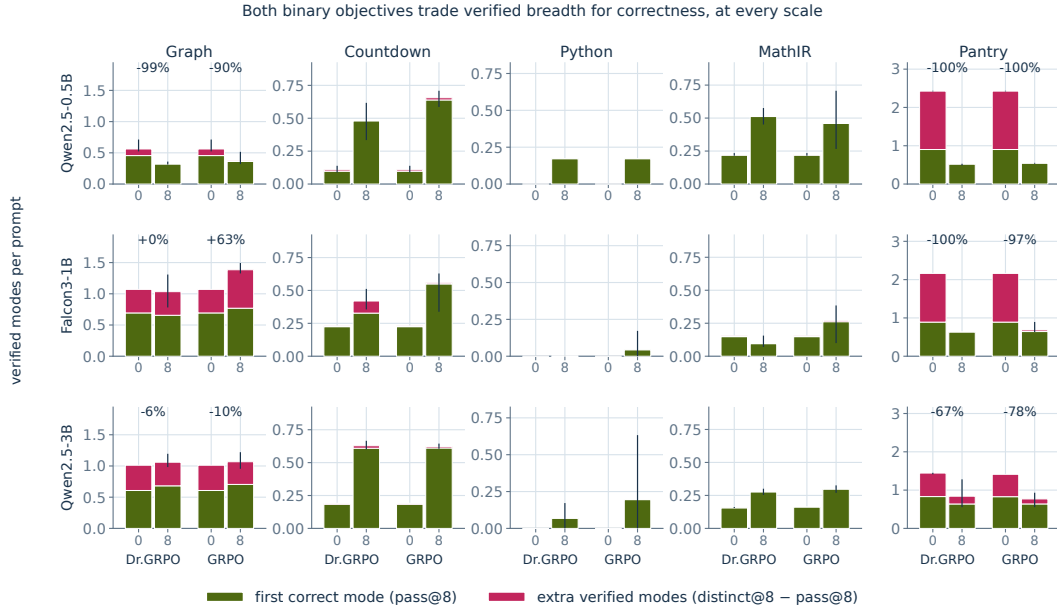


Figure 10: **Both verifier-only objectives reduce verified solution breadth, at every scale.** Bars split `distinct@8` into the first correct mode and every verified mode beyond it, at training pass 0 and pass 8, for unmodified Dr.GRPO then GRPO in each panel. All 150 evaluations are complete at five seeds and four fixed $K = 8$ draws; whiskers are the seed range and pass 0 is measured per method. Badges give the change in extra verified modes, printed only where pass-0 breadth exceeds .05 of them; the other three domains start at .011 or less.

Table 7: **Absolute references for raw `distinct@8`.** *Uniform* samples the exposed action representation uniformly and is defined only for Graph (27 color strings) and PantryPlan (64 bit masks). Frozen is the matched pass-0 model; the final two columns are pass-8 means. Dashes avoid pretending that an open-ended program or derivation space has a canonical uniform distribution.

Model	Domain	Uniform	Frozen	Dr.GRPO	ReplayDr.GRPO
Qwen2.5-0.5B	Graph coloring	1.633	.561	.325	2.438
	PantryPlan	2.153	2.426	.522	1.539
Falcon3-1B	Graph coloring	1.633	1.070	1.034	1.717
	PantryPlan	2.153	2.162	.633	1.576
Qwen2.5-3B	Graph coloring	1.633	1.012	1.060	1.637
	PantryPlan	2.153	1.445	.841	1.514

temperature/top- p decoding sweep; that inexpensive robustness check would distinguish recoverable low-probability tails from collapse that persists under wider decoding.

B.6 COLLAPSE IS REGRESSION, NOT A SMALLER GAIN

Aligned methods share a pass-0 reference wherever they inherit the same initial model and the same fixed evaluation requests. That holds for every Qwen2.5-0.5B and Falcon3-1B domain and for three of the five Qwen2.5-3B domains. On Qwen2.5-3B MathIR and PantryPlan, E80-R1 seeds 71–74 carry marginally different pass-0 requests from the plain-GRPO method (.154 against .162, and .832 against .828 `pass@8`), so Figure 10 measures pass 0 within each method rather than drawing one shared origin. Table 8 gives the macro-average final values the main text quotes.

At Qwen2.5-0.5B, PantryPlan’s bit-uniform `distinct@8` reference is 2.153 (2.714 on instruction-following support), above every final model; Graph’s reference is 1.633, which ReplayDr.GRPO reaches or exceeds. We therefore use PantryPlan as an enumerable stress test of support concentration, not as a claim of complete coverage. The table also identifies .108 as Countdown’s frozen pass-0 `distinct@8`; its terminal Dr.GRPO control is .487.

Table 8: **Problem precheck, five-domain macro-averages.** Both verifier-only objectives at pass 0 and pass 8, all three scales, five seeds and four fixed $K = 8$ draws per model–domain–method comparison. “Extra modes” is `distinct@8-pass@8`; “removed” is the fraction of that quantity lost between pass 0 and pass 8. Breadth loss is present at every scale and is far from uniform across scales, and at Falcon3-1B Dr.GRPO macro `pass@8` falls as well.

Model	Dr.GRPO			GRPO		
	pass@8	extra modes	removed	pass@8	extra modes	removed
Qwen2.5-0.5B	0.335 → 0.402	0.329 → 0.002	99.4%	0.335 → 0.435	0.329 → 0.006	98.2%
Falcon3-1B	0.391 → 0.342	0.332 → 0.094	71.6%	0.391 → 0.454	0.332 → 0.135	59.4%
Qwen2.5-3B	0.356 → 0.455	0.205 → 0.120	41.5%	0.356 → 0.490	0.199 → 0.101	49.4%

At Qwen2.5-0.5B, both verifier-only estimators regress on the two domains with substantial initial breadth: Dr.GRPO moves from .561 → .325 on Graph and 2.426 → .522 on PantryPlan; ordinary GRPO moves from .561 → .372 and 2.426 → .545, respectively. The other three domains begin with at most .011 extra verified modes per prompt on average, leaving little initial breadth to lose.

Across the 15 evaluable Dr.GRPO panels, six finish below their pass-0 `distinct@8`: the two Qwen2.5-0.5B comparisons above; Falcon3-1B Graph 1.070 → 1.034, MathIR .154 → .097, and PantryPlan 2.162 → .633; and Qwen2.5-3B PantryPlan 1.445 → .841. Training on verified reward alone can therefore destroy correct-mode breadth already present in the base model. ReplayDr.GRPO ends below its pass-0 reference in only two of the 15 available domain summaries, both PantryPlan; the Falcon Countdown summary uses four admissible seeds (Appendix B.1).

The main-text precheck reads the matched Dr.GRPO controls (E78, E79, E80-R1) against the plain-GRPO cohort (E95 at all three scales, with the E114 seed extension supplying Qwen2.5-3B seeds 71–74), requiring complete four-draw evaluations at steps 0 and 3,072 for all five domains and all five registered seeds in each of the six method-by-scale blocks. All 150 evaluations are present, so no panel of Figure 10 is partial. A retained-breadth percentage is reported only where pass-0 extra modes exceed .05 per prompt: Graph and PantryPlan clear that floor at every scale, and Countdown, MathIR, and Python factors never reach .011, where one additional verified mode in one draw of one prompt already moves the four-draw mean by .002.

C SEMANTIC MAXENT

Policy entropy can collapse during RL for reasoning, motivating explicit exploration control (Cui et al., 2025). Outcome-based exploration uses answer frequencies to encourage rarely sampled outputs (Song et al., 2025). Semantic MaxEnt applies this motivation to validator-positive execution keys: an entropy-inspired bonus favors underrepresented verified outcomes in fresh rollouts.

The distinction between fresh exploration and retention motivates our main method. The Semantic MaxEnt bonus acts only on outcomes present in the current rollouts; it supplies no direct reinforcement for a remembered mode that is absent. ReplayMaxRL assigns these roles to separate mechanisms: MaxRL targets success across sampling budgets (Tajwar et al., 2026), while uniform replay revisits verified exemplars independently of their current sampling frequency. The Dr.GRPO/MaxRL × replay factorial tests whether this persistent memory adds value beyond the fresh-rollout objective. Semantic MaxEnt remains a direct comparison with entropy-inspired exploration.

C.1 SCORE-FUNCTION MOTIVATION

For prompt x , let $q_\theta(a) = p_\theta^+(a | x)$ denote the policy conditioned on validator-positive canonical outcomes. If this distribution were known, its entropy gradient would satisfy

$$\nabla_\theta H(q_\theta) = \mathbb{E}_{a \sim q_\theta} [(-\log q_\theta(a)) \nabla_\theta \log q_\theta(a)]. \quad (33)$$

The implementation cannot observe q_θ and instead uses a detached, smoothed empirical predictor. Equation 33 therefore motivates the redistribution direction; it is not a claim that the implemented estimator is an unbiased entropy gradient.

C.2 ESTIMATOR

For each prompt, the estimator counts validator-positive canonical keys in prior policy history and successful leave-one-out peers. If K observed keys have total count N , a unit pseudocount and one reserved unseen bucket define

$$\hat{p}(a) = \frac{n(a) + 1}{N + K + 1}, \quad \hat{p}(\text{UNSEEN}) = \frac{1}{N + K + 1}. \quad (34)$$

For an eligible successful row, the added detached advantage is

$$A_{i,t}^{\text{sem}} = \eta \frac{\min\{-\log \hat{p}(a_i), C\} - \mathbb{E}_{\hat{p}}[\min\{-\log \hat{p}(a), C\}]}{C}, \quad \eta = .10, C = 5. \quad (35)$$

Invalid and unsuccessful rows receive zero. The centered score is bounded by η in magnitude. The unseen bucket assigns finite score to a newly observed key without identifying an undiscovered solution. The fixed estimator has heterogeneous effects across domains and scales; these comparisons do not establish a general entropy or support-retention guarantee.

Predictor centering can change correctness as well as diversity. Let $q = p(\cdot \mid \mathcal{C})$ and freeze a predictor score s_c and baseline b independently of the scored action (for example, conditional on its leave-one-out peers). Then

$$\mathbb{E}_{A \sim p}[\mathbf{1}\{A \in \mathcal{C}\}(s_A - b)\nabla_z \log p_A] = P\mathbb{E}_{c \sim q}[s_c \nabla_z \log q_c] + (\mathbb{E}_q[s] - b)\nabla_z P.$$

Indeed, on correct actions $\nabla \log p_c = \nabla \log q_c + \nabla \log P$, and $\mathbb{E}_q[\nabla \log q_c] = 0$. The second term vanishes when the baseline equals the current conditional mean score, which a historical predictor need not satisfy. Even with unclipped $s_c = -\log q_c$ and $b = H(q)$, averaging over all unconditional rollout rows gives $P\nabla H(q)$, rather than $\nabla H(q)$.

For an exact code-matched counterexample, suppose that there is one correct key and at least one incorrect category. Conditional verified-key entropy is identically zero. With an explicit historical-plus-peer count $N \geq 1$, the implemented open-set predictor has seen mass $u = (N + 1)/(N + 2)$ and unseen mass $v = 1/(N + 2)$. At the fixed $C = 5$ its advantage on the seen key is

$$-\frac{\eta v}{C}(\min\{\log(N + 2), C\} + \log u) < 0.$$

Thus an all-success same-key group receives nonzero negative semantic pressure despite a zero conditional-entropy gradient. This establishes estimator mismatch without determining the long-run behavior of the resulting stochastic policy updates.

C.3 VERIFIED-SUPPORT DISCOVERY COMPARATOR

The discovery comparison in Figure 11 augments the row-local support with already verified replay keys and keys found by an isolated proposal sampler. For row i , let H_i be successful policy history, P_{-i} its successful leave-one-out peers, and R_i the verified replay/proposal support. Then

$$S_i = \text{supp}(H_i) \cup \text{supp}(P_{-i}) \cup R_i \cup \{a_i\}, \quad n_i(z) = H_i(z) + P_{-i}(z),$$

and, with $N_i = \sum_z n_i(z)$,

$$q_i(z) = \frac{n_i(z) + 1}{N_i + |S_i|}, \quad A_i^{\text{sem}} = .10 \left[\frac{\min\{-\log q_i(a_i), 5\}}{5} - \sum_{z \in S_i} q_i(z) \frac{\min\{-\log q_i(z), 5\}}{5} \right]. \quad (36)$$

Proposal-only keys contribute support but not counts; proposal rows never enter PPO or policy history. A response enters replay only after the ordinary task verifier accepts it. This bundled comparison adds both proposal sampling and semantic regularization to replay. It does not isolate their effects or directly measure discovery or survival of individual modes. Adaptive controllers, group-centering diagnostics, and open-bank development branches are excluded.

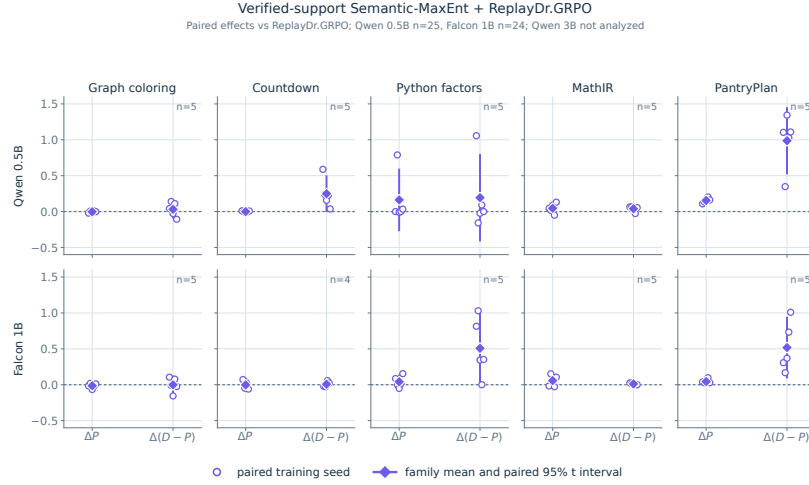


Figure 11: **Bundled verified-support comparison.** Effects compare ReplayDr.GRPO augmented with semantic regularization and proposal sampling against ReplayDr.GRPO after eight passes. Qwen uses $n = 5$; Falcon uses $n = 5$ except Countdown ($n = 4$). The historical exploratory amendment retains its $n = 4$ interval policy, separately from the primary analyses (Appendix B.1). This bundled historical comparison provides exploratory support for discovery interventions and does not isolate the components of the primary ReplayMaxRL method.

Across 49 integrity-valid pairs, the mean correctness/adjusted-breadth gains are $+0.072/+0.300$ for Qwen2.5-0.5B and $+0.026/+0.217$ for Falcon3-1B. PantryPlan is the only domain whose intervals exclude zero on both measures at both scales; the remaining effects are heterogeneous.

D EXTERNAL PYTHON VERIFICATION CONTRACT

Main-body link. This appendix expands the Python Factors contract in Section 4 and its limitation in Section 8, where observed behavior rather than source-code identity defines a Python mode.

D.1 ACCEPTED LANGUAGE AND STATIC BOUNDS

Every candidate is stripped, limited to 240 characters, and parsed by `ast.parse` in expression mode. It must be exactly one `lambda n: EXPR`: positional-only arguments, defaults, keyword arguments, variadic arguments, additional parameters, and any parameter name other than `n` are rejected. The complete AST is capped at 64 nodes. Task specifications contain two to eight distinct integer inputs in $[4, 1000]$ and are rejected unless their exact behavior space contains at least two valid output vectors.

The expression grammar admits integer literals with absolute value at most 10,000; the input variable `n`; `+`, `-`, `*`, floor division, and remainder; integer comparisons; Boolean `and/or`; unary plus, minus, and `not`; and conditional expressions. Boolean and non-integer literals are rejected. Calls, imports, attributes, comprehensions, containers, subscripts, assignment, loops, and every identifier except `n` have no accepted AST node and therefore cannot reach execution. The grammar is defined once in parent-importable contract code, and the external worker applies it immediately before compilation.

D.2 EXTERNAL PROCESS AND FAIL-CLOSED SEMANTICS

The trainer never evaluates model-written code. A locked parent client starts the same Python executable with isolated mode `-I`, a fresh process session, and a persistent JSON-lines worker. Each request contains only the restricted source and trusted task JSON. The worker reparses the candidate, compiles it in expression mode, evaluates it with `__builtins__={}`, and runs every frozen input under a 0.25-second real-time alarm. The parent imposes a separate one-second response deadline.

The worker returns one JSON object containing either `valid=true`, the integer output vector, and its canonical key, or `valid=false` and an exception class. A worker timeout, parent timeout, empty or malformed output, broken pipe, non-integer output, or failed divisor check yields $\perp$ and reward zero. Broken pipes, operating-system errors, and malformed responses terminate the worker and permit one clean restart; a timed out request is failed immediately. Worker stdout is never interpreted as a candidate answer, and no exception path manufactures a key.

D.3 CORRECTNESS AND BEHAVIORAL IDENTITY

For each frozen input n_j , the returned value d_j must be an integer but not a Boolean and must satisfy $1 < d_j < n_j$ and $n_j \bmod d_j = 0$. Correctness is conjunctive across the complete input list. The canonical mode key is the ordered vector

$$\text{python_factor}: d_1, \dots, d_m.$$

Consequently, syntactically different functions that execute to the same vector merge, while a difference on any frozen input creates a different mode. The exact support size used only for dataset validation is $\prod_j |\{d : 1 < d < n_j, d \mid n_j\}|$; neither that catalogue nor its size is given to the policy, bank, scheduler, or stopping rule.

For inputs $[6, 10, 15]$,

```
lambda n: 2 if n % 2 == 0 else 3
```

returns $(2, 2, 3)$, while

```
lambda n: 3 if n % 3 == 0 else 5
```

returns $(3, 5, 3)$. Both verify and receive different behavior keys.

E DOMAIN PROMPTS

Main-body link. This appendix reproduces the exact prompt the policy receives in each of the five MODEBENCH domains of Section 4 and Table 9.

Every domain is single-turn: one system message, one user message, and no few-shot examples. The user message is the dataset `problem` field; the wrapper is the chat template the launcher exported through `OAT_ZERO_PROMPT_TEMPLATE`, defined in `src/oat_drgro/templates.py`. Both methods of every domain use the identical string, and the same template is applied at training and at evaluation, so the prompt is never a difference between methods. Each block below is the first row of that cohort’s frozen evaluation split, rendered by that template; `ops/check_paper_domain_prompts.py` regenerates these blocks from the datasets and fails closed if the typeset text drifts from them. Lines longer than the column are wrapped here with a two-space continuation indent; joining each indented continuation to its predecessor with one space recovers the string byte for byte, and the generator asserts that round trip. The only real newlines are the ones inside the PantryPlan pantry listing.

E.1 GRAPH COLORING

Template `qwen_boxed`; 384 training and 128 evaluation
rows in `var/data/graph_coloring_modebench_v2`.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
Color this graph with vertices 1 through 6. The undirected edges are: (1,2), (1,3), (3,5),
(3,6), (4,6), (5,6). Assign each vertex one color from {1, 2, 3} so that no edge has the
same color on both endpoints. A partial coloring is 2??1?2, where ? means uncolored.
Keep every shown digit fixed. There are exactly 3 question marks. Return exactly one
final answer and no explanation: exactly 3 digits, each one 1, 2, or 3, for the missing
positions from left to right, inside \boxed{}.<|im_end|>
<|im_start|>assistant
```

E.2 COUNTDOWN

Template `qwen_boxed`; 384 training and 128 evaluation
rows in `var/data/exact_countdown_easy3_probe`.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
Using the numbers [3, 6, 9], create an arithmetic expression that equals 18. Use each
given number exactly once. You may use +, -, *, /, and parentheses. Return exactly one
final answer and no explanation: the expression inside \boxed{}.<|im_end|>
<|im_start|>assistant
```

E.3 PYTHON FACTORS

Template `qwen_boxed`; 384 training and 128 evaluation
rows in `var/data/python_factor_modebench_v1`.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
Write one pure Python function with the exact form lambda n: EXPR. An external Python tool
will call it once for each n in [18, 82, 91, 93]. For every call, return an integer d
satisfying 1 < d < n and n % d == 0. Different valid return vectors are different
solution modes. You may use integer literals, n, +, -, *, /, //, %, comparisons, Boolean
operators, and conditional expressions; calls, imports, attributes, containers, and
other names are forbidden. Return exactly the one-line lambda inside \boxed{} and no
explanation.<|im_end|>
<|im_start|>assistant
```

E.4 MATHIR ACTION MENU

Template gwen_boxed; 384 training and 128 evaluation rows in var/data/mathir_action_menu_v1.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
MathIR linear-menu-v1. Solve for x by choosing an executable action sequence. Every
selected action is applied exactly to BOTH sides of the current equation, followed by
exact simplification. Bindings: a=2, b=-9, c=-1. Initial equation: x/a + b = c. Action
menu: A means div(a); B means add(b); C means sub(b); D means sub(c); E means
sub(mul(a,b)); F means mul(a). Choose 1-4 action IDs. Illegal operations, repeated
states, and paths that do not finish with x isolated are rejected. Return only the IDs
separated by semicolons inside the answer box; for example formatting only: \boxed{A;C}.
Do not return x's numeric value.<|im_end|>
<|im_start|>assistant
```

E.5 PANTRYPLAN

Template gwen_pantry_support_mask; 384 training and 128 evaluation rows in var/data/pantry_plan_modebench_v2.

```
<|im_start|>system
Return only six binary digits and no other text. Each digit maps to the corresponding
Pantry row in printed order. A trusted environment will choose exact quantities on
precisely the selected rows.<|im_end|>
<|im_start|>user
Create one feasible high fiber snack from this frozen pantry.
Nutrition values are per 100 g. Quantities are exact integer grams.

Pantry:
- almonds: available=100g, step=25g, minimum-if-used=50g; energy=584kcal, protein=21.5g,
  fiber=10.8g, sodium=0.0mg; tags=tree_nut
- banana: available=125g, step=25g, minimum-if-used=50g; energy=97.0kcal, protein=0.74g,
  fiber=4.62g, sodium=0.0mg; tags=none
- carrots: available=150g, step=25g, minimum-if-used=50g; energy=45.0kcal, protein=0.941g,
  fiber=3.1g, sodium=86.6mg; tags=none
- grape_tomatoes: available=100g, step=25g, minimum-if-used=50g; energy=27.0kcal,
  protein=0.83g, fiber=2.1g, sodium=6.0mg; tags=none
- navel_orange: available=150g, step=25g, minimum-if-used=50g; energy=47.0kcal,
  protein=0.91g, fiber=2.0g, sodium=9.0mg; tags=none
- sunflower_seeds: available=125g, step=25g, minimum-if-used=50g; energy=571kcal,
  protein=18.9g, fiber=7.22g, sodium=0.0mg; tags=none

Use 2 to 4 ingredients.
Forbidden tags: none
Targets:
- mass_g: minimum 125, maximum 200
- energy_kcal: minimum 239.4, maximum 432.25
- protein_g: minimum 6.423
- fiber_g: minimum 3.478
- sodium_mg: maximum 37.6

Choose the ingredient support as a six-bit mask in the exact Pantry row order. Bit 1
includes that row and bit 0 excludes it.<|im_end|>
<|im_start|>assistant
```

F DATASET AND IDENTITY VALIDATION

Main-body link. This appendix documents the materialization and identity claims in Section 4 and Table 9, including the checks that bind correctness and identity to one execution.

F.1 EXECUTABLE CORRECTNESS AND IDENTITY INVENTORY

Table 9: **MODEBENCH**. Correctness and identity come from one executable path; training starts without the mode catalogue. *Valid modes* is the mean number of correct execution modes per evaluation prompt, exact over the released multi-answer split. Every domain trains on 384 prompts and evaluates on a fixed 128-prompt split; the exact prompt each policy receives is reproduced verbatim in Appendix E for a direct, byte-level audit of the evaluation interface.

Domain	Generated answer / check	Executed key	Aliases that merge	valid modes
Graph coloring	Complete assignment; check fixed colors and edges	Color vector	Spacing only	6.27
Countdown	Parse operands; execute exactly	Normalized AST	Commutative order	4.52
Python factors	Execute restricted lambda externally	Return vector	Source programs with same behavior	229.44
MathIR	Execute bounded action program	State trajectory	Programs with the same states	5.00
PantryPlan	Exact ingredient allocation; check all bounds	Ingredient support	Quantity and ordering aliases	18.19

F.2 DETERMINISTIC SPLIT MATERIALIZATION

Table 10: Deterministic materialization identities. For every Python row, two independently executed functions must verify with different keys. Every MathIR certified route is executed by the same interpreter used for model outputs. Together these checks bind both acceptance and identity to deterministic execution for every row in the released split.

	Python factors	MathIR action menu
Training rows	384	384
Evaluation rows	128	128
Exact modes per prompt	16–3,600	5
Training-row SHA-256	bcfa9acc...37cf7	bfc98362...5d3835
Evaluation-row SHA-256	be0f621c...94d183	df5783dd...328b96
Gold catalogue supplied to policy	no	no
Training-bank initialization	empty	empty

F.3 MATCHED LEVEL-2 CONSTRUCTION

ModeBench currently releases two levels for each domain. Level 2 changes the problem structure while preserving the Level-1 verifier, canonicalizer, split sizes, and exact per-split histogram of valid-mode counts. Its 384/128/128 train/development/evaluation identities are disjoint from every Level-1 and Level-2 split. Consequently, a level comparison changes difficulty without silently changing the amount of valid support available to be found.

Domain	Level 1	Level 2 structural change
Graph coloring	Three hidden vertices	Four hidden vertices
Countdown	Three operands	Four operands, with values up to 14
Python factors	Four executed cases, all at most 96	Four cases, including a value above 96
MathIR	One-sided linear families	Variable-on-both-sides families
PantryPlan	Base dietary/composition constraints	More active dietary exclusions and tighter composition

Before treatment training, frozen Qwen2.5-0.5B evaluation admitted a Level-2 domain only if its `pass@8` was in $[.10, .90]$ and strictly below the paired Level-1 value under identical decod-

ing. The released manifest `var/data/modebench_harder_v2_matched_r5/identity.json` records the support histograms, row hashes, construction seeds, disjointness checks, and admission receipts. The same generator-and-admission recipe can create further levels; the benchmark definition is not restricted to the two instantiated here.

F.4 PANTRYPLAN SPLIT VALIDATION

The deterministic v2 release contains 384 train, 64 development, and 128 evaluation prompts across four formulation families; exact exhaustive support counts range from 8 to 45. The reported comparison uses the full 384-prompt training split and the fixed 128-row evaluation split, whose 2,328 feasible supports average 18.1875 per prompt. The train/evaluation row hashes are `a4883c5e...a757f` and `5d81ba8d...5923`. The terminal validation status is `pass` with decision `pantry_terminal_eligible`. Correctness checks exact inventory, serving steps, mass, nutrient bounds, and dietary tags; identity is ingredient support. The benchmark does not claim that a verified allocation is a recipe, palatable, cookable, or safe.

G ALGORITHMIC DETAILS

Main-body link. This appendix formalizes the replay state transition summarized in Section 5.

For each prompt x , let $\mathcal{B}_x$ be the capacity- M verified-exemplar bank and let τ be the recurrent replay cursor, which makes the next bank selection exactly resumable after interruption.

G.1 STATE AND OPTIMIZER UPDATE

Algorithm 1 One ReplayMaxRL optimizer update

Require: Policy π_θ ; prompt x and verifier specification s_x ; group size G ; fresh objective $o \in \{\text{MaxRL}, \text{Dr.GRPO}\}$; replay weight λ_{replay} ; states $(\mathcal{B}, \tau)$

Ensure: Atomically updated $(\theta', \mathcal{B}', \tau')$

- 1: Draw the on-policy group $\mathcal{G} = \{y_i \sim \pi_\theta(\cdot | x)\}_{i=1}^G$
- 2: **for** $i = 1, \dots, G$ **do**
- 3: Execute $\tilde{c}_i \leftarrow V(y_i, s_x)$ and observe task reward r_i
- 4: Set $c_i \leftarrow \tilde{c}_i$ iff $r_i > 0 \wedge \tilde{c}_i \neq \perp$; otherwise set $c_i \leftarrow \perp$
- 5: **end for**
- 6: Compute fresh task advantages A_i^o from $\{r_j\}_{j=1}^G$; for ReplayMaxRL use the formula in Section 5.1
- 7: After the full group is validated, deterministically retain one policy exemplar for each new $c_i \neq \perp$ in $\mathcal{B}_x$, up to capacity M
- 8: $(z, \tau') \leftarrow \text{NEXTBANK}(\{\mathcal{B}_u\}_u, \tau)$ by a fixed round-robin cursor
- 9: **if** a replay bank $\mathcal{B}_z$ exists **then**
- 10: For each retained exemplar e_c , compute $s_c = |e_c|^{-1} \log \pi_\theta(e_c | z)$
- 11: $L_{\text{replay}} \leftarrow -|\mathcal{B}_z|^{-1} \sum_{c \in \mathcal{B}_z} s_c$
- 12: **else**
- 13: $L_{\text{replay}} \leftarrow 0$
- 14: **end if**
- 15: $w_{\text{rep}} \leftarrow (G - 1)/G^2$ for one replay pseudo-group per G on-policy rows
- 16: $\theta' \leftarrow \text{UPDATE}(\theta, L_{\text{PPO}}(A^o) + w_{\text{rep}} \lambda_{\text{replay}} L_{\text{replay}})$
- 17: Checkpoint $(\theta', \mathcal{B}', \tau')$ as one atomic state

G.2 BANK ADMISSION, DETERMINISM, AND CAPACITY

A bank is keyed by the SHA-256 digest of the exact prompt-token sequence. Only loss-active rows with positive task reward and a nonempty validator key are eligible. The full rollout group is validated before new keys are committed, so candidate admission uses the bank state from before the group.

For each newly observed key, candidates are normalized to token tuples and the lexicographically smallest eligible response is stored. New keys are processed in sorted-key order, making admission invariant to rollout-row order. The bank holds at most one exemplar per canonical key and at most $M = 16$ exemplars per prompt. Existing exemplars are never evicted or overwritten by a later surface form; once capacity is full, later keys may remain in diagnostic counts but cannot enter the replay support. A prompt-hash collision or a changed token sequence under an existing prompt hash is a hard error.

G.3 RECURRENT SCHEDULING AND VERIFIED-LIKELIHOOD LOSS

At each optimizer update the global scheduler sorts the nonempty prompt-bank hashes and selects one by a round-robin cursor. Singleton banks are eligible: verified likelihood can preserve a discovered success before a second mode is found. The cursor advances independently of reward, evaluation, desired support, or bank entropy and is serialized with the bank, so an interruption cannot silently resample the schedule.

For the selected bank, stored prompt and response tokens are materialized unchanged into a right-padded teacher-forcing batch. Prompt and padding tokens are masked; each response receives its mean token log probability s_c . Uniform verified likelihood averages $-s_c$ first across modes within the selected prompt, then applies the registered replay coefficient and the per-rollout scaling shown in Algorithm 1. Thus each stored mode has equal target weight even if its on-policy frequency was high when discovered. Exhaustive support and evaluation outputs never enter the loss.

G.4 MATCHED CONTROL, TELEMETRY, AND RESUME STATE

The matched control uses the same validation, bank-update, scheduling, replay-materialization, and score-pass code, but replaces the replay score-gradient vector by exact zeros. The treatment changes that derivative from zero to live. Policy-dependent samples, bank contents, selected banks, exemplar lengths, and realized computation can subsequently differ; sharing the code does not hold those quantities fixed. Both methods log available and replayed banks, retained modes, realized token counts, raw and applied replay losses, score-gradient sums and norms, scheduler phase, and whether the exact-zero switch was active.

Model, optimizer, bank counts, exemplar tokens, replay cursor, data cursor, and request-stream state are checkpointed as one resume contract. The loader checks the schema, replay capacity, scheduler settings, estimator constants, token types, prompt hashes, exemplar/support consistency, and cursor bounds before accepting a checkpoint.

H REPLAY MECHANISM CHECKS

The intervention changes whether the replay derivative is applied. Treatment and control share verifier, bank-admission, round-robin scheduling, replay-materialization, and score-pass code. The control zeros the replay derivative; the treatment applies it. The resulting policies can build different banks and incur different replay workloads. ReplayMaxRL and ReplayDr.GRPO differ only in the fresh-rollout advantage. No semantic-entropy term, proposal model, novelty reward, adaptive replay coefficient, reference KL, or gold support is active.

H.1 BANK AND SCHEDULER TELEMETRY

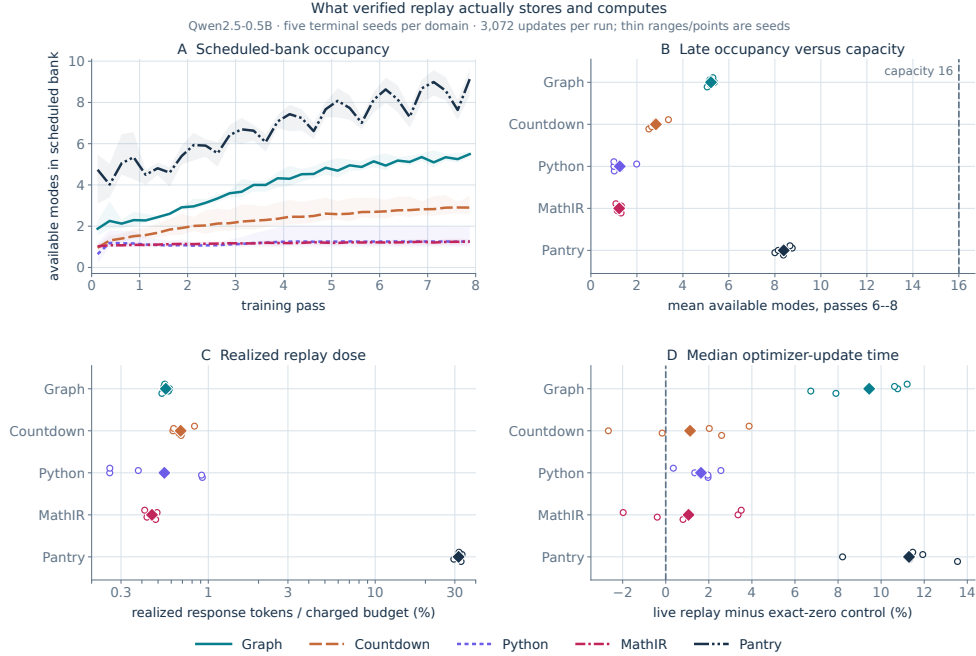


Figure 12: The verified replay path is active and bounded. Telemetry summarizes all 25 terminal Qwen2.5-0.5B ReplayDr.GRPO runs and their matched controls. It records bank availability and occupancy, capacity hits, realized replay-token dose, and paired optimizer time. These measurements verify the replay intervention. Testing individual-mode preservation additionally requires longitudinal scores for the same stored exemplars throughout training.

The bank stores at most one surface exemplar per verified canonical key and at most 16 keys per prompt. Identity-level survival histories were not retained, so occupancy cannot be interpreted as a mode-survival curve and Theorem A.12’s individual-mode prediction is not tested by the experiments

in this paper. Existing fingerprints verify key/weight alignment but do not record each fixed exemplar’s longitudinal policy score; the disjoint evaluation prompts cannot supply that missing measurement. The registered E121 fixed-bank survival study addresses this gap by freezing a Graph bank after one pass and recording per-exemplar score trajectories (paper/preregistration/e121_fixed_bank_survival_telemetry_20260903.md). It is planned mechanism evidence; no E121 empirical result is reported here. The present telemetry is used only to verify that replay was scheduled, nonzero in treatment, exactly zero in controls, and within its registered resource envelope. Earlier bank-normalization, starvation-fallback, and open-bank experiments altered multiple mechanisms and are excluded from the appendix because they do not identify the ReplayMaxRL effect reported here.

I ALIGNED SCALE EXTENSIONS

I.1 MATCHED DESIGN

The scale extensions preserve the five ModeBench domains, paired initialization and data identities, eight-pass horizon, evaluation grid, verifier, bank capacity, replay scheduling code, and recovery rules. Registered seeds are 55–59 for Falcon3-1B and 70–74 for Qwen2.5-3B. The Falcon Countdown ReplayDr.GRPO source at seed 59 is excluded from efficacy summaries (Appendix B.1). Hardware and optimizer settings are matched within a model–domain–seed comparison. Results are reported separately by model and domain; no missing result is imputed and no cross-model aggregate is interpreted as a scaling law.

I.2 TERMINAL PAIRED EVIDENCE

Figure 4 in the main text gives the paired ReplayDr.GRPO effects. Table 11 records the corresponding exact matched final results.

Table 11: **Matched results after eight training passes.** Columns compare Dr.GRPO with ReplayDr.GRPO after exactly eight training passes. Entries are means over the displayed paired-seed intersection; $n@pass$ gives its size and training pass. P, M, and D denote $pass@8$, $mean@8$, and $distinct@8$. All source-admissible terminal pairs are included: Falcon Countdown has $n = 4$, other blocks $n = 5$ (Appendix B.1). No missing endpoint is imputed. Pale green marks a ReplayDr.GRPO mean that is strictly higher than its matched Dr.GRPO mean.

Model	Domain	$n@pass$	Dr.GRPO			ReplayDr.GRPO		
			P	M	D	P	M	D
Qwen2.5-0.5B	Graph coloring	5@8.0	.323	.323	.325	.969	.404	2.438
	Countdown	5@8.0	.480	.465	.487	.672	.597	1.637
	Python factors	5@8.0	.172	.172	.172	.528	.507	.571
	MathIR	5@8.0	.512	.464	.513	.796	.688	.829
	PantryPlan	5@8.0	.522	.522	.522	.729	.545	1.539
Falcon3-1B	Graph coloring	5@8.0	.656	.300	1.034	.809	.334	1.717
	Countdown	4@8.0	.325	.267	.415	.632	.343	.838
	Python factors	5@8.0	.000	.000	.000	.979	.722	1.107
	MathIR	5@8.0	.095	.068	.097	.736	.292	.756
	PantryPlan	5@8.0	.633	.633	.633	.751	.560	1.576
Qwen2.5-3B	Graph coloring	5@8.0	.682	.360	1.060	.836	.349	1.637
	Countdown	5@8.0	.609	.569	.628	.746	.505	.962
	Python factors	5@8.0	.069	.069	.069	.659	.603	.852
	MathIR	5@8.0	.278	.168	.279	.686	.580	.706
	PantryPlan	5@8.0	.638	.558	.841	.744	.546	1.514

I.3 MAXRL FACTORIAL PER-DOMAIN RESULTS

For a common paired seed j and endpoint Y , the factorial interaction is

$$I_j = (Y_j^{\text{ReplayMaxRL}} - Y_j^{\text{MaxRL}}) - (Y_j^{\text{ReplayDr.GRPO}} - Y_j^{\text{Dr.GRPO}}).$$

A positive replay effect under each fresh objective need not give $I_j > 0$. Here “complementary” means that replay adds value under both objectives; it does not assert a positive interaction. Negative interactions can coexist with those gains, including when a bounded endpoint approaches its ceiling. Interaction uncertainty must use these within-seed differences of differences, not differences between separately calculated confidence limits.

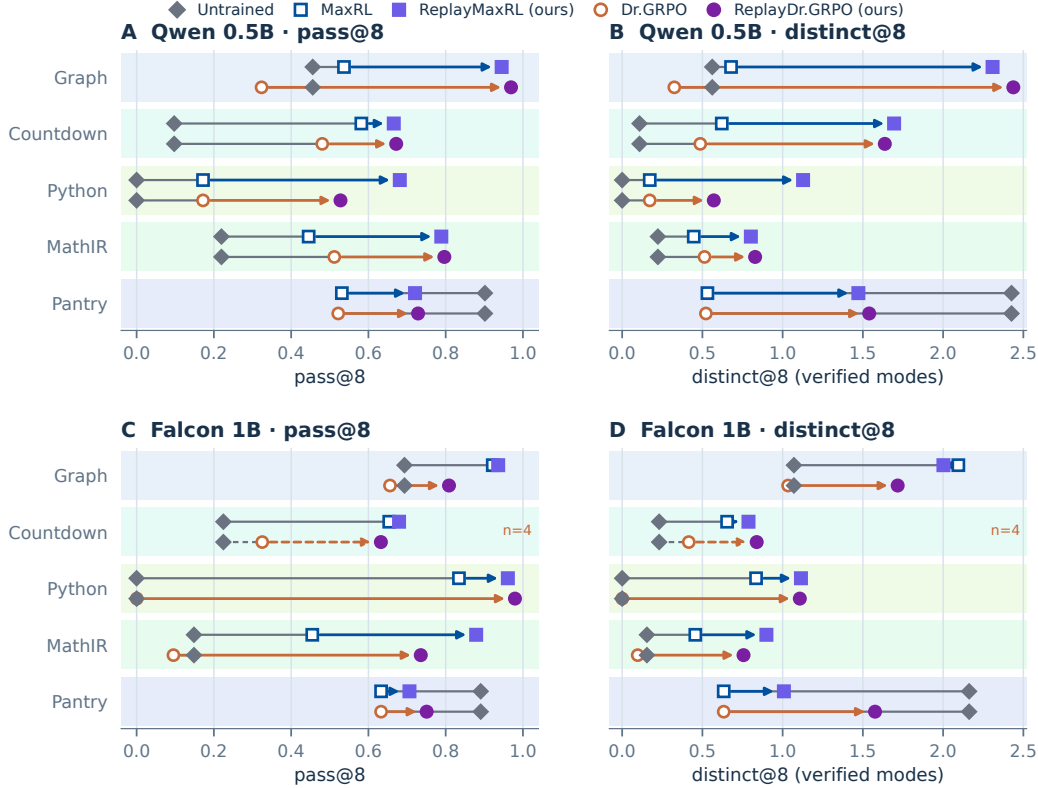


Figure 13: **Per-domain results behind the cross-domain averages in Figure 5.** Panels A–B show Qwen2.5-0.5B and panels C–D show Falcon3-1B. MaxRL–ReplayMaxRL pairs retain five terminal seeds in every domain. Dr.GRPO–ReplayDr.GRPO uses four admissible seeds for Falcon Countdown and five elsewhere. Domain tints match Figure 2. Replay arrows, trained-arm markers, and untrained references use the same encoding as the main figure.

All five Qwen2.5-0.5B ReplayMaxRL–MaxRL blocks are complete. Both endpoint effects are positive in every domain: pass@8 effects range from $+0.084$ to $+0.509$, and raw distinct@8 effects range from $+0.356$ to $+1.630$. All five Falcon MaxRL–ReplayMaxRL blocks are also complete. ReplayMaxRL has a positive mean pass@8 effect in every Falcon domain and a positive raw distinct@8 effect in four of five. Graph is the exception on raw support, with -0.093 modes (unadjusted 95% interval $[-0.229, +0.043]$) and a small correctness effect of $+0.014$ ($[-0.016, +0.044]$). MathIR has the largest Falcon effects: $+0.424$ on pass@8 ($[+0.310, +0.539]$) and $+0.443$ on distinct@8 ($[+0.317, +0.569]$). The within-seed cross-domain effects are $+0.133$ ($[+0.043, +0.223]$) and $+0.228$ ($[+0.094, +0.361]$), respectively. Falcon Python remains inconclusive despite positive means: $+0.127$ ($[-0.372, +0.625]$) and $+0.280$ ($[-0.209, +0.769]$).

The completed Qwen2.5-3B Python block extends the breadth result. All five seeds, 70–74, now have valid pass-8 endpoints for all four arms (Table 12). ReplayMaxRL raises raw distinct@8 relative to MaxRL by $+0.391$ (unadjusted paired 95% interval $[+0.079, +0.704]$); every paired seed has a positive raw-support effect. The mean correctness gain is $+0.277$, but its interval $[-0.037, +0.590]$

includes zero. This supports a domain-specific breadth benefit at the larger model, without establishing a correctness gain or a model-size trend. The other four Qwen2.5-3B MaxRL blocks remain incomplete; no Qwen2.5-3B MaxRL cross-domain average is reported.

Table 12: **Newly completed Qwen2.5-3B Python factorial.** Means use all five registered seeds, 70–74, at exactly pass 8. The paired ReplayMaxRL–MaxRL effects and their unadjusted intervals are reported in the text; arm means alone are not paired effects.

Method	n	pass@8	distinct@8
Dr.GRPO	5	0.069	0.069
ReplayDr.GRPO	5	0.659	0.852
MaxRL	5	0.356	0.356
ReplayMaxRL	5	0.633	0.748

I.4 INTERPRETATION BOUNDARY

At terminal pass 8, the ReplayDr.GRPO contrast has 24 admissible Falcon3-1B pairs and 25 Qwen2.5-3B pairs. Falcon Countdown uses four seeds; the other blocks use five. Raw `distinct@8` is higher in every admissible pair. Correctness-adjusted breadth is strongest on Graph coloring, Countdown, and PantryPlan; Python factors and MathIR more often improve correctness without a consistent breadth effect. These are per-domain, within-scale comparisons. They do not establish monotonic improvement with parameter count. MaxRL–ReplayMaxRL pair results use their own admissible seed sets; full factorial interactions use the four-arm intersection. Incomplete Level-2 blocks remain descriptive and do not support cross-domain generality.

J REPRODUCIBILITY AND CLAIM BOUNDARIES

J.1 INTERIM MATCHED-LEVEL COMPARISON

Figure 6B uses the September 6 frozen evaluation snapshot for Qwen2.5-0.5B-Instruct. For each domain and seed, we select the latest nonnegative training step with complete, valid evaluations for all eight combinations of level and method. Each evaluation contains all four fixed-seed draws of eight samples on 128 held-out prompts. Conflicting repeated evaluations invalidate that checkpoint; no outcome threshold or performance ranking enters selection. Each displayed point first averages draws, then the selected seeds within a domain, then the five domain means with equal weights.

There are 21 matched domain–seed cells. Graph, Countdown, Python, and MathIR use all five seeds, 43–47; PantryPlan uses only seed 45 at step 0 in both levels and all four methods. For seeds 43–47, Graph uses step 3,072 throughout; Countdown uses 960, 192, 96, 192, and 96; Python uses 1,728, 1,248, 3,072, 960, and 3,072; MathIR uses 3,072, 3,072, 1,920, 2,304, and 1,632. These means describe the matched checkpoints available in the snapshot. The complete eight-pass Graph block is reported separately in the text.

The admitted evaluations and exact source-line provenance are frozen in `paper/results/modebench_level_comparison_snapshot.json`. `ops/exp_scaling/plot_paper_modebench_levels.py` regenerates the figure; its adjacent JSON records every selected checkpoint, domain mean, and final average. The initial Pantry values are observed evaluations; missing runs are never assigned a numerical value or replaced by admission measurements.

J.2 COMPARISON REGISTRY

Table 13: **Frozen figure coverage and evidence boundaries for the three experiments.** A complete block contains all five registered paired terminal seeds. Partial or source-excluded intersections are not counted as complete blocks. E120 uses its September 4, 15:40:26 UTC snapshot.

	Question	Registered unit	Complete	Still needed
Exp. 1	Does canonical replay retain verified support?	Level 1; 3 scales $\times$ 5 domains; Dr./ReplayDr.	14/15 blocks; one $n = 4$	retain Falcon Countdown as descriptive after source exclusion
Exp. 2	Does memory add value beyond MaxRL?	Level 1; 3 scales $\times$ 5 domains; MaxRL/ReplayMaxRL	11/15 blocks	four Qwen-3B blocks; full factorial has ten complete blocks and one $n = 4$
Exp. 3	Does the factorial transfer to a harder matched level?	Level 2; Qwen-0.5B; 5 domains $\times$ 4 methods	1/5 blocks; Graph complete; MathIR/Python $n = 2$	finish the other four domains
Ablation	Does fresh-frequency replay differ from uniform key replay?	Level 1; 9 model-domain blocks; 45 cells	5/9 blocks; 25/45 cells	larger-model blocks absent from the frozen E120 snapshot

The primary registry is the 2×2 factorial over fresh objective (Dr.GRPO or MaxRL) and verified replay (off or on). Level 1 covers five domains at Qwen2.5-0.5B, Falcon3-1B, and Qwen2.5-3B; Level 2 repeats the full factorial on a separately generated, support-matched set at Qwen2.5-0.5B. A result enters the paper only at a registered checkpoint shared by the paired methods. Seed prefixes are labeled with exact n and are never promoted to five-seed estimates. GRPO, UCPO, sparse RLEP-Dr, and fixed Semantic MaxEnt remain direct comparators rather than components of the primary factorial. The eleven complete MaxRL–ReplayMaxRL blocks remain valid at five seeds each. Requiring all four arms leaves ten complete five-seed blocks and one four-seed Falcon Countdown block. Cross-objective interactions require the common admissible seed intersection; they are not differences of aggregates with different seed sets.

J.3 DATED ENDPOINT UPDATE: SEPTEMBER 6, 2026

Table 14 records a fresh audit of the registered E118, E119, and E120 ledgers, collected from 16:56:42 to 17:09:35 UTC. Completion receipts and admitted endpoints are counted separately: an endpoint requires the exact step-3,072 evaluation, all four fixed-seed sampled draws, and no conflicting retry values. The accompanying `latest_results_20260906.json` records the collection interval, ledger hashes, admitted seed sets, and evidence boundaries. These counts describe progress across the registered cells; incomplete intersections do not acquire five-seed estimates.

Table 14: **Latest audited terminal coverage.** Complete blocks require five paired MaxRL/ReplayMaxRL seeds for E118, five four-method seeds for E119, and five frequency-replay seeds with their registered uniform comparator for E120. Counts in the frozen-figure registry above refer to their original snapshots and remain fixed for figure reproducibility.

Campaign	Receipts	Endpoints	Complete blocks
E118 MaxRL factorial	116/150	116/150	11/15
E119 Level 2	52/100	52/100	1/5
E120 replay weights	33/45	33/45	6/9

The complete MaxRL/ReplayMaxRL block set and the sole complete Level-2 Graph factorial are unchanged. E118 now has 57 matched terminal pairs, including single Qwen2.5-3B Graph (seed 73) and MathIR (seed 74) pairs; these partial blocks are descriptive and receive no interval. E120 now adds a complete Falcon3-1B Graph block, reported with all paired seeds and uncertainty in Appendix B.4. The original Qwen2.5-0.5B primary weighting analysis and Figure 6’s interim matched-level comparison retain their disclosed frozen inputs. Newly available partial runs remain progress evidence.

J.4 TRAINING AND EVALUATION CONTRACT

Table 15: **Core matched run contract.** Scale-specific systems changes are applied to every method within a paired comparison, so systems changes cannot define a treatment arm.

Component	Frozen setting
Data	384 training and 128 disjoint evaluation prompts per domain and level; no evaluation row enters training or replay.
Rollouts	$G = 16$, temperature 1, top- $p = 1$; one prompt group per update.
Training	Eight passes (3,072 updates), one PPO epoch, $\beta_{KL} = 0$; paired methods share initialization, prompt stream, optimizer, and checkpoint cadence.
Evaluation	Passes 0, .5, . . . , 8; a separate temperature-0 greedy trace and four fixed temperature-1, top- $p = 1$, $K = 8$ draws per prompt for pass@8, mean@8, and distinct@8; no best-checkpoint selection.
Replay	One capacity-16 bank per prompt; one bank selected by a serialized round-robin cursor per update; $\lambda_{\text{replay}} = .10$; controls use the same replay code with an exact-zero derivative; realized banks and workloads can differ.

The pinned revisions are listed separately to keep model identity explicit:

- Qwen2.5-0.5B-Instruct: 7ae557604adf67be50417f59c2c2f167def9a775;
- Falcon3-1B-Instruct: 28ba2251970a01dd1edc7ba7dad2eb71216ccfd;
- Qwen2.5-3B-Instruct: aa8e72537993ba99e69dfaafa59ed015b17504d1.

Training seeds are 43–47, 55–59, and 70–74, respectively. Qwen2.5-0.5B uses a constant 2×10^{-7} learning rate; Falcon3-1B uses the same rate with optimizer state offload; Qwen2.5-3B uses a 10^{-7} peak cosine schedule, 10^{-8} floor, and 10% warmup. Every paired comparison uses the same hardware class within its model–domain–seed block.

Graph coloring and Countdown allow 192 response tokens; Python factors uses 192/512/192 tokens at 0.5B/1B/3B; MathIR uses 64/128/64; PantryPlan uses eight. Evaluation base seeds are 610100, 610200, 610300, 610400, and 76299 in domain order. Sampled draw d uses base seed plus d , with independent responses within each draw. Metrics average over all 128 prompts and four draws; raw prompt responses, per-draw endpoints, and their aggregate spread are retained. Evaluation outputs never enter training, replay, checkpoint choice, or stopping.

J.5 RECOVERY AND FAIL-CLOSED RULES

Model and optimizer state, bank contents, replay cursor, data cursor, and random request-stream state are checkpointed atomically. Resume checks schema, capacity, scheduler settings, objective constants, token types, prompt hashes, exemplar/key consistency, and cursor bounds. Non-finite values, state mismatch, missing completion markers, changed data identities, or aggregate drift are hard failures. Placement-only recovery amendments may change hardware or wall time but not the scientific update.

J.6 ARTIFACTS AND IMPLEMENTATION MAP

Every reported figure has an adjacent machine-readable record containing its source paths, exact seed set, checkpoint rule, displayed values, and SHA-256 hashes. `paper/FIGURE_MANIFEST.md` maps figures to builders and versioned records. The ModeBench example renderer re-executes every displayed coloring, factorization, arithmetic transition, and pantry allocation. Result builders abort on missing matched comparisons or source-hash drift rather than carrying a shallower checkpoint forward.

Under `src/oat_drgrp/`, benchmark environments are implemented in `canonical_actions.py`, `python_modebench.py`, `mathir.py`, and `pantry_plan.py`; isolated Python execution uses `python_modebench_process.py` and `python_modebench_worker.py`. Verified replay and bank state are implemented in `canonical_replay.py` and `online_canonical_bank.py`; learner integration is in `learner/grpo.py` and `learner/run.py`. The post-hoc raw-collapse and token-entropy audit is regenerated by `ops/exp_scaling/build_paper_e78_python_collapse_diagnostics.py`

into paper/results/e78_python_collapse_diagnostics.json. E119
all-domain terminal progress is stored beside Figure 6; E120-R1 is regenerated
by ops/exp_scaling/build_paper_e120_frequency_progress.py into
paper/results/e120_frequency_progress.json and its table body. The reg-
istered E120 breadth and correctness contrasts are reproduced from that frozen snap-
shot by ops/exp_scaling/build_paper_e120_primary_breadth.py. Its
paper/results/e120_primary_breadth.json records source hashes, all paired
seed values, exact bootstrap conventions, and the within-seed five-domain summary;
the two adjacent table bodies reproduce Tables 2 and 3. Dataset manifests under
var/data/ bind every train/evaluation identity and level.